

QFT of free relativistic particles

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1 Relativistic one-particle states and Lorentz transformations

1.1 The physical question

In non-relativistic quantum mechanics, a particle species is specified by a mass, a spin, and possibly internal labels such as charge. In relativistic quantum theory this statement becomes sharper: a one-particle state must carry an irreducible unitary representation of the Poincare group. This is the group generated by spacetime translations and Lorentz transformations.

The Poincare group element is denoted by (Λ, a) , where Λ is a proper orthochronous Lorentz transformation and a^μ is a spacetime translation. It acts on spacetime points as

$$x^\mu \mapsto x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu. \quad (1)$$

The corresponding action on the Hilbert space is implemented by a unitary operator $U(\Lambda, a)$.

The defining covariance condition is

$$U(\Lambda, a)^{-1} P^\mu U(\Lambda, a) = \Lambda^\mu{}_\nu P^\nu, \quad (2)$$

or equivalently

$$U(\Lambda, a)P^\mu U(\Lambda, a)^{-1} = (\Lambda^{-1})^\mu{}_\nu P^\nu, \quad (3)$$

depending on whether one thinks of the transformation as active or passive. What matters physically is that the four-momentum eigenvalue transforms as a Lorentz vector.

1.2 Momentum eigenstates

Let $|p, \sigma\rangle$ denote a one-particle state with four-momentum p^μ and spin label σ . For a massive particle of mass $m > 0$,

$$p^2 = p_\mu p^\mu = m^2, \quad p^0 > 0, \quad (4)$$

where we use the mostly-minus metric

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (5)$$

Thus the physical momenta lie on the positive-energy mass shell

$$\mathcal{H}_m^+ = \{p^\mu : p^2 = m^2, p^0 > 0\}. \quad (6)$$

The momentum operator satisfies

$$P^\mu |p, \sigma\rangle = p^\mu |p, \sigma\rangle. \quad (7)$$

A convenient relativistic normalization is

$$\langle p', \sigma' | p, \sigma \rangle = (2\pi)^3 2p^0 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma}. \quad (8)$$

This normalization is natural because the measure

$$\frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} \quad (9)$$

is Lorentz invariant on the mass shell.

It is useful to keep track of one normalization point, because Weinberg often first writes the one-particle states in a non-covariant, or δ^3 -normalized, basis. I will denote those intermediate states by a subscript W:

$$\langle p', \sigma' | p, \sigma \rangle_W = (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma}.$$

The invariantly normalized states used above are then, up to the harmless common factor $(2\pi)^{3/2}$,

$$|p, \sigma\rangle = \sqrt{2p^0} |p, \sigma\rangle_W.$$

This distinction is the source of the square-root factors that appear in Weinberg's derivation.

1.3 How translations act

Translations are simple because momentum states diagonalize the translation generators. Since

$$U(\mathbf{1}, a) = \exp(+ia_\mu P^\mu), \quad (10)$$

we have

$$U(\mathbf{1}, a) |p, \sigma\rangle = e^{+ia \cdot p} |p, \sigma\rangle. \quad (11)$$

This phase is the quantum version of the plane wave factor. The nontrivial problem is not translations but Lorentz transformations, because Lorentz transformations change the momentum and may also rotate the spin indices.

1.4 The standard momentum and standard boosts

For a massive particle all physical momenta are related by Lorentz transformations to the rest momentum

$$k^\mu = (m, 0, 0, 0). \quad (12)$$

Choose, for every $p \in \mathcal{H}_m^+$, a standard Lorentz transformation $L(p)$ such that

$$L(p)^\mu{}_\nu k^\nu = p^\mu. \quad (13)$$

A common choice is the rotation-free boost taking the rest frame to momentum p . Explicitly, with

$$p^0 = E_{\mathbf{p}} = \sqrt{m^2 + \mathbf{p}^2}, \quad \beta = \frac{\mathbf{p}}{E_{\mathbf{p}}}, \quad \gamma = \frac{E_{\mathbf{p}}}{m}, \quad (14)$$

one may take the following rotation-free boost, written with ordinary Euclidean spatial components $\mathbf{p}$:

$$L(p)^0{}_0 = \frac{p^0}{m}, \quad L(p)^0{}_i = \frac{p^i}{m}, \quad (15)$$

$$L(p)^i{}_0 = \frac{p^i}{m}, \quad L(p)^i{}_j = \delta^i_j + \frac{p^i p^j}{m(p^0 + m)}. \quad (16)$$

Here $i, j = 1, 2, 3$ label spatial directions, and repeated spatial indices in this displayed formula are treated with the Euclidean metric. The important point is simply

$$L(p)k = p. \quad (17)$$

We now use the covariantly normalized one-particle states introduced above. With this convention the natural definition is

$$|p, \sigma\rangle = U(L(p)) |k, \sigma\rangle. \quad (18)$$

There is no momentum-dependent normalization factor in this equation. The factor that appears in Weinberg's δ^3 -normalized basis is a change of basis effect, not an intrinsic part of the Lorentz transformation of covariantly normalized states.

Let us derive this directly from the covariant norm. Put

$$q = \Lambda p, \quad q' = \Lambda p'. \quad (19)$$

The mass-shell measure is invariant:

$$\frac{d^3 \mathbf{q}}{2q^0} = \frac{d^3 \mathbf{p}}{2p^0}. \quad (20)$$

Equivalently, at the point selected by the delta function,

$$d^3 \mathbf{q} = \frac{q^0}{p^0} d^3 \mathbf{p}. \quad (21)$$

Therefore the three-dimensional delta function on the mass shell transforms as

$$\delta^3(\mathbf{q}' - \mathbf{q}) = \frac{p^0}{q^0} \delta^3(\mathbf{p}' - \mathbf{p}). \quad (22)$$

Multiplying by the energy factor in the covariant normalization gives

$$2q^0 \delta^3(\mathbf{q}' - \mathbf{q}) = 2p^0 \delta^3(\mathbf{p}' - \mathbf{p}). \quad (23)$$

Thus

$$\langle q', \sigma' | q, \sigma \rangle = (2\pi)^3 2q^0 \delta^3(\mathbf{q}' - \mathbf{q}) \delta_{\sigma'\sigma} \quad (24)$$

$$= (2\pi)^3 2p^0 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma} \quad (25)$$

$$= \langle p', \sigma' | p, \sigma \rangle. \quad (26)$$

This is the promised derivation: the covariant norm is already Lorentz invariant. Hence the boosted states in Eq. (18) have the same covariant normalization as the rest states. A momentum-dependent factor such as $\sqrt{m/p^0}$ would spoil this convention; it belongs to a different, noncovariant normalization.

For comparison, define the δ^3 -normalized states by

$$|p, \sigma\rangle_W = \frac{1}{\sqrt{2p^0}} |p, \sigma\rangle. \quad (27)$$

Then

$$\langle p', \sigma' | p, \sigma \rangle_W = (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma}. \quad (28)$$

Since $|p, \sigma\rangle = U(L(p)) |k, \sigma\rangle$ and $k^0 = m$, this Weinberg basis obeys

$$|p, \sigma\rangle_W = \sqrt{\frac{m}{p^0}} U(L(p)) |k, \sigma\rangle_W. \quad (29)$$

The point of this formula is not that the Lorentz transformation has become nonunitary. The operator $U(L(p))$ is unitary. The square-root factor appears because the states carrying the subscript W are normalized with $\delta^3(\mathbf{p}' - \mathbf{p})$, and this three-dimensional delta function is not Lorentz invariant by itself.

Let us check this explicitly. Temporarily write

$$|p, \sigma\rangle_W = A(p) U(L(p)) |k, \sigma\rangle_W. \quad (30)$$

To test the norm, fix the boost $B = L(p)$ and write

$$p = Bk, \quad p' = Bk'. \quad (31)$$

By unitarity,

$${}_W \langle k', \sigma' | U(B)^\dagger U(B) | k, \sigma \rangle_W = {}_W \langle k', \sigma' | k, \sigma \rangle_W \quad (32)$$

$$= (2\pi)^3 \delta^3(\mathbf{k}' - \mathbf{k}) \delta_{\sigma'\sigma}. \quad (33)$$

But the desired norm of the final states is written with $\delta^3(\mathbf{p}' - \mathbf{p})$. From the Jacobian derived above, evaluated at $k^0 = m$ and $p^0 = (Bk)^0$,

$$\delta^3(\mathbf{p}' - \mathbf{p}) = \frac{m}{p^0} \delta^3(\mathbf{k}' - \mathbf{k}), \quad \delta^3(\mathbf{k}' - \mathbf{k}) = \frac{p^0}{m} \delta^3(\mathbf{p}' - \mathbf{p}). \quad (34)$$

Therefore

$${}_W \langle p', \sigma' | p, \sigma \rangle_W = |A(p)|^2 (2\pi)^3 \delta^3(\mathbf{k}' - \mathbf{k}) \delta_{\sigma'\sigma} \quad (35)$$

$$= |A(p)|^2 \frac{p^0}{m} (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma}. \quad (36)$$

To obtain the Weinberg normalization

$${}_W \langle p', \sigma' | p, \sigma \rangle_W = (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \delta_{\sigma'\sigma}, \quad (37)$$

we must choose

$$|A(p)|^2 = \frac{m}{p^0}, \quad A(p) = \sqrt{\frac{m}{p^0}}, \quad (38)$$

where the positive phase convention has been chosen. Thus the square-root in the previous equation is precisely the factor that makes the definition compatible with the unitarity of $U(L(p))$ when the noncovariant δ^3 norm is used. It is only the Jacobian needed after replacing the covariant norm by the noncovariant δ^3 norm.

1.5 The Wigner rotation

Now apply a Lorentz transformation Λ to a state $|p, \sigma\rangle$. Since $p = L(p)k$, the transformed momentum is

$$\Lambda p = \Lambda L(p)k. \quad (39)$$

On the other hand, the standard boost $L(\Lambda p)$ also carries k to Λp . Therefore the product

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p) \quad (40)$$

leaves k invariant:

$$W(\Lambda, p)k = k. \quad (41)$$

This transformation $W(\Lambda, p)$ is called the Wigner little-group element. It is the part of the Lorentz transformation that remains after removing the change of momentum. In words:

A Lorentz transformation first moves the particle to a new momentum, and then induces a momentum-dependent rotation of the spin labels.

We can now derive the transformation law rather than just state it. Using

$$\Lambda L(p) = L(\Lambda p)W(\Lambda, p),$$

we find in Weinberg's δ^3 -normalized basis

$$\begin{aligned} U(\Lambda) |p, \sigma\rangle_W &= \sqrt{\frac{m}{p^0}} U(\Lambda L(p)) |k, \sigma\rangle_W \\ &= \sqrt{\frac{m}{p^0}} U(L(\Lambda p)) U(W(\Lambda, p)) |k, \sigma\rangle_W \\ &= \sqrt{\frac{m}{p^0}} \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) U(L(\Lambda p)) |k, \sigma'\rangle_W \\ &= \left(\frac{(\Lambda p)^0}{p^0} \right)^{1/2} \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle_W. \end{aligned}$$

This is Weinberg's form of the one-particle transformation law. If instead we use the invariantly normalized states

$$|p, \sigma\rangle = \sqrt{2p^0} |p, \sigma\rangle_W,$$

then the square-root factor cancels:

$$\begin{aligned} U(\Lambda) |p, \sigma\rangle &= \sqrt{2p^0} U(\Lambda) |p, \sigma\rangle_W \\ &= \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) \sqrt{2(\Lambda p)^0} |\Lambda p, \sigma'\rangle_W. \end{aligned}$$

Therefore, in the invariant normalization used in these notes,

$$U(\Lambda) |p, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle, \quad (42)$$

which is the cleanest form of the Wigner transformation law.

Including translations gives

$$U(\Lambda, a) |p, \sigma\rangle = e^{+ia \cdot (\Lambda p)} \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle. \quad (43)$$

This is the central one-particle transformation law. The entire construction of relativistic quantum fields begins by asking: which matrices $D(W)$ are allowed?

2 Massive little group and arbitrary spin

2.1 The little group

Given a fixed standard momentum k^μ , the little group is the subgroup of the Lorentz group that leaves k^μ unchanged:

$$G_k = \{W \in \text{SO}^+(1, 3) : Wk = k\}. \quad (44)$$

For a massive particle we choose

$$k^\mu = (m, 0, 0, 0). \quad (45)$$

Then $Wk = k$ implies that W cannot mix the time direction with the spatial directions. Hence W must be a purely spatial rotation:

$$W^0_0 = 1, \quad W^0_i = W^i_0 = 0, \quad W^i_j = R^i_j, \quad (46)$$

where

$$R \in \text{SO}(3). \quad (47)$$

Therefore the massive little group is

$$G_k \cong \text{SO}(3). \quad (48)$$

Quantum mechanically we use projective representations of $\text{SO}(3)$, or equivalently ordinary representations of its double cover:

$$\text{SU}(2) \longrightarrow \text{SO}(3). \quad (49)$$

Thus the massive one-particle states are classified by unitary irreducible representations of $\text{SU}(2)$.

2.2 Spin- j representations

The irreducible representations of $\text{SU}(2)$ are labelled by

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots \quad (50)$$

For fixed j , the spin label takes the $2j + 1$ values

$$\sigma = -j, -j + 1, \dots, j - 1, j. \quad (51)$$

The rest-frame states are therefore

$$|k, \sigma\rangle, \quad \sigma = -j, \dots, j. \quad (52)$$

Under a pure rotation R , they transform as

$$U(R) |k, \sigma\rangle = \sum_{\sigma'=-j}^j D_{\sigma'\sigma}^{(j)}(R) |k, \sigma'\rangle. \quad (53)$$

Here $D^{(j)}(R)$ is the spin- j representation matrix of $\text{SU}(2)$ associated with the spatial rotation R .

The angular momentum generators J_i satisfy

$$[J_i, J_j] = i\epsilon_{ijk} J_k. \quad (54)$$

In the spin- j representation,

$$\mathbf{J}^2 |k, \sigma\rangle = j(j + 1) |k, \sigma\rangle, \quad J_3 |k, \sigma\rangle = \sigma |k, \sigma\rangle. \quad (55)$$

For infinitesimal rotations,

$$D^{(j)}(R) = \exp(-i\boldsymbol{\theta} \cdot \mathbf{J}^{(j)}), \quad (56)$$

where $\mathbf{J}^{(j)}$ are the $(2j + 1) \times (2j + 1)$ spin matrices.

2.3 Casimir interpretation

The same classification can be stated using the two Poincare Casimirs. The first is

$$P^2 = m^2. \quad (57)$$

The second uses the Pauli-Lubanski vector

$$W_{\text{PL}}^\mu = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} P_\nu J_{\rho\sigma}. \quad (58)$$

For a massive spin- j representation,

$$W_{\text{PL}\mu} W_{\text{PL}}^\mu = -m^2 j(j+1). \quad (59)$$

In the rest frame this statement reduces to the familiar angular momentum statement

$$\mathbf{J}^2 = j(j+1). \quad (60)$$

Thus Weinberg's little-group construction reproduces the usual spin quantum number, but now in a fully relativistic language.

2.4 General massive transformation law

Combining the previous section with the little-group representation, the one-particle states of a massive spin- j particle transform as

$$U(\Lambda) |p, \sigma\rangle = \sum_{\sigma'=-j}^j D_{\sigma'\sigma}^{(j)}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle. \quad (61)$$

The Wigner rotation is

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p). \quad (62)$$

Since $W(\Lambda, p)$ leaves $k = (m, 0, 0, 0)$ invariant, it belongs to the massive little group. Therefore the matrix $D^{(j)}(W)$ is exactly an ordinary spin- j rotation matrix.

This formula has an important conceptual meaning. A Lorentz transformation does not merely send

$$p \mapsto \Lambda p. \quad (63)$$

It also changes the observer's notion of spin quantization axis. For a generic boost, the induced spin transformation is not simply the spatial part of Λ ; it is the Wigner rotation $W(\Lambda, p)$, which depends on the particle's momentum.

2.5 Why this is enough to classify massive particles

A massive one-particle representation is specified by two invariant labels:

$$m > 0, \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots \quad (64)$$

The mass m fixes the orbit of momenta, namely the positive-energy hyperboloid $p^2 = m^2$. The spin j fixes the irreducible representation of the little group $\text{SU}(2)$. Thus the Hilbert space of one-particle states has the schematic form

$$\mathcal{H}_{m,j} \simeq L^2(\mathcal{H}_m^+, d^3\mathbf{p}/2p^0) \otimes \mathbb{C}^{2j+1}. \quad (65)$$

The first factor describes momentum. The second factor describes the spin degeneracy.

For the electron, the relevant values are

$$m = m_e, \quad j = \frac{1}{2}, \quad \sigma = \pm \frac{1}{2}. \quad (66)$$

The one-particle electron states therefore transform according to

$$U(\Lambda) |p, \sigma\rangle = \sum_{\sigma'=\pm 1/2} D_{\sigma'\sigma}^{(1/2)}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle. \quad (67)$$

The antiparticle, the positron, carries the same mass and spin but the opposite electric charge. The spin part of the Poincare representation is the same.

3 The particle-field connection: How all classical field theories emerge from locality and quantum mechanics

3.1 The abstract field transformation law

A quantum field is an operator-valued distribution. It is not itself a one-particle state. We write a completely general field multiplet as

$$\Psi_\ell(x), \quad \ell = 1, \dots, N. \quad (68)$$

The label ℓ is deliberately abstract. It may be a scalar label, a vector label, a spinor label, a tensor-spinor label, or a collection of internal and spacetime labels. At this stage we do not assume any particular spin or any particular type of field.

The field components transform in a finite-dimensional representation of the Lorentz group. We denote this representation by

$$S_{\ell\ell'}(\Lambda). \quad (69)$$

This notation is important. The matrix $S_{\ell\ell'}(\Lambda)$ acts on the field index ℓ . It should not be confused with the unitary little-group matrix $D_{\sigma\sigma'}(W)$, which acts on the physical one-particle spin label σ .

Our convention for the Poincare transformation of the field is

$$U(\Lambda, a) \Psi_\ell(x) U(\Lambda, a)^{-1} = \sum_{\ell'} S_{\ell\ell'}(\Lambda^{-1}) \Psi_{\ell'}(\Lambda x + a). \quad (70)$$

For a pure Lorentz transformation this becomes

$$U(\Lambda) \Psi_\ell(x) U(\Lambda)^{-1} = \sum_{\ell'} S_{\ell\ell'}(\Lambda^{-1}) \Psi_{\ell'}(\Lambda x). \quad (71)$$

Equivalently,

$$U(\Lambda)^{-1} \Psi_\ell(x) U(\Lambda) = \sum_{\ell'} S_{\ell\ell'}(\Lambda) \Psi_{\ell'}(\Lambda^{-1}x). \quad (72)$$

The unitary operator $U(\Lambda)$ acts on the Hilbert space. The finite-dimensional matrix $S(\Lambda)$ acts on the field components. These are different representations of the same Lorentz transformation. In general $S(\Lambda)$ is not unitary, because the Lorentz group is noncompact.

3.2 The particle labels are also abstract

Now suppose that the field is meant to annihilate and create a particle species whose one-particle states are

$$|p, \sigma\rangle. \quad (73)$$

Here p lies on a massive positive-energy mass shell, and σ labels a basis of a unitary representation of the massive little group. For an irreducible massive particle this representation is the spin- j representation of $SU(2)$, but in this section we do not need to choose j . We only need the fact that Lorentz transformations act as

$$U(\Lambda) |p, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle, \quad (74)$$

where

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p) \quad (75)$$

is the Wigner little-group element. The matrix $D(W)$ is unitary, because it acts on physical one-particle spin states.

Thus there are two different index spaces:

$$\ell : \text{field-component index, acted on by } S(\Lambda), \quad (76)$$

$$\sigma : \text{one-particle little-group index, acted on by } D(W). \quad (77)$$

The purpose of the coefficient functions $u_\ell(p, \sigma)$ and $v_\ell(p, \sigma)$ is to connect these two spaces.

3.3 The coefficient functions as intertwiners

Assume that the vacuum is Poincare invariant:

$$U(\Lambda, a) |0\rangle = |0\rangle. \quad (78)$$

Define the particle coefficient functions by the vacuum-to-one-particle matrix element

$$u_\ell(p, \sigma) := \langle 0 | \Psi_\ell(0) | p, \sigma \rangle. \quad (79)$$

By translation covariance,

$$\langle 0 | \Psi_\ell(x) | p, \sigma \rangle = e^{-ip \cdot x} u_\ell(p, \sigma). \quad (80)$$

So $u_\ell(p, \sigma)$ is the momentum-space wavefunction carried by the field operator.

We now derive the compatibility condition. Starting from Eq. (79), insert $U(\Lambda)^{-1}U(\Lambda) = 1$ and use vacuum invariance:

$$u_\ell(p, \sigma) = \langle 0 | U(\Lambda) \Psi_\ell(0) U(\Lambda)^{-1} U(\Lambda) | p, \sigma \rangle. \quad (81)$$

Using Eq. (71) at $x = 0$ gives

$$U(\Lambda) \Psi_\ell(0) U(\Lambda)^{-1} = \sum_{\ell'} S_{\ell\ell'}(\Lambda^{-1}) \Psi_{\ell'}(0). \quad (82)$$

Using the one-particle transformation law (74), we obtain

$$u_\ell(p, \sigma) = \sum_{\ell'} S_{\ell\ell'}(\Lambda^{-1}) \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) \langle 0 | \Psi_{\ell'}(0) | \Lambda p, \sigma' \rangle \quad (83)$$

$$= \sum_{\ell', \sigma'} S_{\ell\ell'}(\Lambda^{-1}) u_{\ell'}(\Lambda p, \sigma') D_{\sigma'\sigma}(W(\Lambda, p)). \quad (84)$$

Multiplying by $S(\Lambda)$ gives Weinberg's consistency condition in the form

$$\boxed{\sum_{\ell'} S_{\ell\ell'}(\Lambda) u_{\ell'}(p, \sigma) = \sum_{\sigma'} u_{\ell}(\Lambda p, \sigma') D_{\sigma'\sigma}(W(\Lambda, p))}. \quad (85)$$

This is the central abstract result. The left-hand side transforms the field index by $S(\Lambda)$. The right-hand side transforms the one-particle state: p becomes Λp , and the little-group index is rotated by $D(W)$.

For antiparticle coefficients one obtains an analogous relation, but one must be more careful about which side of the matrix element carries the one-particle representation. With the convention that the antiparticle term in the field expansion is proportional to

$$v_{\ell}(p, \sigma) b^{\dagger}(p, \sigma) e^{+ip \cdot x},$$

the coefficient $v_{\ell}(p, \sigma)$ may be read as the vacuum-to-antiparticle creation matrix element

$$v_{\ell}(p, \sigma) = \langle \bar{p}, \sigma | \Psi_{\ell}(0) | 0 \rangle, \quad |\bar{p}, \sigma\rangle = b^{\dagger}(p, \sigma) | 0 \rangle.$$

The antiparticle state itself transforms with the same massive spin matrix,

$$U(\Lambda) |\bar{p}, \sigma\rangle = \sum_{\rho} D_{\rho\sigma}(W(\Lambda, p)) |\bar{\Lambda p}, \rho\rangle.$$

Taking the adjoint gives the transformation of the bra,

$$\langle \bar{p}, \sigma | U(\Lambda)^{-1} = \sum_{\rho} D_{\rho\sigma}^*(W(\Lambda, p)) \langle \bar{\Lambda p}, \rho |.$$

This is the origin of the complex conjugate matrix. Repeating the same insertion of $U(\Lambda)^{-1}U(\Lambda) = 1$ gives

$$\begin{aligned} v_{\ell}(p, \sigma) &= \langle \bar{p}, \sigma | U(\Lambda)^{-1} U(\Lambda) \Psi_{\ell}(0) U(\Lambda)^{-1} U(\Lambda) | 0 \rangle \\ &= \sum_{\rho, \ell'} D_{\rho\sigma}^*(W(\Lambda, p)) S_{\ell\ell'}(\Lambda^{-1}) v_{\ell'}(\Lambda p, \rho). \end{aligned}$$

Multiplying by $S(\Lambda)$ gives

$$\sum_{\ell'} S_{\ell\ell'}(\Lambda) v_{\ell'}(p, \sigma) = \sum_{\sigma'} v_{\ell}(\Lambda p, \sigma') D_{\sigma'\sigma}^*(W(\Lambda, p)). \quad (86)$$

Thus D^* does not mean that the antiparticle has a different spin. It appears because v_{ℓ} is tied to an antiparticle bra, or equivalently to a creation operator in the field expansion, and bras transform in the complex-conjugate representation.

3.4 The rest-frame matching condition

We now specialize to the standard rest momentum

$$k = (m, 0, 0, 0). \quad (87)$$

The simplest interpretation of Eq. (85) comes from setting $p = k$ and taking $\Lambda = R$ to be an element of the massive little group. Then

$$Rk = k, \quad W(R, k) = R. \quad (88)$$

The consistency condition reduces to

$$\boxed{\sum_{\ell'} S_{\ell\ell'}(R) u_{\ell'}(k, \sigma) = \sum_{\sigma'} u_{\ell}(k, \sigma') D_{\sigma'\sigma}(R)}. \quad (89)$$

In matrix notation this is

$$S(R)u(k) = u(k)D(R). \quad (90)$$

Thus $u(k)$ is an intertwiner from the little-group representation carried by the particle states into the restriction of the Lorentz field representation to the rotation subgroup. In this sense Eq. (89) says that the relevant part of $S(R)$ is derived from $D(R)$. If $u(k)$ is square and invertible on the field-component space under discussion, then

$$S(R) = u(k)D(R)u(k)^{-1}.$$

With all indices displayed, this same statement is

$$\boxed{S_{\ell\ell'}(R) = \sum_{\sigma, \sigma'} u_{\ell}(k, \sigma) D_{\sigma\sigma'}(R) [u(k)^{-1}]_{\sigma'\ell'}} \quad (91)$$

Here $u_{\ell}(k, \sigma)$ is regarded as a matrix with row index ℓ and column index σ , while its inverse has row index σ and column index ℓ . Thus

$$\sum_{\sigma} u_{\ell}(k, \sigma) [u(k)^{-1}]_{\sigma\ell'} = \delta_{\ell\ell'}, \quad \sum_{\ell} [u(k)^{-1}]_{\sigma\ell} u_{\ell}(k, \sigma') = \delta_{\sigma\sigma'}.$$

More generally, if $u(k)$ embeds the particle spin space into a larger field-component space, then $S(R)$ restricted to the image of $u(k)$ is equivalent to $D(R)$:

$$S(R)|_{\text{Im } u(k)} \simeq D(R).$$

The antiparticle coefficient satisfies the analogous rest-frame condition

$$\boxed{\sum_{\ell'} S_{\ell\ell'}(R) v_{\ell'}(k, \sigma) = \sum_{\sigma'} v_{\ell}(k, \sigma') D_{\sigma'\sigma}^*(R).} \quad (92)$$

Thus Section 3.4 gives the first practical step in constructing a field: solve the finite-dimensional rest-frame intertwiner equations (89) and (92) for $u_{\ell}(k, \sigma)$ and $v_{\ell}(k, \sigma)$. Once these rest-frame objects are known, Section 3.5 boosts them to arbitrary momentum.

This is the precise mathematical statement that an arbitrary field cannot describe an arbitrary particle. To describe a particle with a specified spin representation D , one must choose a field whose Lorentz transformation law $S(\Lambda)$ has the correct rotational content. The full Lorentz representation $S(\Lambda)$ is an extension of this rotational content away from the little group. For spin $\frac{1}{2}$, this extension cannot be achieved with an ordinary real tensor representation of the Lorentz group; one must use a complex spinor representation of the double cover, such as $(\frac{1}{2}, 0)$, $(0, \frac{1}{2})$, or their direct sum. The Dirac field is precisely such a choice.

3.5 Boosting the rest-frame coefficients

Now take $\Lambda = L(p)$ and start from the rest momentum k . With $L(k) = \mathbf{1}$, the corresponding Wigner element is trivial:

$$W(L(p), k) = \mathbf{1}. \quad (93)$$

Therefore Eq. (85) gives

$$u_{\ell}(p, \sigma) = \sum_{\ell'} S_{\ell\ell'}(L(p)) u_{\ell'}(k, \sigma). \quad (94)$$

The same logic gives

$$v_{\ell}(p, \sigma) = \sum_{\ell'} S_{\ell\ell'}(L(p)) v_{\ell'}(k, \sigma). \quad (95)$$

Thus all momentum-dependent wavefunctions are obtained by boosting rest-frame intertwiners. More importantly, this gives the most general momentum dependence compatible with Lorentz covariance, once the standard boosts $L(p)$ have been chosen and the rest-frame solutions $u(k)$ and $v(k)$ have been fixed. We can now verify explicitly that these boosted coefficients satisfy the general condition (85). Since

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p),$$

we have

$$\Lambda L(p) = L(\Lambda p) W(\Lambda, p).$$

Therefore

$$\begin{aligned} [S(\Lambda)u(p)]_{\ell\sigma} &= [S(\Lambda)S(L(p))u(k)]_{\ell\sigma} \\ &= [S(\Lambda L(p))u(k)]_{\ell\sigma} \\ &= [S(L(\Lambda p))S(W(\Lambda, p))u(k)]_{\ell\sigma} \\ &= [S(L(\Lambda p))u(k)D(W(\Lambda, p))]_{\ell\sigma} \\ &= \sum_{\sigma'} u_{\ell}(\Lambda p, \sigma') D_{\sigma'\sigma}(W(\Lambda, p)). \end{aligned}$$

In the fourth line we used the rest-frame intertwiner condition (89), because $W(\Lambda, p)$ belongs to the massive little group. This proves Eq. (85). The antiparticle relation (86) follows in the same way, using the rest-frame condition with $D^*(R)$ for $v(k)$.

3.6 General free-field expansion and the emergence of classical field theories

With invariant one-particle normalization, the free field has the schematic form

$$\Psi_{\ell}(x) = \sum_{\sigma} \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} \left[u_{\ell}(p, \sigma) a(p, \sigma) e^{-ip \cdot x} + v_{\ell}(p, \sigma) b^{\dagger}(p, \sigma) e^{ip \cdot x} \right]. \quad (96)$$

The operators $a^{\dagger}(p, \sigma)$ create particles, and $b^{\dagger}(p, \sigma)$ create antiparticles. At this point the functions u and v are already known in principle: first solve the rest-frame intertwiner equations, then boost. Therefore the differential equation obeyed by the field is not guessed independently. It is determined by the algebraic equations obeyed by the momentum-space coefficients $u(p, \sigma)$ and $v(p, \sigma)$. Section 4 will show this explicitly: for the scalar field the mass-shell condition gives the Klein–Gordon equation, while for the spin- $\frac{1}{2}$ field the boosted intertwiners give the Dirac equation.

The statistics are not fixed by the Lorentz transformation law alone. They are fixed by locality together with the spin-statistics theorem.

Section 4 now applies this abstract construction first to spin 0 and then to the electronic Dirac field.

4 How Klein-Gordon and Dirac equation emerge from quantum field theory

4.1 The free scalar field

The scalar field is the simplest special case of the Section 3 construction. We do not begin by postulating the Klein–Gordon equation. Instead, we begin with the representation-theoretic data of a massive spin-zero particle and then see which differential equation the resulting field satisfies.

The scalar field carries the trivial Lorentz representation

$$S(\Lambda) = 1. \quad (97)$$

It describes spin-zero particles, so the little-group representation is also trivial:

$$D(W) = 1. \quad (98)$$

There is no spin label σ . Weinberg's consistency condition (85) becomes

$$u(\Lambda p) = u(p). \quad (99)$$

Since the massive mass shell is one Lorentz orbit, $u(p)$ is a constant. The same is true for $v(p)$. We may absorb the constants into the definitions of a and b , and choose

$$u(p) = v(p) = 1. \quad (100)$$

The free complex scalar field is then

$$\phi(x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} \left[a(p) e^{-ip \cdot x} + b^\dagger(p) e^{+ip \cdot x} \right], \quad (101)$$

where $p^0 = \sqrt{m^2 + \mathbf{p}^2}$. For a spin-zero field, microcausality is implemented by commutators, not anticommutators. The required oscillator algebra is

$$[a(p), a^\dagger(p')] = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{p}'), \quad (102)$$

$$[b(p), b^\dagger(p')] = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{p}'), \quad (103)$$

with all other commutators vanishing. With these rules one obtains

$$[\phi(x), \phi^\dagger(y)] = \Delta(x - y), \quad (104)$$

where the Pauli–Jordan function is explicitly

$$\Delta(z) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} (e^{-ip \cdot z} - e^{+ip \cdot z}), \quad p^0 = +\sqrt{\mathbf{p}^2 + m^2}, \quad (105)$$

$$= \int \frac{d^4p}{(2\pi)^3} \text{sgn}(p^0) \delta(p^2 - m^2) e^{-ip \cdot z}. \quad (106)$$

The second form makes Lorentz invariance manifest: d^4p , $\delta(p^2 - m^2)$, $\text{sgn}(p^0)$ for proper orthochronous Lorentz transformations, and $p \cdot z$ are invariant. Hence

$$\Delta(\Lambda z) = \Delta(z). \quad (107)$$

Now take $z = x - y$ spacelike. Then there exists a proper orthochronous Lorentz transformation which sends z to $-z = y - x$: boost to a frame where $z^0 = 0$, rotate the spatial vector by π , and boost back. Lorentz invariance gives $\Delta(z) = \Delta(-z)$, while the explicit expression (105) gives $\Delta(-z) = -\Delta(z)$. Therefore

$$\Delta(z) = 0, \quad z^2 < 0. \quad (108)$$

Thus the commutator vanishes at spacelike separation and microcausality is established. Had we used anticommutators for the scalar oscillators, the field anticommutator would instead contain

$$\Delta^{(1)}(z) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} (e^{-ip \cdot z} + e^{+ip \cdot z}), \quad (109)$$

which is even under $z \mapsto -z$ and is not forced to vanish by the above Lorentz-invariance argument. Indeed it is nonzero for spacelike separation. Thus causality selects the bosonic commutation relations for the scalar field, not anticommutation relations. For a real scalar field the particle is its own antiparticle, and one sets $b = a$:

$$\phi(x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} \left[a(p) e^{-ip \cdot x} + a^\dagger(p) e^{+ip \cdot x} \right]. \quad (110)$$

Now the field equation follows. Each plane wave in (101) is supported on the massive one-particle orbit $p^2 = m^2$. Therefore

$$(\partial^2 + m^2) e^{-ip \cdot x} = 0, \quad (\partial^2 + m^2) e^{+ip \cdot x} = 0,$$

and hence

$$\boxed{(\partial^2 + m^2)\phi(x) = 0.} \quad (111)$$

This is the Klein–Gordon equation, but in the present logic it has been derived from the mass-shell support of the field expansion; it was not imposed by hand as an independent starting point. This scalar example shows the logic in its most transparent form: trivial field representation plus trivial little-group representation gives a Lorentz scalar field creating spin-zero particles, and the on-shell condition gives the Klein–Gordon equation.

There are two more standard ways of obtaining exactly the same equation. They are useful because the same logic will be repeated for the spin- $\frac{1}{2}$ field: first one finds the field equation from the particle construction, and then one checks that it is also reproduced by an action and by the Hamiltonian formalism.

For the complex scalar field, the Klein–Gordon action is

$$S_{\text{KG}} = \int d^4x \mathcal{L}_{\text{KG}}, \quad \mathcal{L}_{\text{KG}} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi. \quad (112)$$

In varying this action, ϕ and $\phi^\dagger$ are treated as independent fields. Variation with respect to $\phi^\dagger$ gives

$$\delta S_{\text{KG}} = \int d^4x \left[\partial_\mu (\delta \phi^\dagger) \partial^\mu \phi - m^2 \delta \phi^\dagger \phi \right] \quad (113)$$

$$= - \int d^4x \delta \phi^\dagger (\partial_\mu \partial^\mu + m^2) \phi, \quad (114)$$

where we integrated by parts and dropped the boundary term. Since $\delta \phi^\dagger$ is arbitrary, the Euler–Lagrange equation is again

$$(\partial^2 + m^2)\phi = 0. \quad (115)$$

Variation with respect to ϕ gives the conjugate equation

$$(\partial^2 + m^2)\phi^\dagger = 0. \quad (116)$$

For a real scalar field one imposes $\phi^\dagger = \phi$ and writes instead

$$\mathcal{L}_{\text{KG}}^{\text{real}} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2. \quad (117)$$

The factor of $1/2$ simply avoids double counting when the field is real.

The Hamiltonian formulation gives the same equation once more. For the complex scalar field the canonical momenta are

$$\pi(x) = \frac{\partial \mathcal{L}_{\text{KG}}}{\partial \dot{\phi}(x)} = \dot{\phi}^\dagger(x), \quad \pi^\dagger(x) = \frac{\partial \mathcal{L}_{\text{KG}}}{\partial \dot{\phi}^\dagger(x)} = \dot{\phi}(x). \quad (118)$$

Therefore

$$\mathcal{H}_{\text{KG}} = \pi \dot{\phi} + \pi^\dagger \dot{\phi}^\dagger - \mathcal{L}_{\text{KG}} \quad (119)$$

$$= \pi^\dagger \pi + \nabla \phi^\dagger \cdot \nabla \phi + m^2 \phi^\dagger \phi. \quad (120)$$

The equal-time Poisson brackets are

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\}_{\text{P}} = \delta^3(\mathbf{x} - \mathbf{y}), \quad \{\phi^\dagger(t, \mathbf{x}), \pi^\dagger(t, \mathbf{y})\}_{\text{P}} = \delta^3(\mathbf{x} - \mathbf{y}), \quad (121)$$

with all other elementary Poisson brackets vanishing. The Hamiltonian is

$$H_{\text{KG}} = \int d^3\mathbf{x} \mathcal{H}_{\text{KG}}. \quad (122)$$

Using the mode expansion in Eq. (101), the same operator can be written in terms of particle and antiparticle number operators. After normal ordering, one obtains

$$: H_{\text{KG}} := \int d\Pi_p p^0 \left[a^\dagger(p) a(p) + b^\dagger(p) b(p) \right], \quad d\Pi_p := \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0}. \quad (123)$$

The derivation of the above from the mode expansion in Eq. (101) is left as an exercise. For a real scalar field the particle is its own antiparticle, so the second term is absent and one has simply

$$: H_{\text{KG}}^{\text{real}} := \int d\Pi_p p^0 a^\dagger(p) a(p). \quad (124)$$

The colons remind us that the infinite zero-point constant has been subtracted. This is the operator statement that the Klein–Gordon field is a collection of independent harmonic oscillators, one for each momentum mode. Hamilton’s equations give

$$\dot{\phi} = \{\phi, H_{\text{KG}}\}_{\text{P}} = \pi^\dagger, \quad \dot{\pi}^\dagger = \{\pi^\dagger, H_{\text{KG}}\}_{\text{P}} = \nabla^2 \phi - m^2 \phi. \quad (125)$$

Combining these two equations gives

$$\ddot{\phi} - \nabla^2 \phi + m^2 \phi = 0, \quad (126)$$

which is precisely

$$(\partial^2 + m^2)\phi = 0. \quad (127)$$

Quantization replaces Poisson brackets by equal-time commutators according to the correspondence rule

$$\{F, G\}_{\text{P}} \longrightarrow \frac{1}{i} [F, G]. \quad (128)$$

Thus Eq. (121) becomes

$$[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad [\phi^\dagger(t, \mathbf{x}), \pi^\dagger(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad (129)$$

with all other elementary equal-time commutators vanishing. The Hamilton equations can then be written directly in Heisenberg form,

$$i\dot{\phi} = [\phi, H_{\text{KG}}] = i\pi^\dagger, \quad i\dot{\pi}^\dagger = [\pi^\dagger, H_{\text{KG}}] = i(\nabla^2 \phi - m^2 \phi). \quad (130)$$

These are the quantum version of the classical Hamilton equations and again imply the Klein–Gordon equation.

There is an important consistency check with the action-free construction above. Since

$$[\phi(x), \phi^\dagger(y)] = \Delta(x - y), \quad (131)$$

and since $\pi(y) = \dot{\phi}^\dagger(y)$, we find at equal times

$$[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = \partial_{y^0} \Delta(x - y) \big|_{x^0=y^0=t} \quad (132)$$

$$= -\partial_{z^0} \Delta(z) \big|_{z^0=0, \mathbf{z}=\mathbf{x}-\mathbf{y}}. \quad (133)$$

From Eq. (105),

$$\Delta(0, \mathbf{z}) = 0, \quad \partial_{z^0} \Delta(z) \big|_{z^0=0} = -i\delta^3(\mathbf{z}). \quad (134)$$

Therefore

$$[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad (135)$$

which is exactly the canonical commutator. Thus the equal-time Hamiltonian commutator is not an additional arbitrary postulate: it is the equal-time derivative of the Lorentz-covariant Pauli–Jordan commutator obtained from the Weinberg construction.

Thus the same Klein–Gordon equation has appeared in three equivalent ways: from the Weinberg particle-field construction, from the Euler–Lagrange equation of the scalar action, and from the Hamiltonian equations of motion. In the next subsections we repeat this pattern for spin $\frac{1}{2}$: the particle-field construction produces the Dirac equation, and Section 5 will then show how the Dirac action and Hamiltonian reproduce it.

4.2 Clifford algebra and spinor representations of the Lorentz group

The most economical way to introduce spinors is not to start in four dimensions. It is to start from the Clifford algebra, because this construction works in any spacetime dimension. Let d be the spacetime dimension, with tangent-space indices

$$a, b = 0, 1, \dots, d-1,$$

and metric η_{ab} . The Clifford algebra is generated by matrices γ^a satisfying

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbf{1}. \quad (136)$$

A spinor is, by definition, an object on which these gamma matrices act.

The spinor representation of the Lorentz group is obtained from the Clifford algebra. Define

$$\Sigma^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]. \quad (137)$$

Let us pause to connect this definition with the Lorentz algebra in arbitrary dimension. In the same convention as Eq. (137), the abstract Lorentz generators $J^{ab} = -J^{ba}$ obey

$$[J^{ab}, J^{cd}] = \eta^{bc}J^{ad} - \eta^{ac}J^{bd} - \eta^{bd}J^{ac} + \eta^{ad}J^{bc}. \quad (138)$$

This convention has no explicit factor of i in the algebra, because the finite spinor transformation will be written as $\exp(\frac{1}{2}\omega_{ab}\Sigma^{ab})$. If one instead uses Hermitian quantum-mechanical Lorentz generators $M^{ab} = iJ^{ab}$, Eq. (138) becomes the more familiar formula with an overall factor of i on the right-hand side.

The Clifford matrices give a representation of this algebra. Indeed, using only $\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbf{1}$, one obtains

$$[\Sigma^{ab}, \Sigma^{cd}] = \eta^{bc}\Sigma^{ad} - \eta^{ac}\Sigma^{bd} - \eta^{bd}\Sigma^{ac} + \eta^{ad}\Sigma^{bc}. \quad (139)$$

Thus the assignment

$$J^{ab} \longmapsto \Sigma^{ab} \quad (140)$$

is a spinor representation of the Lorentz algebra. It is this statement, not a special four-dimensional accident, that makes the Clifford algebra the natural starting point for spinors.

Using Eq. (136), one finds

$$[\Sigma^{ab}, \gamma^c] = \eta^{bc}\gamma^a - \eta^{ac}\gamma^b. \quad (141)$$

Therefore, if

$$S(\Lambda) = \exp\left(\frac{1}{2}\omega_{ab}\Sigma^{ab}\right), \quad \omega_{ab} = -\omega_{ba}, \quad (142)$$

then the gamma matrices transform as a Lorentz vector:

$$S(\Lambda)^{-1}\gamma^aS(\Lambda) = \Lambda^a{}_b\gamma^b. \quad (143)$$

This equation is the basic definition of the spinor transformation law. The spinor matrix $S(\Lambda)$ is the operator that implements the Lorentz transformation on the Clifford module.

Let us also record why the dimension of spinors is forced. Antisymmetrized products of gamma matrices are defined by

$$\gamma^{a_1\cdots a_r} := \gamma^{[a_1}\gamma^{a_2}\cdots\gamma^{a_r]}, \quad r = 0, 1, \dots, d, \quad (144)$$

where $r = 0$ denotes the identity matrix. Because of the Clifford relation, every product of gamma matrices can be reduced to a linear combination of these antisymmetrized products. The number of independent antisymmetric tensors of rank r in d dimensions is $\binom{d}{r}$, and hence the Clifford algebra has vector-space dimension

$$\sum_{r=0}^d \binom{d}{r} = 2^d. \quad (145)$$

This is the clean counting argument: the Clifford algebra is spanned by scalar, vector, bivector, and higher antisymmetric gamma products.

Now suppose the gamma matrices are represented on an N -component complex spinor space. In even dimension $d = 2n$, the complex Clifford algebra is the full matrix algebra on an irreducible spinor space. Since the full algebra of $N \times N$ complex matrices has dimension N^2 , Eq. (145) gives

$$N^2 = 2^{2n}, \quad \Rightarrow \quad N = 2^n. \quad (146)$$

Thus in even spacetime dimension the minimal complex spinor dimension is

$$N_{\min} = 2^{d/2}. \quad (147)$$

In odd dimension $d = 2n + 1$, the product of all gamma matrices is central. In an irreducible representation it is therefore proportional to the identity, with one of two possible signs. After this sign is fixed, the irreducible Clifford module has dimension

$$N_{\min} = 2^n = 2^{(d-1)/2}. \quad (148)$$

Combining the two cases,

$$\boxed{N_{\min} = 2^{\lfloor d/2 \rfloor}}. \quad (149)$$

For $d = 4$, this gives $N_{\min} = 4$. Thus four-dimensional relativistic spinors naturally have four Clifford components before any chiral decomposition is made.

We now specialize to four dimensions. Introduce

$$\sigma^\mu = (\mathbf{1}, \sigma^1, \sigma^2, \sigma^3), \quad \bar{\sigma}^\mu = (\mathbf{1}, -\sigma^1, -\sigma^2, -\sigma^3), \quad (150)$$

where σ^i are the Pauli matrices. In the chiral basis the gamma matrices are

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix}. \quad (151)$$

The corresponding chiral projectors are

$$P_L = \frac{1 - \gamma^5}{2}, \quad P_R = \frac{1 + \gamma^5}{2}. \quad (152)$$

The words left-handed and right-handed refer to these two chirality eigenspaces: P_L projects onto the $\gamma^5 = -1$ subspace, and P_R projects onto the $\gamma^5 = +1$ subspace. In a massless theory these chiralities coincide with the usual handedness, or helicity, for particle states.

The four-dimensional Lorentz algebra has a special structure. Define

$$A_i = \frac{1}{2}(J_i + iK_i), \quad B_i = \frac{1}{2}(J_i - iK_i). \quad (153)$$

Then

$$[A_i, A_j] = i\epsilon_{ijk}A_k, \quad [B_i, B_j] = i\epsilon_{ijk}B_k, \quad [A_i, B_j] = 0. \quad (154)$$

Thus, at the level of the complexified Lie algebra,

$$\mathfrak{so}(1, 3)_{\mathbb{C}} \simeq \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C}). \quad (155)$$

Finite-dimensional complex representations are therefore labelled by two spins (j_L, j_R) . The two Weyl representations are

$$\left(\frac{1}{2}, 0\right), \quad \left(0, \frac{1}{2}\right). \quad (156)$$

It is useful to describe these representations directly in terms of $\mathrm{SL}(2, \mathbb{C})$. A Lorentz vector can be encoded in the Hermitian matrix

$$X = x_\mu \sigma^\mu. \quad (157)$$

For $A \in \mathrm{SL}(2, \mathbb{C})$, define

$$X' = AXA^\dagger. \quad (158)$$

Since

$$\det X = x^2, \quad (159)$$

this transformation preserves the Minkowski norm and therefore defines a proper orthochronous Lorentz transformation $\Lambda(A)$. The map $A \mapsto \Lambda(A)$ is two-to-one: A and $-A$ define the same Lorentz transformation.

Let V_L have basis

$$e_\alpha, \quad \alpha = 1, 2, \quad (160)$$

and let V_R have basis

$$\bar{e}_{\dot{\alpha}}, \quad \dot{\alpha} = 1, 2. \quad (161)$$

The Weyl representation matrices are

$$S_L(\Lambda(A)) = A, \quad (162)$$

$$S_R(\Lambda(A)) = (A^\dagger)^{-1}. \quad (163)$$

Equivalently, their infinitesimal rotation and boost generators are

$$V_L : \quad J_i = \frac{\sigma_i}{2}, \quad K_i = -\frac{i\sigma_i}{2}, \quad (164)$$

$$V_R : \quad J_i = \frac{\sigma_i}{2}, \quad K_i = +\frac{i\sigma_i}{2}. \quad (165)$$

Thus left and right Weyl representations behave identically under spatial rotations but oppositely under boosts. This is why a massive spin- $\frac{1}{2}$ construction will need both chiralities: rotations alone see spin $\frac{1}{2}$, but Lorentz boosts distinguish the two Weyl representations.

The four-component Dirac representation is the direct sum

$$V_D = V_L \oplus V_R, \quad S_D(\Lambda) = S_L(\Lambda) \oplus S_R(\Lambda). \quad (166)$$

At this point this is only a Lorentz representation, not yet a quantum field. In the ordered basis

$$(e_{L,+}, e_{L,-}, e_{R,+}, e_{R,-}), \quad (167)$$

where $+$ and $-$ denote spin up and spin down with respect to a chosen z -axis, the generators are

$$J_i = \frac{1}{2} \begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}, \quad (168)$$

$$K_i = \frac{i}{2} \begin{pmatrix} -\sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}. \quad (169)$$

For a pure rotation,

$$S_D(R) = D^{(1/2)}(R) \oplus D^{(1/2)}(R). \quad (170)$$

This is the rotational content that will allow the representation to describe a massive spin- $\frac{1}{2}$ particle.

4.3 Solving the spin- $\frac{1}{2}$ intertwiner and deriving the Dirac equation

We now use the abstract result of Section 3 in the order emphasized there: first solve the rest-frame intertwiners of Section 3.4, then boost them as in Section 3.5, and only at the end read off the field equation from the resulting momentum-space coefficients. The particle little-group representation is

$$D(W) = D^{(1/2)}(W), \quad \sigma = \pm \frac{1}{2}. \quad (171)$$

The field representation is the four-component representation $S_D(\Lambda)$ in Eq. (166). Thus the field-component index is

$$\ell = 1, 2, 3, 4, \quad (172)$$

corresponding to the ordered basis

$$\ell = 1 : e_{L,+}, \quad \ell = 2 : e_{L,-}, \quad \ell = 3 : e_{R,+}, \quad \ell = 4 : e_{R,-}. \quad (173)$$

The four complex components of a Dirac spinor should not be confused with four independent spin states. A massive spin- $\frac{1}{2}$ particle has two spin states. The four components appear because the finite-dimensional Lorentz representation used for the field is

$$V_D = V_L \oplus V_R,$$

a direct sum of two two-component Weyl spaces. The upper two entries are coordinates in V_L and the lower two entries are coordinates in V_R . Thus $(1, 0, 0, 0)^T$ means: left-chiral field component with spin up along the chosen z -axis. Similarly,

$$(0, 1, 0, 0)^T : \text{left-chiral field component, spin down}, \quad (174)$$

$$(0, 0, 1, 0)^T : \text{right-chiral field component, spin up}, \quad (175)$$

$$(0, 0, 0, 1)^T : \text{right-chiral field component, spin down}. \quad (176)$$

These are field-component basis vectors, not yet one-particle states.

Let

$$\xi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (177)$$

be the usual spin- $\frac{1}{2}$ basis. It satisfies

$$D^{(1/2)}(R)\xi_\sigma = \sum_{\sigma'} \xi_{\sigma'} D_{\sigma'\sigma}^{(1/2)}(R). \quad (178)$$

Here and below the symbols $+$ and $-$ on the explicit spinors denote the two rest-frame spin labels $+\frac{1}{2}$ and $-\frac{1}{2}$.

At rest, the abstract intertwiner equation (89) says

$$S_D(R)u(k) = u(k)D^{(1/2)}(R), \quad k^\mu = (m, 0, 0, 0). \quad (179)$$

We now solve this equation explicitly. We use Schur's lemma in the following elementary form: if ρ_1 and ρ_2 are irreducible complex representations of a group and a linear map T satisfies

$$\rho_1(g)T = T\rho_2(g) \quad \text{for all } g, \quad (180)$$

then $T = 0$ if the two irreducible representations are inequivalent, while T is proportional to the identity map if they are equivalent.

Because

$$S_D(R) = D^{(1/2)}(R) \oplus D^{(1/2)}(R), \quad (181)$$

the rest intertwiner $u(k)$ may be written as a column of two 2×2 blocks,

$$u(k) = \begin{pmatrix} U_L \\ U_R \end{pmatrix}, \quad (182)$$

where U_L maps the particle spin space into the left Weyl copy and U_R maps it into the right Weyl copy. Eq. (179) then becomes

$$D^{(1/2)}(R)U_L = U_L D^{(1/2)}(R), \quad D^{(1/2)}(R)U_R = U_R D^{(1/2)}(R). \quad (183)$$

Each block is an intertwiner from the irreducible spin- $\frac{1}{2}$ representation to an equivalent copy of itself. Schur's lemma therefore implies

$$U_L = c_L \mathbf{1}_2, \quad U_R = c_R \mathbf{1}_2. \quad (184)$$

Thus

$$u(k, \sigma) = \begin{pmatrix} c_L \xi_\sigma \\ c_R \xi_\sigma \end{pmatrix}. \quad (185)$$

This is the first important point. The rest spinor is not guessed. Rotational covariance fixes its tensor structure. The same two-component rest spinor ξ_σ must be embedded once in the left Weyl copy and once in the right Weyl copy. Only the constants c_L and c_R remain.

For the antiparticle coefficient the rest-frame equation is the corresponding specialization of Eq. (86):

$$S_D(R)v(k) = v(k)D^{(1/2)*}(R). \quad (186)$$

The complex conjugate appears for the reason explained in Section 3: it is the coefficient multiplying an antiparticle creation operator. To solve it, use the pseudoreality of the spin- $\frac{1}{2}$ representation. Define

$$\epsilon = -i\sigma^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \eta_\sigma := \epsilon \xi_\sigma^*. \quad (187)$$

Then

$$D^{(1/2)}(R)\eta_\sigma = \sum_{\sigma'} \eta_{\sigma'} D_{\sigma'\sigma}^{(1/2)*}(R). \quad (188)$$

Therefore the general rest-frame solution for v is

$$v(k, \sigma) = \begin{pmatrix} d_L \eta_\sigma \\ d_R \eta_\sigma \end{pmatrix}. \quad (189)$$

The remaining constants are not fixed by rotations. They are fixed by normalization and, for the standard massive Dirac field, by imposing parity invariance. Let us state only what is needed here; the full operator transformations under C , P and T will be discussed later. Parity is the spacetime reflection

$$P : (t, \mathbf{x}) \mapsto (t, -\mathbf{x}). \quad (190)$$

On Lorentz representations it exchanges the two Weyl representations:

$$\left(\frac{1}{2}, 0\right) \longleftrightarrow \left(0, \frac{1}{2}\right). \quad (191)$$

Therefore a parity-invariant massive spinor field cannot live in only one Weyl representation. It must contain both V_L and V_R , and the parity matrix on $V_D = V_L \oplus V_R$ must exchange the two summands. In the chiral basis used above this matrix is

$$\Pi_D = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix} = \gamma^0. \quad (192)$$

Representation-theoretically, this is the matrix which implements the equivalence between the Lorentz representation and its parity-reflected partner.

Since the rest momentum $k = (m, 0, 0, 0)$ is unchanged by parity, a rest spin state may be chosen to have definite intrinsic parity. We choose the particle coefficient $u(k, \sigma)$ to have positive intrinsic parity and the antiparticle coefficient $v(k, \sigma)$ to have the opposite intrinsic parity:

$$\Pi_D u(k, \sigma) = +u(k, \sigma), \quad \Pi_D v(k, \sigma) = -v(k, \sigma). \quad (193)$$

The deep reason for this choice is that without it the Dirac field will NOT obey a local equation. This is explained in Section 4.4. Eq. (193) implies

$$c_L = c_R, \quad d_L = -d_R. \quad (194)$$

With the covariant normalization convention used throughout these notes, we choose

$$c_L = c_R = \sqrt{m}, \quad d_L = \sqrt{m}, \quad d_R = -\sqrt{m}. \quad (195)$$

This gives the explicit rest spinors

$$u(k, +) = \sqrt{m} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u(k, -) = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad (196)$$

$$v(k, +) = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \quad v(k, -) = \sqrt{m} \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}. \quad (197)$$

For example,

$$u(k, -) = \sqrt{m}(0, 1, 0, 1)^T$$

because $\xi_- = (0, 1)^T$ and the positive-parity condition puts the same two-component spinor in the left and right Weyl slots:

$$u(k, -) = \sqrt{m} \begin{pmatrix} \xi_- \\ \xi_- \end{pmatrix}.$$

Thus the column vector in Eq. (196) represents spin down at rest, not an additional chirality state. Its four entries are the coordinates of the same spin-down rest state inside $V_L \oplus V_R$.

Now boost the rest-frame intertwiners exactly as in Section 3. For any momentum $p = L(p)k$,

$$u(p, \sigma) = S_D(L(p))u(k, \sigma), \quad v(p, \sigma) = S_D(L(p))v(k, \sigma). \quad (198)$$

For the rotation-free boost one may write this explicitly. Let

$$p \cdot \sigma := p_\mu \sigma^\mu, \quad p \cdot \bar{\sigma} := p_\mu \bar{\sigma}^\mu. \quad (199)$$

On the massive shell,

$$(p \cdot \sigma)(p \cdot \bar{\sigma}) = m^2 \mathbf{1}. \quad (200)$$

Choosing the positive square roots,

$$S_L(L(p)) = \left(\frac{p \cdot \sigma}{m} \right)^{1/2}, \quad S_R(L(p)) = \left(\frac{p \cdot \bar{\sigma}}{m} \right)^{1/2}. \quad (201)$$

Therefore

$$u(p, \sigma) = \begin{pmatrix} (p \cdot \sigma)^{1/2} \xi_\sigma \\ (p \cdot \bar{\sigma})^{1/2} \xi_\sigma \end{pmatrix}, \quad (202)$$

$$v(p, \sigma) = \begin{pmatrix} (p \cdot \sigma)^{1/2} \eta_\sigma \\ -(p \cdot \bar{\sigma})^{1/2} \eta_\sigma \end{pmatrix}. \quad (203)$$

These explicit formulae are useful, but the construction itself is basis independent: the boosted spinors are simply the rest intertwiners transported by the finite-dimensional Lorentz representation $S_D(L(p))$.

Let us now check explicitly that the general Weinberg equations are solved. Since

$$\Lambda L(p) = L(\Lambda p)W(\Lambda, p), \quad (204)$$

we have

$$S_D(\Lambda)u(p) = S_D(\Lambda)S_D(L(p))u(k) \quad (205)$$

$$= S_D(L(\Lambda p))S_D(W(\Lambda, p))u(k) \quad (206)$$

$$= S_D(L(\Lambda p))u(k)D^{(1/2)}(W(\Lambda, p)) \quad (207)$$

$$= u(\Lambda p)D^{(1/2)}(W(\Lambda, p)). \quad (208)$$

In indices this is precisely the general equation (85):

$$\sum_{\ell'} S_{D, \ell \ell'}(\Lambda) u_{\ell'}(p, \sigma) = \sum_{\sigma'} u_\ell(\Lambda p, \sigma') D_{\sigma' \sigma}^{(1/2)}(W(\Lambda, p)). \quad (209)$$

Similarly,

$$\sum_{\ell'} S_{D, \ell \ell'}(\Lambda) v_{\ell'}(p, \sigma) = \sum_{\sigma'} v_\ell(\Lambda p, \sigma') D_{\sigma' \sigma}^{(1/2)*}(W(\Lambda, p)), \quad (210)$$

which is Eq. (86). Thus the boosted spinors solve the abstract field-particle matching equations of Section 3.

We can now write the field. This step is exactly Weinberg's logic: once the one-particle states, the field representation, and the intertwiners u and v are known, the free field is the operator which annihilates a particle with coefficient u and creates an antiparticle with coefficient v . With the covariant one-particle normalization used in these notes,

$$\langle p', \sigma' | p, \sigma \rangle = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{\sigma \sigma'}, \quad (211)$$

the invariant integration measure is

$$d\Pi_p := \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0}. \quad (212)$$

The corresponding special case of the general expansion (96) is

$$\psi_\ell(x) = \sum_{\sigma=\pm 1/2} \int d\Pi_p \left[u_\ell(p, \sigma) a(p, \sigma) e^{-ip \cdot x} + v_\ell(p, \sigma) b^\dagger(p, \sigma) e^{+ip \cdot x} \right]. \quad (213)$$

Equivalently,

$$\psi_\ell(x) = \sum_{\sigma=\pm 1/2} \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} \left[u_\ell(p, \sigma) a(p, \sigma) e^{-ip \cdot x} + v_\ell(p, \sigma) b^\dagger(p, \sigma) e^{+ip \cdot x} \right]. \quad (214)$$

Here $a^\dagger(p, \sigma)$ creates an electron and $b^\dagger(p, \sigma)$ creates a positron. The nonzero anticommutators are

$$\{a(p, \sigma), a^\dagger(p', \sigma')\} = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{\sigma\sigma'}, \quad (215)$$

$$\{b(p, \sigma), b^\dagger(p', \sigma')\} = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{\sigma\sigma'}. \quad (216)$$

Thus the oscillators are covariantly normalized. If instead one uses δ^3 -normalized oscillators, the same field is written with the more familiar factor $d^3\mathbf{p}/[(2\pi)^{3/2} \sqrt{2p^0}]$ and with correspondingly rescaled creation and annihilation operators.

The Dirac equation now follows as an output. The cleanest rest-frame derivation uses the parity eigenvalue condition above rather than the explicit columns. In the chiral basis $\Pi_D = \gamma^0$, so Eq. (193) says

$$\gamma^0 u(k, \sigma) = u(k, \sigma), \quad \gamma^0 v(k, \sigma) = -v(k, \sigma). \quad (217)$$

Since $k_\mu \gamma^\mu = m\gamma^0$, this is equivalent to

$$(m\gamma^0 - m)u(k, \sigma) = 0, \quad (m\gamma^0 + m)v(k, \sigma) = 0. \quad (218)$$

This is the rest-frame Dirac equation for the coefficient functions. The point is that it follows from the representation theory plus the parity choice; it has not been postulated as a wave equation.

Using the covariance of the gamma matrices,

$$S_D(\Lambda) \gamma^\mu S_D(\Lambda)^{-1} = \Lambda^\mu{}_\nu \gamma^\nu, \quad (219)$$

or, equivalently, $S_D(L(p)) k_\mu \gamma^\mu S_D(L(p))^{-1} = p_\mu \gamma^\mu$, Eq. (218) boosts to

$$(p_\mu \gamma^\mu - m)u(p, \sigma) = 0, \quad (p_\mu \gamma^\mu + m)v(p, \sigma) = 0. \quad (220)$$

Finally apply the differential operator $i\gamma^\mu \partial_\mu - m$ to the field expansion (213). On the positive-frequency term it produces $(p_\mu \gamma^\mu - m)u(p, \sigma)$; on the negative-frequency term it produces $-(p_\mu \gamma^\mu + m)v(p, \sigma)$. Both vanish by Eq. (220). Therefore

$$\boxed{(i\gamma^\mu \partial_\mu - m)\psi(x) = 0.} \quad (221)$$

The logical order is important: first one classifies massive spin- $\frac{1}{2}$ particles, then one chooses a Lorentz field representation with the correct rotational and parity structure, then one solves the intertwiner equations, and only at the end does the Dirac field equation emerge.

4.4 Why the electronic field must be quantized with anticommutators

There is one final point which is essential for causality. In relativistic quantum theory two measurements performed at spacelike separation should not influence each other. For a quantum field this becomes a local condition. For bosonic fields one demands that commutators vanish outside the light cone. For fermionic fields one demands the graded version of the same statement: anticommutators of the basic fields vanish outside the light cone. We now see this explicitly for the Dirac field.

The natural object to pair with ψ is not $\psi^\dagger$ but the Dirac adjoint. For the momentum-space spinors define

$$\bar{u}(p, \sigma) := u(p, \sigma)^\dagger \gamma^0, \quad \bar{v}(p, \sigma) := v(p, \sigma)^\dagger \gamma^0. \quad (222)$$

The reason is Lorentz covariance. The ordinary Hermitian conjugate transforms with $S_D(\Lambda)^\dagger$, whereas the Dirac adjoint transforms with $S_D(\Lambda)^{-1}$:

$$\overline{S_D(\Lambda)u} = \bar{u} S_D(\Lambda)^{-1}. \quad (223)$$

This uses the pseudo-unitarity relation $S_D(\Lambda)^\dagger \gamma^0 S_D(\Lambda) = \gamma^0$, which will be derived systematically in the next section when we discuss Dirac bilinears. Already here it explains why the products $u\bar{u}$ and $v\bar{v}$ are the covariant spin sums.

Using Eq. (213), the adjoint field is

$$\bar{\psi}_{\ell'}(y) = \sum_{\sigma} \int d\Pi_p \left[\bar{u}_{\ell'}(p, \sigma) a^\dagger(p, \sigma) e^{+ip \cdot y} + \bar{v}_{\ell'}(p, \sigma) b(p, \sigma) e^{-ip \cdot y} \right]. \quad (224)$$

The spinors obtained above obey the completeness relations

$$\sum_{\sigma} u(p, \sigma) \bar{u}(p, \sigma) = p_\mu \gamma^\mu + m, \quad \sum_{\sigma} v(p, \sigma) \bar{v}(p, \sigma) = p_\mu \gamma^\mu - m. \quad (225)$$

Let us derive these rather than merely quote them. At rest the explicit spinors give

$$\sum_{\sigma} u(k, \sigma) \bar{u}(k, \sigma) = m(\gamma^0 + \mathbf{1}), \quad (226)$$

$$\sum_{\sigma} v(k, \sigma) \bar{v}(k, \sigma) = m(\gamma^0 - \mathbf{1}). \quad (227)$$

Since $k_\mu \gamma^\mu = m\gamma^0$, this is precisely

$$\sum_{\sigma} u(k, \sigma) \bar{u}(k, \sigma) = k_\mu \gamma^\mu + m, \quad \sum_{\sigma} v(k, \sigma) \bar{v}(k, \sigma) = k_\mu \gamma^\mu - m. \quad (228)$$

Now boost. Because $u(p, \sigma) = S_D(L(p))u(k, \sigma)$ and $\bar{u}(p, \sigma) = \bar{u}(k, \sigma)S_D(L(p))^{-1}$, the u spin sum transforms as

$$\sum_{\sigma} u(p, \sigma) \bar{u}(p, \sigma) = S_D(L(p)) \left[\sum_{\sigma} u(k, \sigma) \bar{u}(k, \sigma) \right] S_D(L(p))^{-1} \quad (229)$$

$$= S_D(L(p))(k_\mu \gamma^\mu + m)S_D(L(p))^{-1} \quad (230)$$

$$= p_\mu \gamma^\mu + m. \quad (231)$$

The derivation for the v spin sum is identical and gives the second relation in Eq. (225). This is the covariant reason why the Dirac adjoint appears.

Now compute the anticommutator step by step. Write $z = x - y$. Only the $a-a^\dagger$ and $b^\dagger-b$ terms contribute:

$$\{\psi_\ell(x), \bar{\psi}_{\ell'}(y)\} = \sum_{\sigma, \tau} \int d\Pi_p d\Pi_q u_\ell(p, \sigma) \bar{u}_{\ell'}(q, \tau) e^{-ip \cdot x} e^{+iq \cdot y} \{a(p, \sigma), a^\dagger(q, \tau)\} \quad (232)$$

$$+ \sum_{\sigma, \tau} \int d\Pi_p d\Pi_q v_\ell(p, \sigma) \bar{v}_{\ell'}(q, \tau) e^{+ip \cdot x} e^{-iq \cdot y} \{b^\dagger(p, \sigma), b(q, \tau)\}. \quad (233)$$

Using

$$\{a(p, \sigma), a^\dagger(q, \tau)\} = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{q}) \delta_{\sigma\tau}, \quad (234)$$

and the identical formula for $b, b^\dagger$, one $d\Pi$ integration is removed and the spin labels are identified. Hence

$$\{\psi_\ell(x), \bar{\psi}_{\ell'}(y)\} = \int d\Pi_p \left[\sum_\sigma u_\ell(p, \sigma) \bar{u}_{\ell'}(p, \sigma) e^{-ip \cdot z} + \sum_\sigma v_\ell(p, \sigma) \bar{v}_{\ell'}(p, \sigma) e^{+ip \cdot z} \right] \quad (235)$$

$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} [(p_\mu \gamma^\mu + m)_{\ell\ell'} e^{-ip \cdot z} + (p_\mu \gamma^\mu - m)_{\ell\ell'} e^{+ip \cdot z}]. \quad (236)$$

This is the desired expression. The relative sign in the mass term is fixed by the antiparticle spin sum, not inserted by hand.

Introduce the same scalar Pauli-Jordan function as in the scalar case,

$$\Delta(z) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} (e^{-ip \cdot z} - e^{+ip \cdot z}) \quad (237)$$

$$= \int \frac{d^4p}{(2\pi)^3} \text{sgn}(p^0) \delta(p^2 - m^2) e^{-ip \cdot z}. \quad (238)$$

The four-dimensional form makes clear that $\Delta(\Lambda z) = \Delta(z)$ for every proper orthochronous Lorentz transformation. For spacelike z , the vector $-z$ is Lorentz-related to z . Since the three-dimensional form also gives $\Delta(-z) = -\Delta(z)$, one obtains

$$\Delta(z) = 0, \quad z^2 < 0. \quad (239)$$

Eq. (236) can then be written compactly as

$$\{\psi_\ell(x), \bar{\psi}_{\ell'}(y)\} = (i\gamma^\mu \partial_\mu + m)_{\ell\ell'} \Delta(x - y). \quad (240)$$

Indeed, applying $i\gamma^\mu \partial_\mu + m$ to the first exponential in Δ gives $(p_\mu \gamma^\mu + m)e^{-ip \cdot z}$, while applying it to the second term $-e^{+ip \cdot z}$ gives $(p_\mu \gamma^\mu - m)e^{+ip \cdot z}$. For the Dirac field the spinor structure has therefore produced a single first-order differential operator acting on the same Pauli-Jordan function. Since Δ vanishes throughout the spacelike region, its first derivative also vanishes there. Hence

$$\boxed{\{\psi_\ell(x), \bar{\psi}_{\ell'}(y)\} = 0 \quad \text{for} \quad (x - y)^2 < 0.} \quad (241)$$

Thus the anticommutator gives precisely the causal cancellation required for spinor fields.

It is useful to see why the commutator would fail. If instead one used commutators for the operators $a, b, a^\dagger, b^\dagger$, the relative sign of the antiparticle term would be reversed:

$$[\psi_\ell(x), \bar{\psi}_{\ell'}(y)] = (i\gamma^\mu \partial_\mu + m)_{\ell\ell'} \Delta^{(1)}(x - y), \quad (242)$$

where

$$\Delta^{(1)}(z) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0} (e^{-ip \cdot z} + e^{+ip \cdot z}). \quad (243)$$

Unlike the Pauli-Jordan function, $\Delta^{(1)}(z)$ does not vanish for spacelike z . Therefore the commutator would not give a causal Dirac field. The anticommutator is not an optional convention: for the spin- $\frac{1}{2}$ field constructed from the Weinberg intertwiner equations, it is precisely what makes the particle and antiparticle contributions cancel outside the light cone.

Why the particle and antiparticle rest spinors must have opposite parity

Let us return to the parity choice made in Eq. (193), because it is not just a decorative convention. It is the point at which the representation-theoretic construction becomes the usual local Dirac field.

Rotations alone do not distinguish the two Weyl components at rest. They only imply that the particle coefficient has the form

$$u(k, \sigma) = \begin{pmatrix} c_L \xi_\sigma \\ c_R \xi_\sigma \end{pmatrix}, \quad (244)$$

and, after using the pseudoreality of spin $\frac{1}{2}$, that the antiparticle coefficient has the form

$$v(k, \sigma) = \begin{pmatrix} d_L \eta_\sigma \\ d_R \eta_\sigma \end{pmatrix}. \quad (245)$$

Here ξ_σ and η_σ both transform as spin- $\frac{1}{2}$ rest spinors, with the second written in the complex-conjugate convention appropriate to the antiparticle creation coefficient. The constants c_L, c_R, d_L, d_R are not fixed by rotations. This is why one needs another physical input.

For a massive spinor field the natural extra input is parity covariance. Parity reverses space,

$$(t, \mathbf{x}) \mapsto (t, -\mathbf{x}), \quad (246)$$

and exchanges the two inequivalent Weyl representations,

$$\left(\frac{1}{2}, 0\right) \longleftrightarrow \left(0, \frac{1}{2}\right). \quad (247)$$

Thus, in the chiral decomposition $V_D = V_L \oplus V_R$, the parity matrix is proportional to the operator which interchanges the two summands. With the normalization used above,

$$\Pi_D = \gamma^0 = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}. \quad (248)$$

Since the rest momentum $k = (m, \mathbf{0})$ is unchanged by parity, rest spinors may be chosen as eigenvectors of Π_D . The two possible parity eigenvectors are

$$\Pi_D \begin{pmatrix} \chi \\ \chi \end{pmatrix} = + \begin{pmatrix} \chi \\ \chi \end{pmatrix}, \quad \Pi_D \begin{pmatrix} \chi \\ -\chi \end{pmatrix} = - \begin{pmatrix} \chi \\ -\chi \end{pmatrix}. \quad (249)$$

So parity can reduce the arbitrary constants in Eqs. (244) and (245) to symmetric or antisymmetric combinations of left and right Weyl components.

The important question is why the particle coefficient u and the antiparticle coefficient v must be assigned opposite parity eigenvalues. The reason is locality. We want a single local first-order differential equation for the whole field

$$\psi(x) = \sum_\sigma \int d\Pi_p \left[u(p, \sigma) a(p, \sigma) e^{-ip \cdot x} + v(p, \sigma) b^\dagger(p, \sigma) e^{+ip \cdot x} \right]. \quad (250)$$

Suppose this local equation is the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0. \quad (251)$$

Acting on the positive-frequency part gives

$$(i\gamma^\mu \partial_\mu - m) \left[u(p, \sigma) e^{-ip \cdot x} \right] = (p_\mu \gamma^\mu - m) u(p, \sigma) e^{-ip \cdot x}. \quad (252)$$

Therefore the particle coefficient must obey

$$(p_\mu \gamma^\mu - m)u(p, \sigma) = 0. \quad (253)$$

On the negative-frequency part, however, the derivative gives the opposite sign:

$$i\partial_\mu e^{+ip \cdot x} = -p_\mu e^{+ip \cdot x}. \quad (254)$$

Hence

$$(i\gamma^\mu \partial_\mu - m) [v(p, \sigma) e^{+ip \cdot x}] = -(p_\mu \gamma^\mu + m)v(p, \sigma) e^{+ip \cdot x}. \quad (255)$$

Thus the antiparticle coefficient must obey

$$(p_\mu \gamma^\mu + m)v(p, \sigma) = 0. \quad (256)$$

The relative sign of the mass term is forced by the opposite frequencies in the local field expansion. It is not an arbitrary sign inserted after the fact.

Now specialize Eqs. (253) and (256) to the rest momentum. Since $k_\mu \gamma^\mu = m\gamma^0$, they become

$$m(\gamma^0 - \mathbf{1})u(k, \sigma) = 0, \quad m(\gamma^0 + \mathbf{1})v(k, \sigma) = 0. \quad (257)$$

Equivalently,

$$\gamma^0 u(k, \sigma) = +u(k, \sigma), \quad \gamma^0 v(k, \sigma) = -v(k, \sigma). \quad (258)$$

But γ^0 is precisely the rest-frame parity matrix Π_D . Therefore the existence of one local first-order Dirac equation for the whole field forces the particle and antiparticle rest spinors to have opposite parity. Conversely, once Eq. (258) is imposed at rest, Lorentz covariance boosts it to Eqs. (253) and (256) for all momenta.

This also explains what would go wrong if one chose the same parity for u and v . For example, if both were chosen with positive parity at rest, then both would satisfy the same rest equation

$$(\gamma^0 - \mathbf{1})w(k, \sigma) = 0. \quad (259)$$

After boosting, both would satisfy

$$(p_\mu \gamma^\mu - m)w(p, \sigma) = 0. \quad (260)$$

That is acceptable for the positive-frequency coefficient u , but it is not acceptable for the negative-frequency coefficient v in Eq. (250). The same local operator $(i\gamma^\mu \partial_\mu - m)$ would then fail to annihilate the antiparticle creation part. One would have to distinguish positive- and negative-frequency pieces by hand. Such a separation is not a local Lorentz-covariant differential operation on $\psi(x)$; it is a nonlocal split into frequency components. Thus the same-parity choice is incompatible with obtaining the usual local Dirac equation for a single four-component field.

The same point reappears in microcausality. The opposite parity assignment gives the spin sums

$$\sum_\sigma u(p, \sigma) \bar{u}(p, \sigma) = p_\mu \gamma^\mu + m, \quad \sum_\sigma v(p, \sigma) \bar{v}(p, \sigma) = p_\mu \gamma^\mu - m. \quad (261)$$

The two mass terms have opposite signs. This is exactly what is needed in Eq. (236): the anticommutator can be written as

$$\{\psi_\ell(x), \bar{\psi}_{\ell'}(y)\} = (i\gamma^\mu \partial_\mu + m)_{\ell\ell'} \Delta(x - y). \quad (262)$$

Thus the spinor anticommutator is a local differential operator acting on the scalar Pauli-Jordan function. Since $\Delta(x - y)$ vanishes at spacelike separation, the Dirac anticommutator also vanishes there.

This is the conceptual role of the opposite parity choice. Rotational covariance constructs the possible rest spinors. Parity covariance tells us that the left and right Weyl components must be combined into parity eigenvectors. Locality then fixes the relative parity: the particle coefficient and antiparticle coefficient must be assigned opposite eigenvalues so that the positive- and negative-frequency parts of one field obey one local first-order Dirac equation and combine into a causal anticommutator.

5 The Weyl and Majorana fermion; C, P, T and the CPT theorem; and why CP violation needs at least three generations

The previous section constructed the free electronic field from representation theory. We now add the dynamical formulation. The important logical point is that the Lagrangian is not used to guess the spinor wave functions. The spinors have already been obtained from Lorentz covariance and the massive spin- $\frac{1}{2}$ intertwiner equations. The Lagrangian now gives a compact dynamical principle, the Hamiltonian, the canonical anticommutation rules, and the symmetry currents.

5.1 Dirac bilinears first: tensors, and pseudo-tensors

Before writing an action, we should understand which local quantities can be used in a Lorentz-invariant Lagrangian. This is the logical reason for discussing bilinears before the Dirac action. The Dirac spinor transforms as

$$\psi'(x) = S(\Lambda)\psi(\Lambda^{-1}x), \quad (263)$$

where $S(\Lambda)$ is the finite-dimensional spinor representation of the Lorentz group. This $S(\Lambda)$ is not the unitary quantum operator $U(\Lambda)$ acting on the Hilbert space. The Hilbert-space operator is unitary for proper Lorentz transformations, while the finite-dimensional spinor matrix $S(\Lambda)$ is generally non-unitary, because the Lorentz group is non-compact.

We do not choose any explicit gamma-matrix representation. We use only the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}, \quad \eta = \text{diag}(+, -, -, -). \quad (264)$$

A useful point should be kept in mind from the beginning. The transpose matrices $(\gamma^\mu)^T$ also obey the same Clifford algebra, and so do $-(\gamma^\mu)^T$. Since the complex irreducible Clifford representation is unique up to similarity, both sets can be related to the original set γ^μ by suitable nonsingular matrices. The matrix used in charge conjugation is not chosen merely because a similarity transformation exists. It is chosen so that the transpose, or contragredient, spinor representation is converted back into the original Dirac spinor representation in a way compatible with the local Dirac equation. This is why the charge-conjugation matrix is taken to obey

$$C(\gamma^\mu)^T C^{-1} = -\gamma^\mu. \quad (265)$$

Indeed, if $\Sigma^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu]$, then (265) implies

$$C(\Sigma^{\mu\nu})^T C^{-1} = -\Sigma^{\mu\nu}. \quad (266)$$

This is precisely the infinitesimal statement that $C\bar{\psi}^T$ transforms as a spinor when $\bar{\psi}$ transforms in the dual representation. Thus the minus sign in (265) is not an arbitrary convention attached to gamma matrices; it is the sign which makes the charge-conjugated field

$$\psi^c := C\bar{\psi}^T \quad (267)$$

transform locally as a Dirac spinor and satisfy the same form of the Dirac equation. We will return to this point when discussing C , P , and T , but it is useful already here because the same invariant spinor pairings underlie all Dirac bilinears. To define a Lorentz-invariant spinor pairing we also need a Hermitizing matrix. Let us first explain why such a matrix exists, rather than simply postulating it. The complex Clifford algebra in four dimensions has a unique irreducible representation up to similarity. Start with any reference representation $\tilde{\gamma}^\mu$ in which the usual Dirac adjoint is already known, so that $\tilde{A}\tilde{\gamma}^\mu$ is Hermitian with $\tilde{A} = \tilde{\gamma}^0$. If another representation is related to it by

$$\gamma^\mu = R\tilde{\gamma}^\mu R^{-1}.$$

then the matrix

$$A := (R^{-1})^\dagger \tilde{A} R^{-1}.$$

is Hermitian and nonsingular, and it satisfies

$$(A\gamma^\mu)^\dagger = \left((R^{-1})^\dagger \tilde{A} \tilde{\gamma}^\mu R^{-1} \right)^\dagger = (R^{-1})^\dagger (\tilde{A} \tilde{\gamma}^\mu) R^{-1} = A\gamma^\mu.$$

Thus the Hermitizing matrix is forced by the equivalence class of the Clifford representation. The clean representation-independent way to state the result is the following. There is a nonsingular Hermitian matrix A such that the four matrices $A\gamma^\mu$ are Hermitian:

$$(A\gamma^\mu)^\dagger = A\gamma^\mu.$$

Equivalently,

$$(\gamma^\mu)^\dagger A = A\gamma^\mu, \quad (\gamma^\mu)^\dagger = A\gamma^\mu A^{-1}.$$

For the usual Dirac spinor representation in mostly-minus signature, one may choose the normalization of this Hermitizing matrix to be

$$A = \gamma^0.$$

This is not a choice of a particular explicit gamma-matrix basis; it is the choice of the invariant spinor pairing. Substituting $A = \gamma^0$ in the preceding relation, and using $(\gamma^0)^2 = \mathbf{1}$, gives

$$(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0, \quad \text{or equivalently} \quad (\gamma^0 \gamma^\mu)^\dagger = \gamma^0 \gamma^\mu. \quad (268)$$

This is sometimes called the pseudo-Hermiticity of the gamma matrices: the gamma matrices are Hermitian with respect to the indefinite Dirac pairing, not with respect to the positive Euclidean pairing. It implies in particular

$$(\gamma^0)^\dagger = \gamma^0, \quad (\gamma^i)^\dagger = -\gamma^i, \quad (\alpha^i)^\dagger = \alpha^i, \quad \alpha^i := \gamma^0 \gamma^i. \quad (269)$$

The word “pseudo” here has a precise meaning: the matrices are Hermitian not with respect to the ordinary positive inner product $\psi^\dagger \chi$, but with respect to the indefinite Dirac pairing $\psi^\dagger \gamma^0 \chi$.

Now introduce the spinor Lorentz generators

$$\Sigma^{\mu\nu} := \frac{1}{4} [\gamma^\mu, \gamma^\nu], \quad S(\Lambda) = \exp\left(\frac{1}{2} \omega_{\mu\nu} \Sigma^{\mu\nu}\right), \quad \omega_{\mu\nu} = -\omega_{\nu\mu}. \quad (270)$$

Using (268), we find

$$(\Sigma^{\mu\nu})^\dagger = \frac{1}{4} [(\gamma^\nu)^\dagger, (\gamma^\mu)^\dagger] = \frac{1}{4} [\gamma^0 \gamma^\nu \gamma^0, \gamma^0 \gamma^\mu \gamma^0] \quad (271)$$

$$= \gamma^0 \frac{1}{4} [\gamma^\nu, \gamma^\mu] \gamma^0 = -\gamma^0 \Sigma^{\mu\nu} \gamma^0. \quad (272)$$

Therefore

$$(\Sigma^{\mu\nu})^\dagger \gamma^0 + \gamma^0 \Sigma^{\mu\nu} = 0. \quad (273)$$

Exponentiating this infinitesimal identity gives

$$\boxed{S(\Lambda)^\dagger \gamma^0 S(\Lambda) = \gamma^0}, \quad \text{equivalently} \quad S(\Lambda)^\dagger \gamma^0 = \gamma^0 S(\Lambda)^{-1}. \quad (274)$$

The above equation should not be viewed as a unitarity condition on $S(\Lambda)$; in general $S(\Lambda)^\dagger S(\Lambda) \neq \mathbf{1}$, especially for boosts. It only says that $S(\Lambda)$ preserves the Dirac bilinear form. The unitary object in quantum theory is the operator $U(\Lambda)$ acting on states in the Hilbert space and fields which we have already studied explicitly.

The same spinor matrices also implement the covariance of the gamma matrices:

$$S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu. \quad (275)$$

At the infinitesimal level this follows from

$$[\Sigma^{\rho\sigma}, \gamma^\mu] = \eta^{\sigma\mu}\gamma^\rho - \eta^{\rho\mu}\gamma^\sigma, \quad (276)$$

and exponentiation gives (275).

This motivates the Dirac adjoint

$$\bar{\psi} := \psi^\dagger \gamma^0. \quad (277)$$

Indeed, using (274),

$$\bar{\psi}'(x) = \psi'(x)^\dagger \gamma^0 = \psi(\Lambda^{-1}x)^\dagger S(\Lambda)^\dagger \gamma^0 \quad (278)$$

$$= \psi(\Lambda^{-1}x)^\dagger \gamma^0 S(\Lambda)^{-1} = \bar{\psi}(\Lambda^{-1}x) S(\Lambda)^{-1}. \quad (279)$$

Therefore

$$\bar{\psi}'(x)\psi'(x) = \bar{\psi}(\Lambda^{-1}x)S(\Lambda)^{-1}S(\Lambda)\psi(\Lambda^{-1}x) = \bar{\psi}\psi(\Lambda^{-1}x). \quad (280)$$

This proves, before writing any action, that $\bar{\psi}\psi$ is a Lorentz scalar. The essential point is not unitarity of $S(\Lambda)$, but preservation of the indefinite spinor pairing.

By contrast, $\psi^\dagger\psi$ is not a Lorentz scalar. It is the time component of the vector current. Define

$$j^\mu := \bar{\psi}\gamma^\mu\psi. \quad (281)$$

Then

$$j'^\mu(x) = \bar{\psi}(\Lambda^{-1}x)S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\psi(\Lambda^{-1}x) \quad (282)$$

$$= \Lambda^\mu{}_\nu j^\nu(\Lambda^{-1}x). \quad (283)$$

Since $\psi^\dagger\psi = \bar{\psi}\gamma^0\psi = j^0$, it transforms as the zeroth component of a four-vector. Under rotations it behaves as a scalar density, but under boosts it mixes with j^i ; therefore it cannot appear by itself as a Lorentz-invariant mass term.

The complete set of independent local Dirac bilinears is obtained from the antisymmetrized products of gamma matrices. In four dimensions the sixteen matrices can be organized as

$$\mathbf{1}, \quad \gamma^5, \quad \gamma^\mu, \quad \gamma^\mu\gamma^5, \quad \sigma^{\mu\nu} := \frac{i}{2}[\gamma^\mu, \gamma^\nu], \quad (284)$$

with

$$\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3 = \frac{i}{4!}\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma. \quad (285)$$

The counting is

$$1 + 1 + 4 + 4 + 6 = 16. \quad (286)$$

Equivalently, the bilinears are

$$S := \bar{\psi}\psi, \quad P := i\bar{\psi}\gamma^5\psi, \quad (287)$$

$$V^\mu := \bar{\psi}\gamma^\mu\psi, \quad A^\mu := \bar{\psi}\gamma^\mu\gamma^5\psi, \quad (288)$$

$$T^{\mu\nu} := \bar{\psi}\sigma^{\mu\nu}\psi, \quad \tilde{T}^{\mu\nu} := \bar{\psi}\sigma^{\mu\nu}\gamma^5\psi. \quad (289)$$

The last bilinear is proportional, up to the conventional normalization and sign choice, to the dual tensor

$$\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}T_{\rho\sigma}. \quad (290)$$

The factor of i in P makes the pseudoscalar Hermitian with the pseudo-Hermiticity convention (268).

Let us now explain the word “pseudo.” Under a proper Lorentz transformation, namely $\det \Lambda = +1$, the Levi-Civita symbol behaves as an invariant tensor. Under an improper Lorentz transformation, such as parity, it acquires an additional sign:

$$\epsilon_{\mu\nu\rho\sigma}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Lambda^\rho{}_\gamma\Lambda^\sigma{}_\delta = (\det \Lambda)\epsilon_{\alpha\beta\gamma\delta}. \quad (291)$$

A pseudo-tensor is therefore an object which transforms like an ordinary tensor under proper Lorentz transformations, but with an extra factor of $\det \Lambda$ under improper transformations. In particular, a pseudoscalar is invariant under proper Lorentz transformations but changes sign under parity.

Using (275), the ordinary tensor bilinears transform as

$$S'(x) = S(\Lambda^{-1}x), \quad (292)$$

$$V'^\mu(x) = \Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x), \quad (293)$$

$$T'^{\mu\nu}(x) = \Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma T^{\rho\sigma}(\Lambda^{-1}x). \quad (294)$$

The bilinears involving one γ^5 are pseudo-objects:

$$P'(x) = (\det \Lambda)P(\Lambda^{-1}x), \quad (295)$$

$$A'^\mu(x) = (\det \Lambda)\Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x), \quad (296)$$

$$\tilde{T}'^{\mu\nu}(x) = (\det \Lambda)\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma \tilde{T}^{\rho\sigma}(\Lambda^{-1}x). \quad (297)$$

Thus the classification is

bilinear	Lorentz type	behavior under parity
$\psi\psi$	scalar	even
$i\bar{\psi}\gamma^5\psi$	pseudoscalar	odd
$\bar{\psi}\gamma^\mu\psi$	vector	ordinary vector
$\bar{\psi}\gamma^\mu\gamma^5\psi$	axial vector	pseudovector
$\bar{\psi}\sigma^{\mu\nu}\psi$	antisymmetric tensor	ordinary tensor
$\bar{\psi}\sigma^{\mu\nu}\gamma^5\psi$	antisymmetric pseudotensor	dual tensor

For example, under parity $x = (t, \mathbf{x}) \mapsto x_P = (t, -\mathbf{x})$,

$$V^0(x) \mapsto +V^0(x_P), \quad V^i(x) \mapsto -V^i(x_P), \quad (298)$$

$$A^0(x) \mapsto -A^0(x_P), \quad A^i(x) \mapsto +A^i(x_P). \quad (299)$$

This is why A^μ is called an axial vector, or pseudovector. These bilinears are the building blocks of Lorentz-invariant interactions. We can form ordinary scalar Lagrangian terms by contracting all ordinary Lorentz indices, for instance $\bar{\psi}\psi$, $\bar{\psi}\gamma^\mu\partial_\mu\psi$, $V_\mu V^\mu$, or $T_{\mu\nu}T^{\mu\nu}$. Pseudoscalar terms can also be formed, such as $i\bar{\psi}\gamma^5\psi$ or $F_{\mu\nu}\tilde{F}^{\mu\nu}$, but they are odd under parity. Therefore, if the theory is required to preserve parity, pseudoscalar terms are excluded; if parity is not imposed, they are allowed Lorentz-invariant densities.

5.2 Dirac Lagrangian, Hamiltonian, and canonical anticommutation rules

We can now write the Dirac action without guessing its form. The preceding subsection showed that $\bar{\psi}\psi$ is a Lorentz scalar and that $\bar{\psi}\gamma^\mu\psi$ is a vector. Since ∂_μ is a covector, the contraction $\bar{\psi}\gamma^\mu\partial_\mu\psi$ is also a scalar. The Hermitian free massive Dirac action may therefore be written as

$$S_D = \int d^4x \mathcal{L}_D, \quad \mathcal{L}_D = \frac{i}{2}\bar{\psi}\gamma^\mu \overleftrightarrow{\partial}_\mu \psi - m\bar{\psi}\psi, \quad (300)$$

where

$$\bar{\psi}\gamma^\mu\overleftrightarrow{\partial}_\mu\psi := \bar{\psi}\gamma^\mu\partial_\mu\psi - (\partial_\mu\bar{\psi})\gamma^\mu\psi. \quad (301)$$

The pseudo-Hermiticity relation (268) ensures that this action is Hermitian, without choosing a special gamma-matrix basis. Integrating the second term by parts gives

$$\int d^4x \frac{i}{2} \bar{\psi}\gamma^\mu\overleftrightarrow{\partial}_\mu\psi = \int d^4x i\bar{\psi}\gamma^\mu\partial_\mu\psi + \text{surface term}. \quad (302)$$

With the usual boundary condition that fields vanish at infinity, the action is equivalently

$$\mathcal{L}_D = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi. \quad (303)$$

This is the form convenient for canonical quantization. It is representation-independent: it uses only the Clifford algebra, the Hermitizing matrix γ^0 , and the covariance identity (275).

Varying the Hermitian action (300), and integrating by parts in the variation, gives

$$(i\gamma^\mu\partial_\mu - m)\psi = 0. \quad (304)$$

Varying with respect to ψ gives the adjoint equation

$$\bar{\psi}(i\overleftarrow{\partial}_\mu\gamma^\mu + m) = 0. \quad (305)$$

Here the arrow means that the derivative acts to the left.

To find the Hamiltonian, use the first-order form (303). Substituting $\bar{\psi} = \psi^\dagger\gamma^0$ gives the time-derivative and spatial-derivative terms explicitly:

$$\mathcal{L}_D = i\psi^\dagger\partial_t\psi + i\psi^\dagger\gamma^0\gamma^i\partial_i\psi - m\psi^\dagger\gamma^0\psi. \quad (306)$$

Define, without choosing any explicit gamma matrices,

$$\alpha^i := \gamma^0\gamma^i, \quad \beta := \gamma^0. \quad (307)$$

The momentum conjugate to ψ_ℓ is therefore

$$\Pi_\ell(x) := \frac{\partial\mathcal{L}_D}{\partial(\partial_t\psi_\ell)} = i\psi_\ell^\dagger(x), \quad (308)$$

while the momentum conjugate to $\psi^\dagger$ vanishes in this first-order form. Equivalently, the independent canonical pair is

$$Q_\ell(t, \mathbf{x}) := \psi_\ell(t, \mathbf{x}), \quad P_\ell(t, \mathbf{x}) := \Pi_\ell(t, \mathbf{x}) = i\psi_\ell^\dagger(t, \mathbf{x}). \quad (309)$$

For fermionic variables the canonical equal-time bracket is an anticommutator:

$$\boxed{\{Q_\ell(t, \mathbf{x}), P_{\ell'}(t, \mathbf{y})\} = i\delta_{\ell\ell'}\delta^3(\mathbf{x} - \mathbf{y})}. \quad (310)$$

Equivalently,

$$\boxed{\{\psi_\ell(t, \mathbf{x}), \psi_{\ell'}^\dagger(t, \mathbf{y})\} = \delta_{\ell\ell'}\delta^3(\mathbf{x} - \mathbf{y})}, \quad \{\psi, \psi\} = \{\psi^\dagger, \psi^\dagger\} = 0. \quad (311)$$

This is the equal-time form of the anticommutation rule whose spacetime consequence was microcausality in Section 4.4. Conversely, the equal-time rule follows from the covariant anticommutator. From the mode expansion one has

$$\{\psi_\ell(x), \bar{\psi}_{\ell'}(y)\} = (i\gamma^\mu\partial_\mu + m)_{\ell\ell'}\Delta(x - y), \quad (312)$$

where the derivative acts on $z = x - y$. Multiplying on the right by γ^0 gives the anticommutator with $\psi^\dagger$:

$$\{\psi_\ell(t, \mathbf{x}), \psi_{\ell'}^\dagger(t, \mathbf{y})\} = [(\mathrm{i}\gamma^\mu \partial_\mu + m)\Delta(z) \gamma^0]_{\ell\ell'} \big|_{z^0=0}. \quad (313)$$

At equal times $\Delta(0, \mathbf{z}) = 0$, so the mass term vanishes. Also $\partial_i \Delta(0, \mathbf{z}) = 0$. The only surviving term is

$$\mathrm{i}\gamma^0 \partial_{z^0} \Delta(z) \big|_{z^0=0} \gamma^0 = \mathrm{i}\gamma^0 (-\mathrm{i}\delta^3(\mathbf{z})) \gamma^0 = \mathbf{1} \delta^3(\mathbf{z}), \quad (314)$$

which is precisely Eq. (311). This is the fermionic analogue of the scalar consistency check: the equal-time canonical rule is the equal-time limit of the Lorentz-covariant causal distribution.

The Hamiltonian density is

$$\mathcal{H}_D = \Pi_\ell \partial_t \psi_\ell - \mathcal{L}_D = \psi^\dagger (-\mathrm{i}\alpha^i \partial_i + m\beta) \psi. \quad (315)$$

Thus

$$H_D = \int \mathrm{d}^3 \mathbf{x} \psi^\dagger (-\mathrm{i}\boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m\beta) \psi. \quad (316)$$

This Hamiltonian is Hermitian. Indeed, using $(\alpha^i)^\dagger = \alpha^i$ and $\beta^\dagger = \beta$ from (269),

$$H_D^\dagger = \int \mathrm{d}^3 \mathbf{x} \left[(\mathrm{i}\partial_i \psi^\dagger) \alpha^i \psi + m \psi^\dagger \beta \psi \right] \quad (317)$$

$$= \int \mathrm{d}^3 \mathbf{x} \psi^\dagger (-\mathrm{i}\alpha^i \partial_i + m\beta) \psi, \quad (318)$$

where in the last line we integrated the spatial derivative by parts and dropped the surface term. This is the integration-by-parts step needed to justify the Hermiticity of the spatial derivative operator.

After normal ordering and using the mode expansion of Section 4, this becomes

$$H_D = \sum_{\sigma=\pm 1/2} \int \frac{\mathrm{d}^3 \mathbf{p}}{(2\pi)^3 2p^0} p^0 \left[a^\dagger(p, \sigma) a(p, \sigma) + b^\dagger(p, \sigma) b(p, \sigma) \right]. \quad (319)$$

The omitted c-number is the usual zero-point energy, removed by normal ordering.

Now the quantum equation of motion follows from the Hamiltonian. Since H_D is bosonic, the Heisenberg equation uses the ordinary commutator,

$$\mathrm{i}\partial_t \psi_\ell(t, \mathbf{x}) = [\psi_\ell(t, \mathbf{x}), H_D]. \quad (320)$$

For a quadratic fermion Hamiltonian one uses the elementary graded identity

$$[\psi_\ell(\mathbf{x}), \psi_m^\dagger(\mathbf{y}) h_{mn} \psi_n(\mathbf{y})] = \{\psi_\ell(\mathbf{x}), \psi_m^\dagger(\mathbf{y})\} h_{mn} \psi_n(\mathbf{y}), \quad (321)$$

where $h = -\mathrm{i}\boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m\beta$ acts on the $\mathbf{y}$ argument. Using (311), one obtains

$$\mathrm{i}\partial_t \psi = (-\mathrm{i}\alpha^i \partial_i + m\beta) \psi. \quad (322)$$

Multiplying by γ^0 gives

$$(\mathrm{i}\gamma^\mu \partial_\mu - m) \psi = 0. \quad (323)$$

Thus the same Dirac equation appears in three logically distinct but consistent ways: from the Lorentz-intertwiner construction of Section 4, from the scalar Dirac Lagrangian, and from the quantum Hamiltonian together with the canonical (anti)commutation relation. In the latter case, we need to choose anticommutation for obtaining microcausality.

5.3 Massless equation, Weyl spinors, Majorana representation, and mass terms

In the chiral basis write

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad \gamma^5 \psi_L = -\psi_L, \quad \gamma^5 \psi_R = +\psi_R. \quad (324)$$

For $m = 0$, the Dirac equation splits into two independent Weyl equations:

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = 0, \quad i\sigma^\mu \partial_\mu \psi_R = 0. \quad (325)$$

Thus a massless left-handed Weyl spinor and a massless right-handed Weyl spinor are independent relativistic fields. In the massless case chirality is conserved. For positive-energy particles, chirality also coincides with helicity: left-handed particles have negative helicity and right-handed particles have positive helicity. For antiparticles the helicity assignment is reversed.

A useful object is the charge-conjugate spinor

$$\psi^c := C\bar{\psi}^T, \quad C^{-1}\gamma^\mu C = -(\gamma^\mu)^T. \quad (326)$$

The matrix C is not a new dynamical object. It is a fixed matrix in spinor space which exists because the matrices $-\gamma^{\mu T}$ obey the same Clifford algebra as the matrices γ^μ . By irreducibility of the complex Clifford representation, the two sets must be related by a similarity transformation; this is precisely the defining relation in Eq. (326). Let us verify two basic facts, because they explain why ψ^c can be used as another Dirac spinor.

First, ψ^c transforms under the same Lorentz representation as ψ . Since

$$\bar{\psi}'(x) = \bar{\psi}(\Lambda^{-1}x)S(\Lambda)^{-1}, \quad (327)$$

we get

$$(\psi')^c(x) = C(\bar{\psi}'(x))^T = C(S(\Lambda)^{-1})^T \bar{\psi}(\Lambda^{-1}x)^T. \quad (328)$$

The defining property of C implies

$$C(S(\Lambda)^{-1})^T = S(\Lambda)C, \quad (329)$$

because the transpose representation has generators $-(\Sigma^{\mu\nu})^T$ and C intertwines it with the original spinor representation. Therefore

$$(\psi')^c(x) = S(\Lambda)\psi^c(\Lambda^{-1}x). \quad (330)$$

So charge conjugation changes internal charges, but not the Lorentz transformation law of the spinor.

Second, if ψ satisfies the Dirac equation, then ψ^c satisfies the same Dirac equation. The adjoint equation is

$$i(\partial_\mu \bar{\psi})\gamma^\mu + m\bar{\psi} = 0. \quad (331)$$

Transposing it gives

$$i(\gamma^\mu)^T \partial_\mu \bar{\psi}^T + m\bar{\psi}^T = 0. \quad (332)$$

Multiplying on the left by C and using $C(\gamma^\mu)^T C^{-1} = -\gamma^\mu$, we find

$$-i\gamma^\mu \partial_\mu \psi^c + m\psi^c = 0, \quad (333)$$

or equivalently

$$(i\gamma^\mu \partial_\mu - m)\psi^c = 0. \quad (334)$$

Thus charge conjugation maps solutions of the Dirac equation to solutions of the same Dirac equation.

A Majorana spinor satisfies

$$\psi^c = \psi. \quad (335)$$

This condition is possible only for a neutral field, because charge conjugation changes the sign of any ordinary $U(1)$ charge.

Let us pause to state the representation-theoretic notion behind this formula. For a group representation with matrices $U(g)$, the complex conjugate representation has matrices $U(g)^*$. Saying that a representation is equivalent to its complex conjugate means that there exists a fixed invertible matrix B such that

$$BU(g)^*B^{-1} = U(g) \quad \text{for all } g. \quad (336)$$

This is a similarity transformation between the representation U and the conjugate representation U^* . If the corresponding antilinear map $J := BK$, where K denotes complex conjugation, obeys $J^2 = +1$, the representation is called real. If $J^2 = -1$, it is called pseudoreal, or quaternionic. Equivalently, the invariant bilinear form is symmetric in the real case and antisymmetric in the pseudoreal case.

The spin- $\frac{1}{2}$ representation of $SU(2)$ is pseudoreal. With

$$\epsilon := i\sigma^2, \quad \epsilon U^* \epsilon^{-1} = U, \quad U \in SU(2), \quad (337)$$

the antilinear map ϵK squares to -1 . This is the same antisymmetric tensor that appears in the time-reversal matrix and in the Weyl Majorana mass below. For the full Lorentz group, however, the two Weyl representations are complex conjugates of each other:

$$\left(\frac{1}{2}, 0\right)^* = \left(0, \frac{1}{2}\right). \quad (338)$$

Thus in four dimensions a Majorana spinor is not a single isolated Weyl representation; it is a field in which the right-handed component is the Lorentz-covariant charge conjugate of the left-handed component.

The Majorana condition may be written explicitly in both four-component and two-component notation. In the chiral basis, using $C = i\gamma^2\gamma^0$ as above,

$$\psi^c = \begin{pmatrix} i\sigma^2\psi_R^* \\ -i\sigma^2\psi_L^* \end{pmatrix}. \quad (339)$$

Therefore $\psi^c = \psi$ is equivalent to

$$\psi_L = i\sigma^2\psi_R^*, \quad \psi_R = -i\sigma^2\psi_L^*. \quad (340)$$

Equivalently, in terms of one independent left-handed Weyl spinor $\chi := \psi_L$,

$$\psi_M^{\text{chiral}} = \begin{pmatrix} \chi \\ -i\sigma^2\chi^* \end{pmatrix}, \quad \psi_R = -i\sigma^2\psi_L^*. \quad (341)$$

Thus a four-component Majorana spinor in four dimensions contains only one independent Weyl spinor. The right-handed component is not a second independent field; it is the Lorentz-covariant charge conjugate of the left-handed one. Conversely, if we start from a right-handed Weyl spinor $\rho := \psi_R$, then the same condition gives

$$\psi_M^{\text{chiral}} = \begin{pmatrix} i\sigma^2\rho^* \\ \rho \end{pmatrix}, \quad \psi_L = i\sigma^2\psi_R^*. \quad (342)$$

In a standard Dirac basis the same Majorana condition can be written in the analogous two-Pauli-spinor form

$$\psi_M^{\text{Dirac}} = \begin{pmatrix} \xi \\ -i\sigma^2 \xi^* \end{pmatrix}, \quad (343)$$

where ξ is an ordinary two-component Pauli spinor. The difference between Eqs. (341) and (343) is a basis change on the gamma matrices and on the spinor components; the invariant statement is Eq. (335). The Weyl basis makes the relation between left and right Lorentz representations transparent, while the Dirac basis often makes the ordinary Dirac mass term visually simplest.

The Majorana representation is a basis of gamma matrices in which charge conjugation is simply complex conjugation on the spinor. Define

$$B_C := C(\gamma^0)^T, \quad \psi^c = B_C \psi^*. \quad (344)$$

In a Majorana basis we want

$$B_C^M = \mathbf{1}, \quad \psi^c = \psi^*. \quad (345)$$

Then the Majorana condition is simply $\psi_M^* = \psi_M$. In terms of the matrix C appearing in $C\bar{\psi}^T$, this means $C_M(\gamma_M^0)^T = \mathbf{1}$. For the explicit matrices below, one may take $C_M = -\gamma_M^0$, up to an irrelevant overall sign convention.

One explicit four-dimensional choice, with mostly-minus metric, is

$$\gamma_M^0 = \begin{pmatrix} 0 & \sigma^2 \\ \sigma^2 & 0 \end{pmatrix}, \quad \gamma_M^1 = \begin{pmatrix} i\sigma^3 & 0 \\ 0 & i\sigma^3 \end{pmatrix}, \quad (346)$$

$$\gamma_M^2 = \begin{pmatrix} 0 & -\sigma^2 \\ \sigma^2 & 0 \end{pmatrix}, \quad \gamma_M^3 = \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix}. \quad (347)$$

All these matrices are purely imaginary and obey

$$\{\gamma_M^\mu, \gamma_M^\nu\} = 2\eta^{\mu\nu} \mathbf{1}. \quad (348)$$

Therefore the operator $i\gamma_M^\mu \partial_\mu - m$ is real. In this representation the Majorana condition can be written as the reality condition

$$\psi_M^* = \psi_M. \quad (349)$$

This is why Majorana fermions are often called real fermions.

The discrete transformations also have a simple form in a Majorana basis. Since charge conjugation is complex conjugation,

$$C\psi_M C^{-1} = \eta_C \psi_M^*. \quad (350)$$

For a Majorana field $\psi_M^* = \psi_M$, so charge conjugation acts only by a possible sign or phase fixed by convention. Parity may be represented by

$$P\psi_M(t, \mathbf{x})P^{-1} = \eta_P \gamma_M^0 \psi_M(t, -\mathbf{x}). \quad (351)$$

A convenient real convention for time reversal is

$$T\psi_M(t, \mathbf{x})T^{-1} = \eta_T B_T^M \psi_M(-t, \mathbf{x}), \quad B_T^M := \gamma_M^1 \gamma_M^3, \quad (352)$$

with T antiunitary. Multiplying B_T^M by an overall phase gives equivalent conventions. The useful point is that these choices preserve the reality condition of the Majorana field.

Let us now list the possible mass terms. The guiding principle is simple: a mass term must be a Lorentz scalar and must also respect whatever internal gauge symmetries are imposed. Lorentz invariance alone is not enough.

For a four-component Dirac spinor the usual Dirac mass is

$$\boxed{\text{Dirac mass: } \mathcal{L}_m^{\text{Dirac}} = -m\bar{\psi}\psi.} \quad (353)$$

In chiral notation it couples left and right components:

$$-m\bar{\psi}\psi = -m(\psi_R^\dagger\psi_L + \psi_L^\dagger\psi_R). \quad (354)$$

Thus a Dirac mass is not a mass for a single Weyl representation alone. It pairs opposite chiralities. In a gauge theory this is allowed only if the two Weyl fields have conjugate gauge quantum numbers, so that their product is a gauge singlet.

For a single left-handed Weyl spinor χ_α , Lorentz invariance permits the antisymmetric contraction

$$\chi\chi := \chi^\alpha\chi_\alpha = \epsilon^{\alpha\beta}\chi_\beta\chi_\alpha. \quad (355)$$

Because the fields anticommute, this expression is not zero. A Majorana mass term is

$$\boxed{\text{Majorana mass for one Weyl spinor: } \mathcal{L}_m^{\text{Maj}} = -\frac{1}{2}\left(m_M\chi\chi + m_M^*\chi^\dagger\chi^\dagger\right).} \quad (356)$$

This term violates any $U(1)$ charge carried by χ by two units, so it is allowed only when the field is neutral under the relevant exact symmetries. In four-component notation, if Ψ_M is the Majorana spinor built from χ as in Eq. (341), the same term is

$$\mathcal{L}_m^{\text{Maj}} = -\frac{1}{2}m_M\bar{\Psi}_M\Psi_M \quad (357)$$

when the phase of χ has been chosen so that m_M is real.

Now introduce flavors. Let χ_i be several left-handed Weyl spinors, with i, j flavor labels. For several Weyl spinors the most general Lorentz-invariant Majorana mass term is

$$\mathcal{L}_m^{\text{Maj}} = -\frac{1}{2}\left(M_{ij}\chi_i\chi_j + M_{ij}^*\chi_i^\dagger\chi_j^\dagger\right), \quad M_{ij} = M_{ji}. \quad (358)$$

Here M_{ij} is a complex symmetric mass matrix in flavor space. It is symmetric because the Lorentz contraction is antisymmetric while the fields anticommute. With spinor indices,

$$M_{ij}\chi_i\chi_j = M_{ij}\epsilon^{\alpha\beta}\chi_{i\alpha}\chi_{j\beta}. \quad (359)$$

Exchanging $i \leftrightarrow j$ gives one minus sign from $\epsilon^{\alpha\beta}$ and one minus sign from moving the fermion fields through each other; the product is therefore symmetric in i, j . Only the symmetric part of M_{ij} contributes.

The same term is often written in modern four-component notation with a charge-conjugation matrix. Introduce four-component spinors Ψ_i whose left-handed projections contain the Weyl fields χ_i :

$$P_L\Psi_i = \begin{pmatrix} \chi_i \\ 0 \end{pmatrix} \quad \text{in the chiral basis.} \quad (360)$$

Define

$$\mathcal{M}_{ij} := M_{ij}, \quad \mathcal{M}^T = \mathcal{M}, \quad (361)$$

where $\mathcal{M}$ acts only on flavor labels, not on Dirac indices. Then Eq. (358) is equivalently

$$\mathcal{L}_m^{\text{Maj}} = -\frac{1}{2}\left[\Psi_i^T C \mathcal{M}_{ij} P_L\Psi_j + \text{h.c.}\right]. \quad (362)$$

This expression is Lorentz invariant. Indeed, under $\Psi \mapsto S\Psi$,

$$\Psi_i^T C P_L\Psi_j \mapsto \Psi_i^T S^T C P_L S\Psi_j. \quad (363)$$

For proper Lorentz transformations chirality is preserved, so $[S, P_L] = 0$, and the charge-conjugation matrix obeys

$$S^T C S = C. \quad (364)$$

Therefore $\Psi_i^T C P_L \Psi_j$ is a Lorentz scalar. In components,

$$\Psi_i^T C \mathcal{M}_{ij} P_L \Psi_j = \sum_{i,j} \mathcal{M}_{ij} \Psi_i^a C_{ab} (P_L \Psi_j)^b. \quad (365)$$

In the chiral basis this reduces to $M_{ij} \chi_i \chi_j$ up to the convention-dependent sign in the definition of C and $\epsilon = i\sigma^2$. This derives Eq. (362) from Eq. (358): the four-component expression is just the same antisymmetric Weyl contraction written with the invariant matrix C .

If we instead introduce four-component Majorana fields $\Psi_{Mi} = \Psi_{Mi}^c$, the mass term can be written as

$$\mathcal{L}_m^{\text{Maj}} = -\frac{1}{2} \bar{\Psi}_{Mi} \mathcal{M}_{ij}^{(M)} \Psi_{Mj}. \quad (366)$$

For real Majorana fields the matrix $\mathcal{M}^{(M)}$ is real and symmetric. Starting from a general complex symmetric Weyl matrix M , one diagonalizes by a Takagi factorization,

$$U^T M U = \text{diag}(m_1, m_2, \dots), \quad m_i \geq 0, \quad (367)$$

with U unitary. The transpose appears because the Majorana bilinear contains two fields of the same chirality:

$$M_{ij} \chi_i \chi_j, \quad \chi_i = U_{ia} \chi'_a \implies M_{ij} U_{ia} U_{jb} \chi'_a \chi'_b = (U^T M U)_{ab} \chi'_a \chi'_b. \quad (368)$$

This is different from the diagonalization of a Hermitian matrix, where one uses $U^\dagger M U$. Here M is complex symmetric, not Hermitian, so the correct diagonalization is Takagi diagonalization.

A minimal gauge-theory interlude: why neutrino masses need indices

The neutrinos provide the main physical example, but to say what mass terms are allowed we need one small piece of Standard Model notation. The electroweak gauge group is

$$G_{\text{EW}} = \text{SU}(2)_L \times U(1)_Y. \quad (369)$$

The subscript L means that the $\text{SU}(2)$ part acts only on left-handed doublets. For each lepton flavor $i = e, \mu, \tau$, the left-handed neutrino and charged lepton are not separate $\text{SU}(2)_L$ singlets. They are the two components of one doublet,

$$L_i^A = \begin{pmatrix} \nu_{Li} \\ e_{Li} \end{pmatrix}^A, \quad A = 1, 2, \quad Y(L_i) = -\frac{1}{2}. \quad (370)$$

Here A is an $\text{SU}(2)_L$ index. Under a local electroweak transformation,

$$L_i^A(x) \longmapsto [U_L(x)]^A_B e^{-i\alpha_Y(x)/2} L_i^B(x), \quad U_L(x) \in \text{SU}(2)_L. \quad (371)$$

The right-handed charged lepton e_{Ri} is an $\text{SU}(2)_L$ singlet. In the minimal Standard Model there is no right-handed neutrino field. The Higgs field is another doublet,

$$H^A = \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}^A, \quad Y(H) = +\frac{1}{2}, \quad \epsilon_{AB} H^A L_i^B \text{ is an } \text{SU}(2)_L \text{ singlet}. \quad (372)$$

We will not discuss symmetry breaking here. The only point needed is this: every allowed term in the Lagrangian must be a singlet of $\text{SU}(2)_L \times U(1)_Y$.

A bare active-neutrino Majorana mass is therefore not allowed. The naive Lorentz scalar

$$M_{ij}\nu_{Li}\nu_{Lj} + \text{h.c.} \quad (373)$$

uses only the upper component of the doublet in Eq. (370); by itself it is not an $SU(2)_L$ invariant expression. Trying to use the full doublet does not fix the problem:

$$\epsilon_{AB}L_i^A L_j^B \quad (374)$$

is an $SU(2)_L$ singlet, but it has hypercharge $Y = -1$ and hence is not invariant under $U(1)_Y$. Thus gauge symmetry forbids a renormalizable Majorana mass for the active neutrinos alone.

There are two standard ways in which neutrino masses can nevertheless be written without violating the gauge symmetry. First, one may add a sterile left-handed Weyl field N_i , equivalently the charge conjugate of a right-handed neutrino. It is a complete electroweak singlet:

$$N_i \sim (\mathbf{1}, 0) \quad \text{under } SU(2)_L \times U(1)_Y. \quad (375)$$

Then a Dirac-type gauge invariant can be formed as

$$y_{ij} \epsilon_{AB} L_i^A H^B N_j + \text{h.c.} \quad (376)$$

In four-component notation this is the same object as $\bar{L}_i \tilde{H} N_{Rj} + \text{h.c.}$, with $\tilde{H}^A := \epsilon^{AB} H_B^*$. Second, since N_i is neutral under all electroweak gauge transformations, it may have a Majorana mass directly:

$$-\frac{1}{2} (M_{N,ij} N_i N_j + \text{h.c.}), \quad M_N^T = M_N. \quad (377)$$

Finally, even without adding N_i , an active-neutrino Majorana mass can be represented at low energy by the dimension-five gauge singlet

$$\frac{c_{ij}}{\Lambda} (\epsilon_{AB} L_i^A H^B) (\epsilon_{CD} L_j^C H^D) + \text{h.c.} \quad (378)$$

The ϵ tensors are the explicit $SU(2)_L$ contractions. The hypercharge also cancels: each factor LH has $Y = -\frac{1}{2} + \frac{1}{2} = 0$. Thus Eq. (378) is gauge invariant. This is the clean way to see why the active-neutrino Majorana mass is possible only as a higher-dimensional operator unless new gauge-singlet fermions are introduced.

For two left-handed Weyl spinors carrying opposite charges, one may write a Dirac mass

$$\mathcal{L}_m^{\text{Dirac, Weyl}} = -(m_D \chi \eta + m_D^* \chi^\dagger \eta^\dagger). \quad (379)$$

The Lorentz contraction is the same kind of $\epsilon^{\alpha\beta}$ contraction as before, but the two fields are distinct. To pass to four-component notation, build a Dirac spinor from the two Weyl spinors by taking one left-handed field and the charge conjugate of the other as the right-handed component:

$$\psi_D = \begin{pmatrix} \chi \\ -i\sigma^2 \eta^* \end{pmatrix}. \quad (380)$$

Then Eq. (379) becomes the familiar four-component Dirac mass

$$\mathcal{L}_m^{\text{Dirac, Weyl}} = -m_D \bar{\psi}_D \psi_D \quad (381)$$

up to the same phase convention used in Eq. (380). This explains why the Dirac basis is often visually simplest for Dirac masses, while the Weyl basis is simplest for Majorana masses.

Experimentally, the classic search for a Majorana neutrino mass is neutrinoless double beta decay, $0\nu\beta\beta$. An observation would show lepton-number violation by two units and would strongly indicate that neutrinos have a Majorana component. Current and next-generation

searches include KamLAND-Zen using ^{136}Xe , LEGEND using ^{76}Ge , CUORE/CUPID using tellurium bolometers, and nEXO using enriched liquid xenon. A Majorana mass by itself does not make the known active neutrinos into the dark matter: active neutrinos are too light and too relativistic to account for cold dark matter. However, a mostly sterile Majorana neutrino with a keV-scale mass is a possible warm-dark-matter candidate in extensions of the Standard Model. This is a model-building possibility, not a consequence of the Majorana mass term alone.

Thus the basic possibilities are: a Dirac mass pairing two conjugate Weyl fields, a Majorana mass for a neutral Weyl field, or both when the gauge symmetries allow it.

5.4 Charge conjugation, parity, and time reversal

The purpose of this subsection is not to memorize three transformation laws. The purpose is to understand why the local laws of the form

$$\mathbf{C}\psi(x)\mathbf{C}^{-1} \propto \psi^c(x), \quad \mathbf{P}\psi(x)\mathbf{P}^{-1} \propto B_P\psi(x_P), \quad \mathbf{T}\psi(x)\mathbf{T}^{-1} \propto B_T\psi(x_T) \quad (382)$$

exist at all, and why the matrices C , B_P , and B_T are not arbitrary.

The starting point is the one-particle Hilbert space. Parity and time reversal have natural actions on spacetime and on the positive-energy momentum label. We write

$$x_P = (t, -\mathbf{x}), \quad x_T = (-t, \mathbf{x}), \quad p^P = p^T := (p^0, -\mathbf{p}), \quad (383)$$

where $p^0 = +\sqrt{\mathbf{p}^2 + m^2}$. Both P and T reverse the spatial momentum of a positive-energy particle. They differ because parity is unitary and leaves time unchanged, while time reversal is antiunitary and reverses the time argument. Spin also transforms: spin is an axial vector, so parity leaves the spin label in a rest-frame basis unchanged, whereas time reversal reverses spin and is represented on spin- $\frac{1}{2}$ indices by the antisymmetric invariant tensor. Charge conjugation, by contrast, does not move the spacetime point or the momentum; it exchanges the particle and antiparticle oscillators.

The nontrivial requirement of a local quantum field theory is stronger than these one-particle statements. A field is local only if the symmetry induced from the particle states can be rewritten as a finite matrix acting on the field at the transformed point, not as a momentum-dependent integral transform. For a scalar field this is almost automatic, because there is no spinor index. For the Dirac field it is highly restrictive: the momentum-space spinors $u(p, \sigma)$ and $v(p, \sigma)$ must conspire with the oscillator transformation laws so that the transformed field is again a local spinor field. This is the guiding principle below.

Warm-up: scalar particles and scalar fields

Consider the complex scalar field of Section 4,

$$\phi(x) = \int d\Pi_p \left[a(p)e^{-ip \cdot x} + b^\dagger(p)e^{+ip \cdot x} \right], \quad d\Pi_p := \frac{d^3\mathbf{p}}{(2\pi)^3 2p^0}. \quad (384)$$

Here $a^\dagger$ creates the particle and $b^\dagger$ creates the antiparticle. The measure $d\Pi_p$ is invariant under $\mathbf{p} \mapsto -\mathbf{p}$.

Charge conjugation. Charge conjugation exchanges the particle and the antiparticle. Choose phases by

$$\mathbf{C}a^\dagger(p)\mathbf{C}^{-1} = \eta_C^{\phi*} b^\dagger(p), \quad \mathbf{C}b^\dagger(p)\mathbf{C}^{-1} = \eta_C^\phi a^\dagger(p), \quad (385)$$

$$\mathbf{C}a(p)\mathbf{C}^{-1} = \eta_C^\phi b(p), \quad \mathbf{C}b(p)\mathbf{C}^{-1} = \eta_C^{\phi*} a(p). \quad (386)$$

Substitution in Eq. (384) gives

$$\mathbf{C}\phi(x)\mathbf{C}^{-1} = \int d\Pi_p \left[\eta_C^\phi b(p) e^{-ip \cdot x} + \eta_C^\phi a^\dagger(p) e^{+ip \cdot x} \right]. \quad (387)$$

This is local because it is precisely the adjoint field at the same point, up to the same phase convention:

$$\boxed{\mathbf{C}\phi(x)\mathbf{C}^{-1} = \eta_C^\phi \phi^\dagger(x)}. \quad (388)$$

For a real neutral scalar, $b = a$, and charge conjugation can only multiply the field by an intrinsic sign.

Parity. Parity maps p to p^P and x to x_P . For a scalar particle define

$$\mathbf{P}a^\dagger(p)\mathbf{P}^{-1} = \eta_P^{\phi*} a^\dagger(p^P), \quad \mathbf{P}b^\dagger(p)\mathbf{P}^{-1} = \eta_P^\phi b^\dagger(p^P), \quad (389)$$

$$\mathbf{P}a(p)\mathbf{P}^{-1} = \eta_P^\phi a(p^P), \quad \mathbf{P}b(p)\mathbf{P}^{-1} = \eta_P^{\phi*} b(p^P). \quad (390)$$

Then

$$\mathbf{P}\phi(x)\mathbf{P}^{-1} = \eta_P^\phi \int d\Pi_p \left[a(p^P) e^{-ip \cdot x} + b^\dagger(p^P) e^{+ip \cdot x} \right]. \quad (391)$$

With $q = p^P$, we have $d\Pi_q = d\Pi_p$ and $p \cdot x = q \cdot x_P$. Therefore

$$\boxed{\mathbf{P}\phi(t, \mathbf{x})\mathbf{P}^{-1} = \eta_P^\phi \phi(t, -\mathbf{x})}. \quad (392)$$

The sign $\eta_P^\phi = +1$ gives an ordinary scalar under parity; $\eta_P^\phi = -1$ gives a pseudoscalar.

Time reversal. Time reversal is antiunitary. Therefore it complex-conjugates every c-number appearing in the mode expansion:

$$\mathbf{T}i\mathbf{T}^{-1} = -i, \quad \mathbf{T}c\mathbf{T}^{-1} = c^*, \quad c \in \mathbb{C}. \quad (393)$$

Choose phases by

$$\mathbf{T}a^\dagger(p)\mathbf{T}^{-1} = \eta_T^{\phi*} a^\dagger(p^T), \quad \mathbf{T}b^\dagger(p)\mathbf{T}^{-1} = \eta_T^\phi b^\dagger(p^T), \quad (394)$$

$$\mathbf{T}a(p)\mathbf{T}^{-1} = \eta_T^\phi a(p^T), \quad \mathbf{T}b(p)\mathbf{T}^{-1} = \eta_T^{\phi*} b(p^T). \quad (395)$$

Since $\mathbf{T}$ is antiunitary, $e^{-ip \cdot x}$ first becomes $e^{+ip \cdot x}$. Changing variables $q = p^T$ then gives $p \cdot x = -q \cdot x_T$. Hence

$$\mathbf{T}\phi(t, \mathbf{x})\mathbf{T}^{-1} = \eta_T^\phi \int d\Pi_q \left[a(q) e^{-iq \cdot x_T} + b^\dagger(q) e^{+iq \cdot x_T} \right], \quad (396)$$

so

$$\boxed{\mathbf{T}\phi(t, \mathbf{x})\mathbf{T}^{-1} = \eta_T^\phi \phi(-t, \mathbf{x})}, \quad (397)$$

with antiunitarity understood. This scalar example displays the basic mechanism: the local spacetime argument of the transformed field is not assumed; it is obtained by applying the symmetry to the oscillators and then changing the momentum variable.

Dirac particles and the Dirac field: the locality problem

We now repeat the same construction for the Dirac field,

$$\psi(x) = \sum_{\sigma} \int d\Pi_p \left[u(p, \sigma) a(p, \sigma) e^{-ip \cdot x} + v(p, \sigma) b^{\dagger}(p, \sigma) e^{+ip \cdot x} \right], \quad d\Pi_p := \frac{d^3 \mathbf{p}}{(2\pi)^3 2p^0}. \quad (398)$$

The oscillators act on the Hilbert space, while the column spinors u and v are c-number intertwiners carrying the finite-dimensional Dirac spinor index. The transformed field will be local only if, after the oscillator transformation and the momentum-variable change, the transformed u and v spinors can be written as one constant spinor matrix multiplying the original field at the transformed point.

This point is worth emphasizing. The discrete symmetry first acts on particles. The field transformation is then derived. The matrices multiplying the field are not guessed. They are precisely the finite-dimensional spinor intertwiners that convert the transformed momentum-space spinors back into the standard u and v spinor basis.

Let

$$|p, \sigma; e^{-}\rangle := a^{\dagger}(p, \sigma)|0\rangle, \quad |p, \sigma; e^{+}\rangle := b^{\dagger}(p, \sigma)|0\rangle. \quad (399)$$

Charge conjugation exchanges these two sectors. Parity sends p to p^P and leaves the spin label fixed in the standard rest-frame spin basis. Time reversal sends p to p^T and reverses spin through the invariant antisymmetric tensor

$$\epsilon := i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (400)$$

For $\mathbf{T}$ one must also remember that the operator is antiunitary, so the c-number spinors and plane waves are complex-conjugated at the intermediate stage.

Charge conjugation: why $C\bar{\psi}^T$ appears

Charge conjugation is unitary and does not change x , p , or the spin label. It only exchanges particle and antiparticle operators. We choose phases by

$$Ca^{\dagger}(p, \sigma)C^{-1} = \eta_C^* b^{\dagger}(p, \sigma), \quad Cb^{\dagger}(p, \sigma)C^{-1} = \eta_C a^{\dagger}(p, \sigma), \quad (401)$$

$$Ca(p, \sigma)C^{-1} = \eta_C b(p, \sigma), \quad Cb(p, \sigma)C^{-1} = \eta_C^* a(p, \sigma). \quad (402)$$

Then Eq. (398) gives

$$C\psi(x)C^{-1} = \eta_C \sum_{\sigma} \int d\Pi_p \left[u(p, \sigma) b(p, \sigma) e^{-ip \cdot x} + v(p, \sigma) a^{\dagger}(p, \sigma) e^{+ip \cdot x} \right]. \quad (403)$$

The question is whether the right-hand side can be written as a local expression made out of the original field. The answer is yes, but only because the Dirac representation has a charge-conjugation intertwiner.

As explained in Section 5.3, the charge-conjugation matrix C is defined by

$$C^{-1}\gamma^{\mu}C = -(\gamma^{\mu})^T. \quad (404)$$

This is not an arbitrary identity. The matrices $-\gamma^{\mu T}$ obey the same Clifford algebra as the matrices γ^{μ} , so irreducibility of the complex Clifford representation implies that the two sets are related by a similarity transformation. That similarity transformation is C .

The local charge-conjugate spinor is

$$\psi^c(x) := C\bar{\psi}(x)^T = C(\gamma^0)^T \psi(x)^*. \quad (405)$$

Here the star denotes componentwise Hermitian conjugation of the operator-valued column. This expression is local: it uses the field operators at the same spacetime point x . The matrix $C(\gamma^0)^T$ is precisely the antilinear spinor-space map that turns a Dirac spinor into another Dirac spinor with conjugate internal charge.

Let us check the mode expansion explicitly. From Eq. (398),

$$C\bar{\psi}(x)^T = \sum_{\sigma} \int d\Pi_p \left[C\bar{u}(p, \sigma)^T a^{\dagger}(p, \sigma) e^{+ip \cdot x} + C\bar{v}(p, \sigma)^T b(p, \sigma) e^{-ip \cdot x} \right]. \quad (406)$$

The spinor identities are

$$C\bar{u}(p, \sigma)^T = v(p, \sigma), \quad C\bar{v}(p, \sigma)^T = u(p, \sigma), \quad (407)$$

up to the same phase convention absorbed into η_C . They follow from Eq. (404): charge conjugation maps a positive-frequency Dirac spinor into the corresponding antiparticle spinor, and maps the antiparticle spinor back into the particle spinor. Substituting Eq. (407) gives

$$\psi^c(x) = \sum_{\sigma} \int d\Pi_p \left[u(p, \sigma) b(p, \sigma) e^{-ip \cdot x} + v(p, \sigma) a^{\dagger}(p, \sigma) e^{+ip \cdot x} \right]. \quad (408)$$

Comparing with Eq. (403), we obtain

$$\boxed{C\psi(x)C^{-1} = \eta_C \psi^c(x) = \eta_C C\bar{\psi}(x)^T.} \quad (409)$$

Thus the appearance of $C\bar{\psi}^T$ is forced by locality and by the Clifford-intertwiner relation. It is the unique local spinor expression whose mode expansion has $a \leftrightarrow b$.

Parity: why the matrix is γ^0

Parity is unitary. Its natural action on a positive-energy momentum label is $p \mapsto p^P = (p^0, -\mathbf{p})$, and its natural action on the spacetime argument is $x \mapsto x_P = (t, -\mathbf{x})$. Since spin is an axial vector, the rest-frame spin label is not reversed by parity.

We now ask for a local field law of the form

$$\mathbf{P}\psi(x)\mathbf{P}^{-1} = \eta_P B_P \psi(x_P), \quad (410)$$

with a constant matrix B_P acting only on the spinor index. To determine B_P , use the Dirac equation. If $u(p, \sigma)$ satisfies

$$(\gamma^{\mu} p_{\mu} - m)u(p, \sigma) = 0, \quad (411)$$

then $B_P u(p, \sigma)$ should be a solution with momentum p^P :

$$(\gamma^{\mu} p_{\mu}^P - m)B_P u(p, \sigma) = 0. \quad (412)$$

This is guaranteed if

$$B_P^{-1} \gamma^{\mu} B_P = P^{\mu}_{\nu} \gamma^{\nu}, \quad P^{\mu}_{\nu} = \text{diag}(1, -1, -1, -1). \quad (413)$$

Equivalently,

$$B_P^{-1} \gamma^0 B_P = \gamma^0, \quad B_P^{-1} \gamma^i B_P = -\gamma^i. \quad (414)$$

The Clifford algebra gives the solution immediately:

$$B_P = \gamma^0 \quad (415)$$

up to an overall phase. Indeed, γ^0 commutes with γ^0 and anticommutes with every spatial γ^i .

The corresponding spinor identities are

$$u(p^P, \sigma) = B_P u(p, \sigma), \quad v(p^P, \sigma) = -B_P v(p, \sigma). \quad (416)$$

The minus sign in the v identity is the same relative parity sign explained at the end of Section 4.4. It is not an arbitrary decoration: without it the transformed particle and antiparticle parts of the field would not combine into one local Dirac field obeying the same first-order equation.

Now define the action on oscillators by

$$P a^\dagger(p, \sigma) P^{-1} = \eta_P^* a^\dagger(p^P, \sigma), \quad P b^\dagger(p, \sigma) P^{-1} = -\eta_P b^\dagger(p^P, \sigma), \quad (417)$$

$$P a(p, \sigma) P^{-1} = \eta_P a(p^P, \sigma), \quad P b(p, \sigma) P^{-1} = -\eta_P^* b(p^P, \sigma). \quad (418)$$

The relative minus sign says that the antiparticle has opposite intrinsic parity relative to the particle in the standard local Dirac field.

Apply these rules to the field expansion and then change variables $q = p^P$:

$$P \psi(x) P^{-1} = \eta_P \sum_\sigma \int d\Pi_p \left[u(p, \sigma) a(p^P, \sigma) e^{-ip \cdot x} - v(p, \sigma) b^\dagger(p^P, \sigma) e^{+ip \cdot x} \right] \quad (419)$$

$$= \eta_P \sum_\sigma \int d\Pi_q \left[u(q^P, \sigma) a(q, \sigma) e^{-iq \cdot x_P} - v(q^P, \sigma) b^\dagger(q, \sigma) e^{+iq \cdot x_P} \right] \quad (420)$$

$$= \eta_P B_P \sum_\sigma \int d\Pi_q \left[u(q, \sigma) a(q, \sigma) e^{-iq \cdot x_P} + v(q, \sigma) b^\dagger(q, \sigma) e^{+iq \cdot x_P} \right]. \quad (421)$$

Therefore

$$\boxed{P \psi(t, \mathbf{x}) P^{-1} = \eta_P B_P \psi(t, -\mathbf{x}), \quad B_P = \gamma^0.} \quad (422)$$

In the chiral basis, γ^0 exchanges the two Weyl components,

$$\gamma_\chi^0 = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}, \quad (423)$$

so parity relates the two inequivalent Weyl representations:

$$P \begin{pmatrix} \psi_L(t, \mathbf{x}) \\ \psi_R(t, \mathbf{x}) \end{pmatrix} P^{-1} = \eta_P \begin{pmatrix} \psi_R(t, -\mathbf{x}) \\ \psi_L(t, -\mathbf{x}) \end{pmatrix}. \quad (424)$$

This is why a single isolated Weyl field cannot be parity invariant by itself.

Time reversal: why an antiunitary spinor matrix is needed

Time reversal is subtler than parity because it is antiunitary. It reverses the three-momentum and spin of a one-particle state, and it also complex-conjugates the c-number wave functions in the mode expansion. We want a local field law of the form

$$T \psi(t, \mathbf{x}) T^{-1} = \eta_T B_T \psi(-t, \mathbf{x}), \quad (425)$$

where antiunitarity of T is understood. The matrix B_T is determined by the requirement that the transformed spinor again obey the Dirac equation.

If ψ solves $(i\gamma^\mu \partial_\mu - m)\psi = 0$, then the time-reversed c-number spinor is built from $\psi^*(x_T)$, not from $\psi(x_T)$ alone. Requiring $B_T \psi^*(x_T)$ to solve the same Dirac equation gives the gamma-matrix intertwining condition

$$B_T^{-1} \gamma^{\mu*} B_T = P^\mu{}_\nu \gamma^\nu, \quad P^\mu{}_\nu = \text{diag}(1, -1, -1, -1). \quad (426)$$

Equivalently,

$$B_T^{-1} \gamma^{0*} B_T = \gamma^0, \quad B_T^{-1} \gamma^{i*} B_T = -\gamma^i. \quad (427)$$

This is the antiunitary analogue of the parity identity. In a standard four-dimensional representation one may take

$$B_T = i\gamma^1\gamma^3, \quad (428)$$

up to an overall phase. In another gamma-matrix basis this matrix is transformed by the same similarity transformation as the gamma matrices.

The spin- $\frac{1}{2}$ label is transformed by the invariant antisymmetric tensor $\epsilon = i\sigma^2$. We choose phases by

$$\mathbb{T}a^\dagger(p, \sigma)\mathbb{T}^{-1} = \eta_T^* \sum_{\sigma'} a^\dagger(p^T, \sigma')\epsilon_{\sigma'\sigma}, \quad \mathbb{T}b^\dagger(p, \sigma)\mathbb{T}^{-1} = \eta_T \sum_{\sigma'} b^\dagger(p^T, \sigma')\epsilon_{\sigma'\sigma}, \quad (429)$$

$$\mathbb{T}a(p, \sigma)\mathbb{T}^{-1} = \eta_T \sum_{\sigma'} \epsilon_{\sigma'\sigma}^* a(p^T, \sigma'), \quad \mathbb{T}b(p, \sigma)\mathbb{T}^{-1} = \eta_T^* \sum_{\sigma'} \epsilon_{\sigma'\sigma}^* b(p^T, \sigma'). \quad (430)$$

The precise placement of complex conjugates is a phase convention tied to the basis of spin states; the invariant content is that time reversal reverses spin and is antiunitary.

The required spinor identities are

$$\sum_{\sigma'} u(p^T, \sigma')\epsilon_{\sigma'\sigma} = B_T u(p, \sigma)^*, \quad \sum_{\sigma'} v(p^T, \sigma')\epsilon_{\sigma'\sigma} = B_T v(p, \sigma)^*. \quad (431)$$

They follow from Eq. (426): the right-hand sides solve the Dirac equations with momentum p^T and transform with the same spin- $\frac{1}{2}$ little-group matrix as the left-hand sides. The remaining proportionality is fixed by the rest-frame normalization and can be absorbed into η_T .

Now apply $\mathbb{T}$ to the field expansion. Because $\mathbb{T}$ is antiunitary, u and v are complex-conjugated and $e^{-ip \cdot x}$ becomes $e^{+ip \cdot x}$. After the change of variables $q = p^T$, the phase becomes the phase at $x_T = (-t, \mathbf{x})$. Using Eq. (431), the transformed field becomes one matrix times the original local field at x_T :

$$\boxed{\mathbb{T}\psi(t, \mathbf{x})\mathbb{T}^{-1} = \eta_T B_T \psi(-t, \mathbf{x}), \quad B_T = i\gamma^1\gamma^3,} \quad (432)$$

with antiunitarity understood. The compact formula hides an important intermediate step: the c-number spinors in the mode expansion have been complex-conjugated by $\mathbb{T}$, and B_T is exactly the matrix which converts those complex-conjugated spinors into the standard spinor basis at reversed momentum.

Summary of the logic

The three transformations can now be summarized in the local form

$$C\psi(x)C^{-1} = \eta_C C\bar{\psi}(x)^T, \quad (433)$$

$$P\psi(t, \mathbf{x})P^{-1} = \eta_P B_P \psi(t, -\mathbf{x}), \quad B_P^{-1}\gamma^\mu B_P = P^\mu{}_\nu \gamma^\nu, \quad (434)$$

$$\mathbb{T}\psi(t, \mathbf{x})\mathbb{T}^{-1} = \eta_T B_T \psi(-t, \mathbf{x}), \quad B_T^{-1}\gamma^{\mu*} B_T = P^\mu{}_\nu \gamma^\nu. \quad (435)$$

The matrices C , B_P , and B_T are therefore not arbitrary additions to the theory. They are the spinor intertwiners which make the one-particle transformations compatible with a local Dirac field. Charge conjugation requires $C\bar{\psi}^T$ because only this local spinor has the mode expansion with $a \leftrightarrow b$. Parity requires $B_P = \gamma^0$ because γ^0 implements $\gamma^0 \mapsto \gamma^0$ and $\gamma^i \mapsto -\gamma^i$. Time reversal requires B_T because antiunitarity complex-conjugates the gamma matrices and the spinor wave functions. These local relations are the input for the CPT discussion below.

5.5 CPT theorem for local Lagrangians built from spin zero and spin one-half fields

The previous subsection did the hard work. It did not merely tell us how the one-particle states transform under C , P , and $\mathbb{T}$. It showed something stronger: the corresponding Hilbert-space

transformations induce *local* transformations of the fields. This is the input needed for the CPT theorem. Once the local transformation of the elementary fields is known, the transformation of every local composite operator follows by ordinary algebra, the chain rule for derivatives, and the gamma-matrix identities derived above.

Let

$$\Theta := \text{CPT}. \quad (436)$$

The spacetime part of Θ is the total inversion

$$I^\mu{}_\nu := -\delta^\mu{}_\nu, \quad x^\mu \mapsto (Ix)^\mu = -x^\mu. \quad (437)$$

In four dimensions

$$\det I = (-1)^4 = +1. \quad (438)$$

This is why pseudo-scalars, pseudo-vectors and pseudo-tensors do not acquire an additional orientation sign under CPT. They do acquire such a sign under parity alone or time reversal alone, because each of those transformations has determinant -1 . But the total inversion has determinant $+1$.

The operator Θ is antiunitary, because it contains T . Therefore

$$\Theta i \Theta^{-1} = -i, \quad \Theta c \Theta^{-1} = c^* \quad (439)$$

for every complex number c . This complex conjugation of coefficients is an essential part of the CPT theorem.

Elementary fields and derivatives

For a complex scalar field the local transformations obtained in Section 5.4 combine to give

$$\Theta \phi_i(x) \Theta^{-1} = \eta_i \phi_i^\dagger(-x), \quad \Theta \phi_i^\dagger(x) \Theta^{-1} = \eta_i^* \phi_i(-x), \quad (440)$$

where η_i is a conventional phase. Derivatives are fixed by the chain rule:

$$\partial_\mu F(-x) = -(\partial_\mu F)(-x). \quad (441)$$

Thus

$$\Theta \partial_{\mu_1} \cdots \partial_{\mu_n} \phi_i(x) \Theta^{-1} = (-1)^n \eta_i \left(\partial_{\mu_1} \cdots \partial_{\mu_n} \phi_i^\dagger \right)(-x), \quad (442)$$

with the analogous formula for $\phi_i^\dagger$. The sign $(-1)^n$ is not a special property of scalar fields; it is the covariant-index factor associated with total inversion.

For a Dirac field, Section 5.4 gives the local CPT law in the form

$$\Theta \psi_A(x) \Theta^{-1} = \eta_A \gamma^5 \psi_A^c(-x) = \eta_A \gamma^5 C \bar{\psi}_A(-x)^T. \quad (443)$$

The charge-conjugate field

$$\psi^c := C \bar{\psi}^T \quad (444)$$

was introduced earlier precisely because it transforms as an ordinary Dirac spinor. The reason is the charge-conjugation identity

$$C^{-1} \gamma^\mu C = -(\gamma^\mu)^T. \quad (445)$$

It converts the transpose, or contragredient, spinor representation back into the original Dirac spinor representation. The remaining matrix γ^5 is the spinor matrix associated with the total inversion. Indeed

$$\gamma^5 \gamma^\mu \gamma^5 = -\gamma^\mu = I^\mu{}_\nu \gamma^\nu. \quad (446)$$

Thus Eq. (443) is not a guess. It is the local field relation forced by the particle-antiparticle exchange of $\mathbf{C}$, the spacetime inversion of $\mathbf{PT}$, and the gamma-matrix intertwiners found in Section 5.4.

The adjoint transforms correspondingly as

$$\Theta \bar{\psi}_A(x) \Theta^{-1} = -\eta_A^* \psi_A(-x)^T C^{-1} \gamma^5. \quad (447)$$

The overall sign follows from the standard identity $\bar{\psi}^c = -\psi^T C^{-1}$ and from moving γ^5 through the Dirac adjoint. This sign is important: it is what makes the scalar bilinear $\bar{\psi}\psi$ CPT even.

Derivatives of spinor fields must also be included. Applying the chain rule to Eqs. (443) and (447) gives

$$\Theta \partial_{\mu_1} \cdots \partial_{\mu_n} \psi_A(x) \Theta^{-1} = (-1)^n \eta_A \gamma^5 C (\partial_{\mu_1} \cdots \partial_{\mu_n} \bar{\psi}_A) (-x)^T, \quad (448)$$

while

$$\Theta \partial_{\mu_1} \cdots \partial_{\mu_n} \bar{\psi}_A(x) \Theta^{-1} = (-1)^n (-\eta_A^*) (\partial_{\mu_1} \cdots \partial_{\mu_n} \psi_A) (-x)^T C^{-1} \gamma^5. \quad (449)$$

These formulas are needed for the Dirac kinetic term and for all derivative interactions.

The universal local tensor rule

We now state the main structural fact. Let $T_{\mu_1 \dots \mu_n}(x)$ be a local Lorentz tensor operator with n covariant vector indices. It may be built in any way from the elementary scalar fields ϕ_i , the Dirac fields ψ_A , their adjoints, and finitely many derivatives, provided it is local and its spinor indices are contracted. If it is a neutral Hermitian tensor, then CPT acts as

$$\boxed{\Theta T_{\mu_1 \dots \mu_n}(x) \Theta^{-1} = (-1)^n T_{\mu_1 \dots \mu_n}(-x).} \quad (450)$$

More generally, if the tensor carries an internal charge, the right-hand side is the corresponding charge-conjugate tensor at $-x$; in a Hermitian local Lagrangian such charged terms always occur together with their Hermitian conjugates. For the neutral tensors entering the real Lagrangian density, Eq. (450) is the form we need.

This is the central point. The sign in Eq. (450) knows only the number of free Lorentz indices. It does *not* remember whether the tensor was made from scalar derivatives, spinor bilinears, tensor bilinears, pseudotensors, or products of all of these. Locality and the field transformations of Section 5.4 are what make this possible.

Let us prove it. The elementary scalar field has no Lorentz index and is mapped to its conjugate at $-x$. Each derivative ∂_μ contributes one covariant index and, by the chain rule, one factor of -1 . Therefore scalar composites satisfy the rule immediately. For example,

$$\Theta (\partial_{\mu_1} \cdots \partial_{\mu_n} \phi_i(x)) \Theta^{-1} = (-1)^n (\partial_{\mu_1} \cdots \partial_{\mu_n} \phi_i^\dagger)(-x). \quad (451)$$

For spinor composites the free Lorentz indices arise from gamma matrices, from $\sigma^{\mu\nu}$, from $\epsilon_{\mu\nu\rho\sigma}$, and from derivatives. Equations (445) and (446) imply that a vector gamma matrix transforms as a vector under total inversion:

$$\gamma^\mu \longmapsto I^\mu{}_\nu \gamma^\nu = -\gamma^\mu. \quad (452)$$

Consequently an antisymmetric product transforms with one factor of I for each index:

$$\sigma^{\mu\nu} \longmapsto I^\mu{}_\rho I^\nu{}_\sigma \sigma^{\rho\sigma}. \quad (453)$$

The orientation tensor satisfies

$$I^\alpha{}_\mu I^\beta{}_\nu I^\gamma{}_\rho I^\delta{}_\sigma \epsilon_{\alpha\beta\gamma\delta} = (\det I) \epsilon_{\mu\nu\rho\sigma} = \epsilon_{\mu\nu\rho\sigma}. \quad (454)$$

Therefore pseudo-tensors obey the same CPT sign rule as ordinary tensors of the same rank. Derivatives of ψ and $\bar{\psi}$ contribute additional factors of I by Eqs. (448) and (449). Products of local tensors transform by multiplying these factors. Finally, if two indices are contracted, the corresponding signs cancel because

$$I^\rho{}_\mu I^\sigma{}_\nu \eta_{\rho\sigma} = \eta_{\mu\nu}, \quad I^\alpha{}_\mu I^\beta{}_\nu I^\gamma{}_\rho I^\delta{}_\sigma \epsilon_{\alpha\beta\gamma\delta} = \epsilon_{\mu\nu\rho\sigma}. \quad (455)$$

Thus only the uncontracted free indices survive, and each of them contributes one factor of -1 . This proves Eq. (450).

Six explicit Dirac bilinears from the field transformations

We now illustrate the theorem directly, without using the tensor rule as a shortcut. Define

$$S := \bar{\psi}\psi, \quad P := i\bar{\psi}\gamma^5\psi, \quad (456)$$

$$V^\mu := \bar{\psi}\gamma^\mu\psi, \quad A^\mu := \bar{\psi}\gamma^\mu\gamma^5\psi, \quad (457)$$

$$T^{\mu\nu} := \bar{\psi}\sigma^{\mu\nu}\psi, \quad \tilde{T}^{\mu\nu} := \bar{\psi}\sigma^{\mu\nu}\gamma^5\psi. \quad (458)$$

Here

$$\sigma^{\mu\nu} := \frac{i}{2}[\gamma^\mu, \gamma^\nu], \quad \gamma^5 := \frac{i}{4!}\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma. \quad (459)$$

We repeatedly use the fermion transpose identity

$$\psi^T M \bar{\psi}^T = -\bar{\psi} M^T \psi, \quad (460)$$

which is simply the minus sign from exchanging two fermionic fields.

For the scalar,

$$\Theta S(x) \Theta^{-1} = (\Theta \bar{\psi} \Theta^{-1})(\Theta \psi \Theta^{-1}) \quad (461)$$

$$= -\psi(-x)^T C^{-1} \gamma^5 \gamma^5 C \bar{\psi}(-x)^T \quad (462)$$

$$= -\psi(-x)^T \bar{\psi}(-x)^T \quad (463)$$

$$= \bar{\psi}(-x) \psi(-x) = S(-x). \quad (464)$$

The first minus sign comes from the adjoint transformation; the last minus sign comes from putting the fermions back in the standard order.

For the Hermitian pseudoscalar, antiunitarity first gives $i \mapsto -i$:

$$\Theta P(x) \Theta^{-1} = (-i)(\Theta \bar{\psi} \Theta^{-1}) \gamma^5 (\Theta \psi \Theta^{-1}) \quad (465)$$

$$= i \psi(-x)^T C^{-1} \gamma^5 \gamma^5 \gamma^5 C \bar{\psi}(-x)^T. \quad (466)$$

Using $C^{-1} \gamma^5 C = (\gamma^5)^T$ and then Eq. (460),

$$\Theta P(x) \Theta^{-1} = i \psi(-x)^T (\gamma^5)^T \bar{\psi}(-x)^T \quad (467)$$

$$= i \bar{\psi}(-x) \gamma^5 \psi(-x) = P(-x). \quad (468)$$

Thus the sign from antiunitarity is exactly compensated by the charge-conjugation transpose of γ^5 .

For the vector,

$$\Theta V^\mu(x) \Theta^{-1} = -\psi(-x)^T C^{-1} \gamma^5 \gamma^\mu \gamma^5 C \bar{\psi}(-x)^T \quad (469)$$

$$= +\psi(-x)^T C^{-1} \gamma^\mu C \bar{\psi}(-x)^T \quad (470)$$

$$= -\psi(-x)^T (\gamma^\mu)^T \bar{\psi}(-x)^T \quad (471)$$

$$= -\bar{\psi}(-x) \gamma^\mu \psi(-x) \quad (472)$$

$$= I^\mu{}_\nu V^\nu(-x). \quad (473)$$

In the second line we used $\gamma^5\gamma^\mu\gamma^5 = -\gamma^\mu$, and in the third line we used $C^{-1}\gamma^\mu C = -(\gamma^\mu)^T$. Since $I^\mu{}_\nu = -\delta^\mu{}_\nu$, the vector is odd under CPT.

For the pseudovector,

$$\Theta A^\mu(x)\Theta^{-1} = -\psi(-x)^T C^{-1}\gamma^5\gamma^\mu\gamma^5 C\bar{\psi}(-x)^T \quad (474)$$

$$= +\psi(-x)^T C^{-1}\gamma^\mu\gamma^5 C\bar{\psi}(-x)^T \quad (475)$$

$$= -\psi(-x)^T (\gamma^\mu\gamma^5)^T \bar{\psi}(-x)^T \quad (476)$$

$$= -\bar{\psi}(-x)\gamma^\mu\gamma^5\psi(-x) \quad (477)$$

$$= I^\mu{}_\nu A^\nu(-x). \quad (478)$$

There is no additional pseudoscalar orientation sign, because the spacetime transformation is I and $\det I = +1$.

For the tensor, remember that $\sigma^{\mu\nu}$ itself contains an explicit factor of i , so antiunitarity contributes a sign. One finds

$$\Theta T^{\mu\nu}(x)\Theta^{-1} = -\psi(-x)^T C^{-1}\gamma^5\sigma^{\mu\nu*}\gamma^5 C\bar{\psi}(-x)^T \quad (479)$$

$$= -I^\mu{}_\rho I^\nu{}_\sigma \psi(-x)^T (\sigma^{\rho\sigma})^T \bar{\psi}(-x)^T \quad (480)$$

$$= I^\mu{}_\rho I^\nu{}_\sigma \bar{\psi}(-x)\sigma^{\rho\sigma}\psi(-x) \quad (481)$$

$$= I^\mu{}_\rho I^\nu{}_\sigma T^{\rho\sigma}(-x). \quad (482)$$

Since $I^\mu{}_\rho I^\nu{}_\sigma = \delta^\mu{}_\rho \delta^\nu{}_\sigma$, a rank-two tensor is CPT even.

For the pseudotensor,

$$\Theta \tilde{T}^{\mu\nu}(x)\Theta^{-1} = -\psi(-x)^T C^{-1}\gamma^5(\sigma^{\mu\nu}\gamma^5)^*\gamma^5 C\bar{\psi}(-x)^T \quad (483)$$

$$= I^\mu{}_\rho I^\nu{}_\sigma \bar{\psi}(-x)\sigma^{\rho\sigma}\gamma^5\psi(-x) \quad (484)$$

$$= I^\mu{}_\rho I^\nu{}_\sigma \tilde{T}^{\rho\sigma}(-x). \quad (485)$$

Equivalently, use

$$\sigma^{\mu\nu}\gamma^5 \propto \epsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma}. \quad (486)$$

The two tensor indices give two factors of I , and the Levi-Civita tensor gives $\det I = +1$. Hence the pseudotensor has the same CPT sign as an ordinary rank-two tensor.

The six examples are therefore summarized by

operator	rank	CPT transformation
$S = \bar{\psi}\psi$	0	$S(x) \mapsto S(-x)$
$P = i\bar{\psi}\gamma^5\psi$	0	$P(x) \mapsto P(-x)$
$V^\mu = \bar{\psi}\gamma^\mu\psi$	1	$V^\mu(x) \mapsto -V^\mu(-x)$
$A^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$	1	$A^\mu(x) \mapsto -A^\mu(-x)$
$T^{\mu\nu} = \bar{\psi}\sigma^{\mu\nu}\psi$	2	$T^{\mu\nu}(x) \mapsto T^{\mu\nu}(-x)$
$\tilde{T}^{\mu\nu} = \bar{\psi}\sigma^{\mu\nu}\gamma^5\psi$	2	$\tilde{T}^{\mu\nu}(x) \mapsto \tilde{T}^{\mu\nu}(-x)$

(487)

This is exactly Eq. (450): rank zero is even, rank one is odd, rank two is even. The pseudo-character does not alter the CPT sign because $\det I = +1$.

Scalars, scalar derivatives, and spinor derivatives in interactions

The same tensor rule immediately controls interactions involving scalar fields and derivatives. For a scalar kinetic term,

$$\Theta \left(\partial_\mu \phi^\dagger \partial^\mu \phi \right) (x) \Theta^{-1} = (-1)(-1) \left(\partial_\mu \phi \partial^\mu \phi^\dagger \right) (-x) \quad (488)$$

$$= \left(\partial_\mu \phi^\dagger \partial^\mu \phi \right) (-x). \quad (489)$$

The two minus signs come from the two derivatives. The Lorentz index is contracted, so the signs cancel.

For a derivative coupling to a vector bilinear,

$$\Theta [(\partial_\mu \varphi) V^\mu] (x) \Theta^{-1} = (-\partial_\mu \varphi)(-V^\mu)(-x) \quad (490)$$

$$= (\partial_\mu \varphi V^\mu)(-x). \quad (491)$$

For a pseudoscalar a , the coupling $(\partial_\mu a) A^\mu$ is treated in the same way. The derivative supplies one factor of -1 , and the pseudovector supplies one factor of -1 ; the contraction is CPT even.

Now include derivatives of spinor fields. The Hermitian Dirac kinetic term is

$$\mathcal{L}_{\text{kin}} = \frac{i}{2} \bar{\psi} \gamma^\mu \overleftrightarrow{\partial}_\mu \psi = \frac{i}{2} \bar{\psi} \gamma^\mu \partial_\mu \psi - \frac{i}{2} (\partial_\mu \bar{\psi}) \gamma^\mu \psi. \quad (492)$$

Using Eqs. (448) and (449), the two unsymmetrized pieces are exchanged:

$$\Theta (i \bar{\psi} \gamma^\mu \partial_\mu \psi) (x) \Theta^{-1} = -i (\partial_\mu \bar{\psi}) \gamma^\mu \psi(-x), \quad (493)$$

$$\Theta [-i (\partial_\mu \bar{\psi}) \gamma^\mu \psi] (x) \Theta^{-1} = +i \bar{\psi} \gamma^\mu \partial_\mu \psi(-x). \quad (494)$$

Therefore

$$\Theta \mathcal{L}_{\text{kin}}(x) \Theta^{-1} = \mathcal{L}_{\text{kin}}(-x). \quad (495)$$

This example is the best way to remember why derivatives of ψ and $\bar{\psi}$ must be included explicitly. The first half of the Hermitian kinetic term does not transform into itself; it transforms into the second half. The local Hermitian combination transforms cleanly.

More generally, a local spinor expression of the form

$$(\partial_{\rho_1} \cdots \partial_{\rho_r} \bar{\psi}) \Gamma_{\mu_1 \cdots \mu_s} (\partial_{\lambda_1} \cdots \partial_{\lambda_t} \psi) \quad (496)$$

transforms with one factor of I for every derivative index and one factor of I for every free Lorentz index carried by Γ . Once all indices are contracted to form a scalar, the factors of I cancel pairwise. If the operator is not Hermitian by itself, CPT maps it to the Hermitian-conjugate operator at $-x$, while antiunitarity complex-conjugates its coefficient.

Complete CPT statement and proof for local Lagrangians

We can now state the theorem in the form used in local relativistic quantum field theory. Let $\mathcal{L}(x)$ be a Hermitian local polynomial built from complex scalar fields ϕ_i , Dirac fields ψ_A , their adjoints, and finitely many spacetime derivatives. Assume that all Lorentz indices are contracted with invariant tensors $\eta_{\mu\nu}$ and $\epsilon_{\mu\nu\rho\sigma}$, so that $\mathcal{L}$ is a Lorentz scalar density. Then

$$\boxed{\Theta \mathcal{L}(x) \Theta^{-1} = \mathcal{L}(-x).} \quad (497)$$

Consequently

$$\Theta \int d^4x \mathcal{L}(x) \Theta^{-1} = \int d^4x \mathcal{L}(-x) = \int d^4x \mathcal{L}(x), \quad (498)$$

where the last equality is the change of variables $x \mapsto -x$.

The proof is now short. Every local term in $\mathcal{L}$ can be decomposed into local Lorentz tensors of the type discussed above, with all free indices contracted. By Eq. (450), a rank- n tensor contributes $(-1)^n$. Each contracted pair of indices removes two such signs. Contractions with $\eta_{\mu\nu}$ are invariant, and contractions with $\epsilon_{\mu\nu\rho\sigma}$ are also invariant because $\det I = +1$. Thus every fully contracted Lorentz scalar operator is mapped to the corresponding CPT-conjugate scalar operator at $-x$.

If the scalar operator $\mathcal{O}$ is Hermitian, this means simply

$$\Theta \mathcal{O}(x) \Theta^{-1} = \mathcal{O}(-x). \quad (499)$$

If $\mathcal{O}$ is not Hermitian, locality and Lorentz covariance still imply

$$\Theta \mathcal{O}(x) \Theta^{-1} = \mathcal{O}^\dagger(-x). \quad (500)$$

up to the conventional internal phases already fixed in the definitions of the fields. A Hermitian Lagrangian contains such an operator in the combination

$$c\mathcal{O}(x) + c^*\mathcal{O}^\dagger(x). \quad (501)$$

Using antiunitarity,

$$\Theta \left[c\mathcal{O}(x) + c^*\mathcal{O}^\dagger(x) \right] \Theta^{-1} = c^*\mathcal{O}^\dagger(-x) + c\mathcal{O}(-x) \quad (502)$$

$$= \left[c\mathcal{O} + c^*\mathcal{O}^\dagger \right] (-x). \quad (503)$$

This proves Eq. (497).

The assumptions are exactly the assumptions of ordinary local relativistic quantum field theory: Lorentz covariance, locality, Hermiticity of the Lagrangian, and the spin-statistics connection. The proof does not assume separate invariance under $\mathbf{C}$, $\mathbf{P}$, $\mathbf{T}$, $\mathbf{CP}$, or $\mathbf{T}$. A theory may violate these symmetries separately and still obey $\mathbf{CPT}$.

5.6 Why CP violation requires three generations: the CKM matrix

We now add one physical application of the preceding discussion. The discrete transformation $\mathbf{CP}$ is not automatically a symmetry of every interaction. In the quark sector of the Standard Model it is violated in the charged weak interaction. The purpose of this subsection is not to give a full course on the Standard Model. We only introduce the minimum gauge-theory notation needed to see why the CKM matrix appears and why a physical $\mathbf{CP}$ -violating phase requires three generations.

The electroweak gauge indices

For the purpose of this subsection we recall only the minimal gauge notation needed for the quark charged-current interaction. The electroweak gauge group is

$$\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y. \quad (504)$$

The subscript L means that the $\mathrm{SU}(2)_L$ gauge transformations act on left-handed doublets. For each quark generation $i = 1, \dots, n$, the left-handed up-type and down-type quarks form one $\mathrm{SU}(2)_L$ doublet,

$$Q'_{Li}{}^A = \begin{pmatrix} u'_{Li} \\ d'_{Li} \end{pmatrix}^A, \quad A = 1, 2. \quad (505)$$

The prime denotes the *weak basis*: this is the basis in which the gauge interaction is diagonal in the generation label. The index A is the $\mathrm{SU}(2)_L$ index. Under an $\mathrm{SU}(2)_L$ gauge transformation,

$$Q'_{Li}{}^A(x) \mapsto [U_L(x)]^A{}_B Q'_{Li}{}^B(x), \quad U_L(x) \in \mathrm{SU}(2)_L. \quad (506)$$

The right-handed quarks are $\mathrm{SU}(2)_L$ singlets.

Including color, the weak-basis quark fields transform under

$$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \quad (507)$$

as

$$Q'_{Li} \sim (\mathbf{3}, \mathbf{2})_{1/6}, \quad u'_{Ri} \sim (\mathbf{3}, \mathbf{1})_{2/3}, \quad d'_{Ri} \sim (\mathbf{3}, \mathbf{1})_{-1/3}. \quad (508)$$

Here $(\mathbf{3}, \mathbf{2})_{1/6}$ means: fundamental of $SU(3)_c$, doublet of $SU(2)_L$, and hypercharge $Y = 1/6$.

We use the convention

$$Q_{\text{em}} = T_3 + Y. \quad (509)$$

Therefore the two components of the left-handed doublet have electric charges

$$u_L : \quad Q_{\text{em}} = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}, \quad d_L : \quad Q_{\text{em}} = -\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}. \quad (510)$$

For the right-handed singlets $T_3 = 0$, so their hypercharges are equal to their electric charges:

$$u_R : \quad Y = \frac{2}{3}, \quad d_R : \quad Y = -\frac{1}{3}. \quad (511)$$

Some older conventions write $Q_{\text{em}} = T_3 + Y/2$. In that convention all the hypercharges above are doubled. In these notes we use Eq. (509).

We will not discuss electroweak symmetry breaking here. We only use the standard notation for the gauge bosons that couple to the quark currents:

$$W_\mu^+, \quad W_\mu^-, \quad Z_\mu^0, \quad A_\mu. \quad (512)$$

The quark electroweak interaction

In terms of these fields, the quark electroweak interaction in the weak basis has the schematic form

$$\begin{aligned} \mathcal{L}_{\text{EW}}^q = & \frac{g}{\sqrt{2}} (\bar{u}'_{Li} \gamma^\mu d'_{Li} W_\mu^+ + \bar{d}'_{Li} \gamma^\mu u'_{Li} W_\mu^-) \\ & + e A_\mu \left[\frac{2}{3} \bar{u}'_i \gamma^\mu u'_i - \frac{1}{3} \bar{d}'_i \gamma^\mu d'_i \right] \\ & + \frac{g}{\cos \theta_W} Z_\mu^0 \left[\bar{u}'_i \gamma^\mu (T_3^u P_L - Q_u \sin^2 \theta_W) u'_i + \bar{d}'_i \gamma^\mu (T_3^d P_L - Q_d \sin^2 \theta_W) d'_i \right]. \end{aligned} \quad (513)$$

Here summation over i is understood, $P_L = (1 - \gamma^5)/2$, and

$$T_3^u = +\frac{1}{2}, \quad T_3^d = -\frac{1}{2}, \quad Q_u = +\frac{2}{3}, \quad Q_d = -\frac{1}{3}. \quad (514)$$

The photon and Z^0 terms are neutral currents: they contain $\bar{u}u$ or $\bar{d}d$. The first line contains the charged currents: W^+ couples an up-type quark to a down-type quark, and W^- couples the Hermitian-conjugate current. For the CKM discussion, we isolate only this first line:

$$\mathcal{L}_{\text{cc}}^q = \frac{g}{\sqrt{2}} (\bar{u}'_{Li} \gamma^\mu d'_{Li} W_\mu^+ + \bar{d}'_{Li} \gamma^\mu u'_{Li} W_\mu^-). \quad (515)$$

This equation is the key. The charged current is diagonal in the generation label i in the weak basis because the gauge field couples the two components of the same $SU(2)_L$ doublet Q'_{Li} .

Weak eigenstates and mass eigenstates

The quark mass terms need not be diagonal in the weak basis. Write the most general mass matrices schematically as

$$-\mathcal{L}_{m,q} = \bar{u}'_L M_u u'_R + \bar{d}'_L M_d d'_R + \text{h.c.} \quad (516)$$

Here M_u and M_d are complex matrices in generation space. They are diagonalized by independent unitary rotations on left- and right-handed fields:

$$U_{uL}^\dagger M_u U_{uR} = D_u, \quad U_{dL}^\dagger M_d U_{dR} = D_d, \quad (517)$$

where D_u, D_d are diagonal with positive entries. The mass eigenstates are defined by

$$u'_L = U_{uL} u_L, \quad d'_L = U_{dL} d_L, \quad u'_R = U_{uR} u_R, \quad d'_R = U_{dR} d_R. \quad (518)$$

Substituting only the left-handed rotations into the charged current gives

$$\bar{u}'_{Li} \gamma^\mu d'_{Li} = (\bar{u}_L U_{uL}^\dagger)_i \gamma^\mu (U_{dL} d_L)_i = \bar{u}_{La} \gamma^\mu (U_{uL}^\dagger U_{dL})_{ab} d_{Lb}. \quad (519)$$

Therefore

$$\mathcal{L}_{cc}^q = \frac{g}{\sqrt{2}} (\bar{u}_{La} \gamma^\mu V_{ab} d_{Lb} W_\mu^+ + \bar{d}_{Lb} \gamma^\mu V_{ab}^* u_{La} W_\mu^-), \quad V := U_{uL}^\dagger U_{dL}. \quad (520)$$

The unitary matrix V is the Cabibbo–Kobayashi–Maskawa, or CKM, matrix. It appears because the up-type and down-type mass matrices are not diagonalized by the same left-handed rotation. Neutral currents remain diagonal because the same unitary matrix appears together with its adjoint, for example $U_{uL}^\dagger U_{uL} = \mathbf{1}$.

How the charged current transforms under CP

We now derive the CP statement from Section 5.4. For a Dirac vector current, the combined action of CP gives

$$\text{CP} [\bar{\psi}_1 \gamma^\mu P_L \psi_2] (t, \mathbf{x}) (\text{CP})^{-1} = -P^\mu{}_\nu [\bar{\psi}_2 \gamma^\nu P_L \psi_1] (t, -\mathbf{x}), \quad (521)$$

up to conventional intrinsic phases of the fields, which cancel in the charged-current interaction. Here $P^\mu{}_\nu = \text{diag}(1, -1, -1, -1)$. The reversal of the order $1 \leftrightarrow 2$ comes from charge conjugation: a current made from a particle of type 2 turning into a particle of type 1 is mapped into the conjugate current.

Applying Eq. (521) gives

$$\bar{u}_{La} \gamma^\mu d_{Lb} \xrightarrow{\text{CP}} -P^\mu{}_\nu \bar{d}_{Lb} \gamma^\nu u_{La}. \quad (522)$$

For the vector boson, the same particle-field logic says that charge conjugation exchanges W^+ and W^- , because they create and annihilate particles of opposite electric charge. Parity supplies the vector index transformation. Thus, in a simple convention,

$$\text{CP} W_\mu^+ (t, \mathbf{x}) (\text{CP})^{-1} = -P_\mu{}^\nu W_\nu^- (t, -\mathbf{x}), \quad \text{CP} W_\mu^- (t, \mathbf{x}) (\text{CP})^{-1} = -P_\mu{}^\nu W_\nu^+ (t, -\mathbf{x}). \quad (523)$$

The two minus signs in Eqs. (522) and (523) multiply to +1, while $P^\mu{}_\nu P_\mu{}^\rho$ contracts the vector index correctly. Hence the operator part of the charged current transforms into its Hermitian conjugate:

$$\bar{u}_{La} \gamma^\mu d_{Lb} W_\mu^+ \xrightarrow{\text{CP}} \bar{d}_{Lb} \gamma^\mu u_{La} W_\mu^-. \quad (524)$$

Since CP is unitary, it does not complex-conjugate ordinary numerical coefficients by anti-linearity. The useful way to state the result is that the operator with coefficient V_{ab} is mapped to the Hermitian-conjugate operator, whose coefficient in the Hermitian Lagrangian is V_{ab}^* . Equivalently,

$$\mathcal{L}_{cc}^q[V] \xrightarrow{\text{CP}} \mathcal{L}_{cc}^q[V^*] \quad (525)$$

in the sense that the transformed operator structure is the Hermitian-conjugate charged current. CP invariance therefore requires that, after allowed field rephasings, V can be made real. If a physical complex phase remains, the interaction distinguishes CP from the identity operation on the charged-current sector.

When is CP conserved?

CP is conserved if one can choose phases of the quark fields so that all physical entries of V are real. The subtle point is that many phases in V are not physical. We can rephase the mass eigenstate quarks as

$$u_a \rightarrow e^{i\alpha_a} u_a, \quad d_b \rightarrow e^{i\beta_b} d_b. \quad (526)$$

The CKM matrix then changes by

$$V_{ab} \rightarrow e^{-i\alpha_a} V_{ab} e^{i\beta_b}. \quad (527)$$

Such a change is only a change of convention for the quark fields. It cannot affect physics.

Now count the physical parameters. An $n \times n$ unitary matrix has n^2 real parameters. These may be organized as

$$\frac{n(n-1)}{2} \text{ angles, } \quad \frac{n(n+1)}{2} \text{ phases.} \quad (528)$$

The $2n$ quark fields can be rephased, but one common phase is baryon number and leaves every V_{ab} unchanged. Hence $2n-1$ phases can be removed. The number of physical CP-violating phases is therefore

$$\frac{n(n+1)}{2} - (2n-1) = \frac{(n-1)(n-2)}{2}. \quad (529)$$

For $n=1$ there is no mixing angle and no phase. For $n=2$ there is one mixing angle, the Cabibbo angle, but no physical phase. For $n=3$ there are three mixing angles and one physical phase. This is why CP violation in the quark charged current requires at least three generations.

For two generations the CKM matrix can always be made real:

$$V_{2 \times 2} = \begin{pmatrix} \cos \theta_C & \sin \theta_C \\ -\sin \theta_C & \cos \theta_C \end{pmatrix}. \quad (530)$$

This matrix can violate flavor conservation, but it cannot violate CP. For three generations a standard parameterization is

$$V = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}, \quad (531)$$

where $s_{ij} = \sin \theta_{ij}$ and $c_{ij} = \cos \theta_{ij}$. The phase δ is the irreducible Kobayashi–Maskawa phase. If $\delta = 0$ or π , the matrix can be chosen real and CP is conserved in the quark mixing sector.

A rephasing-invariant test

Because individual phases of CKM entries depend on convention, CP violation should be diagnosed by a rephasing-invariant quantity. The simplest one is the Jarlskog invariant

$$J = \text{Im} (V_{ij} V_{kl} V_{il}^* V_{kj}^*), \quad i \neq k, \quad j \neq l. \quad (532)$$

For three generations, all such choices have the same magnitude. In the standard parameterization,

$$J = s_{12}s_{23}s_{13}c_{12}c_{23}c_{13}^2 \sin \delta. \quad (533)$$

Thus CP violation requires not only a nonzero phase δ , but also nonzero mixing among all three generations. If any two quarks of the same charge were degenerate, or if any mixing angle vanished, the phase could be removed and J would vanish. A basis-independent statement is

$$\text{Im} \det [M_u M_u^\dagger, M_d M_d^\dagger] \neq 0. \quad (534)$$

This is the Jarlskog criterion for CP violation in the quark sector. It makes the same point in a way that does not depend on the chosen basis of quark fields.

The lesson for us is structural. Gauge interactions are simple in the weak basis because they respect the $SU(2)_L$ doublet index. Mass matrices are simple in the mass basis. The mismatch between these two bases is the CKM matrix, and only with three generations does that mismatch contain a physical complex phase.

6 Optional reading: Clifford algebras and spinors in arbitrary dimensions

The four-dimensional construction above is enough for the electronic field. This optional section records the general-dimensional language, because it clarifies why Majorana, Weyl, and Majorana–Weyl spinors exist only in special dimensions. We use Lorentzian signature

$$\eta_{AB} = \text{diag}(1, -1, \dots, -1), \quad A, B = 0, 1, \dots, D-1, \quad (535)$$

and gamma matrices satisfying

$$\{\Gamma^A, \Gamma^B\} = 2\eta^{AB}\mathbf{1}. \quad (536)$$

The complex Clifford algebra generated by the Γ^A has basis

$$\mathbf{1}, \quad \Gamma^A, \quad \Gamma^{A_1 A_2} := \Gamma^{[A_1} \Gamma^{A_2]}, \quad \dots, \quad \Gamma^{A_1 \dots A_D} := \Gamma^{[A_1} \dots \Gamma^{A_D]}. \quad (537)$$

There are $\binom{D}{r}$ independent antisymmetrized products of rank r , so the algebra has dimension

$$\sum_{r=0}^D \binom{D}{r} = 2^D. \quad (538)$$

A full matrix algebra on an N -dimensional complex spinor space has dimension N^2 . Thus the minimal complex spinor dimension must obey $N^2 \geq 2^D$ when the full Clifford algebra is irreducibly represented. The result is

$$N_{\mathbb{C}} = 2^{\lfloor D/2 \rfloor}. \quad (539)$$

For even $D = 2n$, this is exactly $N_{\mathbb{C}} = 2^n$. For odd $D = 2n + 1$, the complex Clifford algebra has two inequivalent irreducible representations, each again of dimension 2^n ; the product of all gamma matrices is then central and fixed to a sign in each irreducible representation.

6.1 Even dimensions, odd dimensions, and chirality

For even $D = 2n$ there is a chirality matrix

$$\Gamma_* = c_D \Gamma^0 \Gamma^1 \dots \Gamma^{D-1}, \quad \Gamma_*^2 = \mathbf{1}, \quad \{\Gamma_*, \Gamma^A\} = 0, \quad (540)$$

where the phase c_D is chosen so that $\Gamma_*^2 = 1$. The Weyl projectors are

$$P_{\pm} = \frac{1 \pm \Gamma_*}{2}. \quad (541)$$

Thus Weyl spinors exist only in even spacetime dimension.

There is a useful relation between gamma matrices in even dimension $D = 2n$ and odd dimension $D + 1 = 2n + 1$. Starting from the even-dimensional matrices Γ^A and adding one extra spatial direction, define

$$\tilde{\Gamma}^A = \Gamma^A, \quad A = 0, \dots, D-1, \quad \tilde{\Gamma}^D = i\Gamma_*. \quad (542)$$

Because Γ_* anticommutes with every Γ^A and $(i\Gamma_*)^2 = -\mathbf{1}$, these matrices obey the Clifford algebra in $D + 1$ dimensions with one additional spatial direction. Conversely, in odd dimension the product of all gamma matrices is central, so there is no independent chirality operator.

6.2 Real, complex, and pseudoreal spinor representations

The complex spinor representation may have three possible reality types. Let S be the complex spinor space and let

$$\Sigma^{AB} = \frac{1}{4}[\Gamma^A, \Gamma^B] \quad (543)$$

be the spin representation of the Lorentz algebra. A reality structure is an antilinear operator

$$J = B K, \quad (544)$$

where K denotes ordinary complex conjugation and B is a matrix on spinor indices, such that

$$J \Sigma^{AB} J^{-1} = \Sigma^{AB}. \quad (545)$$

Equivalently,

$$B(\Sigma^{AB})^* B^{-1} = \Sigma^{AB}. \quad (546)$$

Then:

- The representation is *real* if such a J exists and $J^2 = +1$. In this case one can impose an ordinary Majorana condition.
- The representation is *pseudoreal*, or quaternionic, if such a J exists and $J^2 = -1$. A single spinor cannot obey $\psi = J\psi$, because applying J twice would give $\psi = -\psi$. One then needs a symplectic Majorana condition on a doublet.
- The representation is *complex* if no such J exists; the complex conjugate spinor is an inequivalent representation.

This is the spinorial version of the same distinction familiar from finite-dimensional group representations: real, complex, and pseudoreal representations differ according to whether the complex conjugate representation can be identified with the original one and, if so, whether the invariant antilinear map squares to $+1$ or -1 .

6.3 Charge conjugation matrices in arbitrary dimension

It is useful to separate two closely related matrices. First, the charge-conjugation matrix C_η is defined by

$$(\Gamma^A)^T = -\eta C_\eta \Gamma^A C_\eta^{-1}, \quad \eta = \pm 1. \quad (547)$$

Equivalently,

$$C_\eta^{-1} \Gamma^A C_\eta = -\eta (\Gamma^A)^T. \quad (548)$$

For even D there are, in general, two possible charge-conjugation matrices, C_+ and C_- , related by multiplication by the chirality matrix:

$$C_- \simeq C_+ \Gamma_*. \quad (549)$$

They have opposite signs in Eq. (547). In odd D only one of these signs is independent in an irreducible representation, because the product of all gamma matrices is central.

Second, the Majorana or complex-conjugation matrix is

$$B_\eta := C_\eta A^T, \quad A := \Gamma^0 \quad \text{for the mostly-minus convention used here.} \quad (550)$$

The Dirac adjoint is $\bar{\psi} = \psi^\dagger A$, so the charge-conjugate spinor is

$$\psi^c = C_\eta \bar{\psi}^T = B_\eta \psi^*. \quad (551)$$

The condition that charge conjugation map a spinor back into the same Lorentz representation is precisely

$$B_\eta(\Sigma^{AB})^* B_\eta^{-1} = \Sigma^{AB}. \quad (552)$$

The square of the antilinear operation is

$$J_\eta^2 = (B_\eta K)(B_\eta K) = B_\eta B_\eta^*. \quad (553)$$

If $B_\eta B_\eta^* = +\mathbf{1}$, the Majorana condition

$$\psi = \psi^c = B_\eta \psi^* \quad (554)$$

is consistent. If $B_\eta B_\eta^* = -\mathbf{1}$, the representation is pseudoreal and the ordinary Majorana condition is inconsistent for a single spinor.

The symmetry of C_η ,

$$C_\eta^T = \epsilon_\eta C_\eta, \quad \epsilon_\eta = \pm 1, \quad (555)$$

is dimension-dependent. This sign determines which bilinears $\psi^T C_\eta \Gamma^{A_1 \dots A_r} \chi$ are symmetric or antisymmetric under interchange of two fermions. For constructing lecture notes, the safest procedure is therefore not to memorize isolated signs, but to keep Eq. (547) and compute the required bilinear symmetry in the chosen dimension and basis.

6.4 Majorana, symplectic Majorana, and Majorana–Weyl spinors

The ordinary Majorana condition is Eq. (554). It exists when the Lorentz spinor representation is real. If the representation is pseudoreal, the correct replacement uses a pair of spinors carrying an extra $\mathrm{Sp}(2N)$ index $i = 1, \dots, 2N$. Let Ω_{ij} be the antisymmetric invariant tensor of $\mathrm{Sp}(2N)$, normalized so that

$$\Omega_{ik} \Omega^{kj} = \delta_i^j, \quad \Omega^* = \Omega. \quad (556)$$

The symplectic Majorana condition is

$$\psi^i = \Omega^{ij} B(\psi_j)^*. \quad (557)$$

If $BB^* = -\mathbf{1}$, the extra factor Ω also squares to $-\mathbf{1}$, so the combined operation squares to $+\mathbf{1}$. This is why a pseudoreal spinor does not admit an ordinary Majorana condition but does admit a symplectic Majorana condition.

In even D , one may also impose a Weyl condition. A Majorana–Weyl spinor is a spinor satisfying both

$$\Gamma_* \psi = \pm \psi, \quad \psi = B\psi^*. \quad (558)$$

This is possible only when the Majorana reality condition preserves a single chirality. A symplectic Majorana–Weyl spinor similarly satisfies

$$\Gamma_* \psi^i = \pm \psi^i, \quad \psi^i = \Omega^{ij} B(\psi_j)^*. \quad (559)$$

This is the structure used, for example, in six-dimensional Lorentzian theories.

For Lorentzian signature $(1, D-1)$, the standard mod-eight pattern is:

$D \bmod 8$	Weyl?	Majorana?	Majorana–Weyl?	symplectic?	comment
0	yes	yes	no	no	real spinors, but the Majorana condition pairs opposite chiralities
1	no	yes	no	no	odd-dimensional real spinor
2	yes	yes	yes	no	ordinary Majorana–Weyl possible, e.g. $D = 10$
3	no	yes	no	no	e.g. $D = 3, 11$
4	yes	yes	no	no	e.g. $D = 4$ Weyl and Majorana spinors
5	no	no	no	yes	symplectic Majorana possible
6	yes	no	no	yes	symplectic Majorana–Weyl possible
7	no	no	no	yes	symplectic Majorana possible

Here “symplectic” means that the ordinary Majorana condition is replaced by Eq. (557). Other signatures are obtained by the usual shift with $p - q \bmod 8$.

6.5 Parity and time reversal in arbitrary dimension

Discrete transformations must also be represented on spinor indices. For full spatial parity,

$$P : \quad x^0 \mapsto x^0, \quad x^i \mapsto -x^i, \quad (560)$$

a standard spinor matrix is

$$S(P) = \eta_P \Gamma^0, \quad (561)$$

where η_P is a conventional phase. It obeys

$$S(P)\Gamma^0 S(P)^{-1} = \Gamma^0, \quad S(P)\Gamma^i S(P)^{-1} = -\Gamma^i. \quad (562)$$

In even D , parity anticommutes with Γ_* and therefore exchanges the two Weyl representations.

Time reversal is antiunitary. Its spinor action has the form

$$\mathbb{T}\psi(x)\mathbb{T}^{-1} = S(T)\psi(Tx), \quad Tx = (-x^0, \mathbf{x}), \quad (563)$$

where the antiunitarity of $\mathbb{T}$ complex-conjugates numerical coefficients. The matrix $S(T)$ is chosen so that

$$S(T)(\Gamma^0)^* S(T)^{-1} = -\Gamma^0, \quad S(T)(\Gamma^i)^* S(T)^{-1} = \Gamma^i. \quad (564)$$

These equations are the arbitrary-dimensional version of the four-dimensional statement that time reversal reverses the sign of the time gamma matrix after complex conjugation. The explicit product of gamma matrices used for $S(T)$ depends on the chosen gamma-matrix basis, but Eq. (564) is the invariant definition.

Charge conjugation, parity, and time reversal are not independent of the reality type. For example, in a Majorana basis charge conjugation may be represented simply by complex conjugation, while in a pseudoreal dimension the corresponding antilinear map squares to -1 on one spinor and must be supplemented by a symplectic index.

6.6 Two dimensions

In $D = 2$ with mostly-minus signature, a convenient real set of gamma matrices is

$$\Gamma^0 = \sigma^1, \quad \Gamma^1 = i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (565)$$

They obey $(\Gamma^0)^2 = +\mathbf{1}$ and $(\Gamma^1)^2 = -\mathbf{1}$. The chirality matrix may be chosen as

$$\Gamma_* = -\Gamma^0 \Gamma^1 = \sigma^3. \quad (566)$$

Thus

$$P_+ = \frac{1 + \sigma^3}{2}, \quad P_- = \frac{1 - \sigma^3}{2}. \quad (567)$$

Since the gamma matrices are real, the Majorana condition is simply

$$\psi^* = \psi. \quad (568)$$

Since $D = 2$ is also even, one may impose Majorana and Weyl conditions simultaneously. A Majorana–Weyl spinor therefore has one real component:

$$\psi_+ = \begin{pmatrix} \xi \\ 0 \end{pmatrix}, \quad \psi_- = \begin{pmatrix} 0 \\ \eta \end{pmatrix}, \quad \xi, \eta \in \mathbb{R}. \quad (569)$$

This is the low-dimensional prototype of the Majorana–Weyl condition used in ten-dimensional Lorentzian theories. Since $10 \equiv 2 \pmod{8}$, ten-dimensional spinors can also be simultaneously Majorana and Weyl. We do not need the ten-dimensional matrices here; the important point is the mod-eight repetition.

6.7 Three dimensions

In $D = 3$ there is no independent chirality matrix. A convenient real two-component representation is

$$\Gamma^0 = i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \Gamma^1 = \sigma^1, \quad \Gamma^2 = \sigma^3, \quad (570)$$

for the mostly-plus convention. To keep the mostly-minus convention of the rest of these notes, multiply all matrices by i and then relabel the time-like direction, or equivalently use

$$\Gamma^0 = \sigma^3, \quad \Gamma^1 = i\sigma^1, \quad \Gamma^2 = i\sigma^2, \quad \eta = \text{diag}(1, -1, -1). \quad (571)$$

These matrices obey

$$\{\Gamma^A, \Gamma^B\} = 2\eta^{AB}\mathbf{1}. \quad (572)$$

The product of all three gamma matrices is proportional to the identity:

$$\Gamma^0\Gamma^1\Gamma^2 = -i\mathbf{1}. \quad (573)$$

Changing the sign of one gamma matrix gives the other inequivalent irreducible representation. Since there is no independent chirality operator, there are no Weyl spinors in three dimensions. Majorana spinors do exist in three dimensions; in a real gamma-matrix convention the Majorana condition again reduces to a two-real-component spinor.

7 Massive spin-one vector field: Out comes the Proca theory

The scalar and Dirac examples illustrate the general method: first classify the one-particle states and solve the field-particle intertwiner equations, then read off the field equation, and only later introduce an action. We now do the same for a massive spin-one particle. This section is deliberately parallel to Section 3.

7.1 Weinberg construction without using the action

A massive spin-one particle has little group $\text{SU}(2)$ and carries the $j = 1$ representation. Let $r = 1, 2, 3$ label a real Cartesian spin basis. In this basis a rotation $R \in \text{SO}(3)$ acts by the ordinary three-dimensional rotation matrix,

$$D_{sr}^{(1)}(R) = R_{sr}. \quad (574)$$

We want a local field whose field index is a Lorentz-vector index. Thus we take

$$V^\mu(x), \quad S^\mu{}_\nu(\Lambda) = \Lambda^\mu{}_\nu. \quad (575)$$

The coefficient functions are polarization vectors,

$$\varepsilon^\mu(p, r) := \langle 0 | V^\mu(0) | p, r \rangle. \quad (576)$$

The general Weinberg condition (85) becomes

$$\Lambda^\mu{}_\nu \varepsilon^\nu(p, r) = \sum_s \varepsilon^\mu(\Lambda p, s) D_{sr}^{(1)}(W(\Lambda, p)). \quad (577)$$

This equation is the spin-one analogue of the scalar and Dirac intertwiner equations. We now solve it at rest and then boost.

Take the standard rest momentum

$$k = (m, 0, 0, 0). \quad (578)$$

For a pure rotation R , the Lorentz-vector representation decomposes under spatial rotations as

$$V^\mu = (V^0, V^i) : \quad \mathbf{4}|_{\text{SO}(3)} = \mathbf{1} \oplus \mathbf{3}. \quad (579)$$

The time component V^0 is a rotational scalar, while the spatial components V^i form a rotational vector. A spin-one particle should be matched to the $\mathbf{3}$, not to the scalar $\mathbf{1}$. Therefore the rest-frame polarization vectors must have zero time component:

$$\varepsilon^0(k, r) = 0. \quad (580)$$

Choose an orthonormal basis $\mathbf{e}_r$ of $\mathbb{R}^3$ and set

$$\varepsilon^\mu(k, r) = (0, \mathbf{e}_r)^\mu, \quad r = 1, 2, 3. \quad (581)$$

Then

$$R^\mu{}_\nu \varepsilon^\nu(k, r) = \sum_s \varepsilon^\mu(k, s) R_{sr}, \quad (582)$$

which is exactly the rest-frame matching condition.

For a general momentum $p = L(p)k$, define

$$\varepsilon^\mu(p, r) = L(p)^\mu{}_\nu \varepsilon^\nu(k, r). \quad (583)$$

This solves Eq. (577) for the same reason as in Section 3.5. In components, for a rotation basis vector $\mathbf{e}_r$,

$$\varepsilon^0(p, r) = \frac{\mathbf{p} \cdot \mathbf{e}_r}{m}, \quad \varepsilon(p, r) = \mathbf{e}_r + \frac{\mathbf{p}(\mathbf{p} \cdot \mathbf{e}_r)}{m(p^0 + m)}. \quad (584)$$

These polarizations obey

$$p_\mu \varepsilon^\mu(p, r) = 0, \quad \varepsilon_\mu(p, r) \varepsilon^\mu(p, s) = -\delta_{rs}, \quad (585)$$

and the completeness relation

$$\sum_{r=1}^3 \varepsilon^\mu(p, r) \varepsilon^\nu(p, r)^* = -\eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}. \quad (586)$$

The field expansion for a neutral massive vector field is then

$$V^\mu(x) = \sum_{r=1}^3 \int d\Pi_p \left[\varepsilon^\mu(p, r) c(p, r) e^{-ip \cdot x} + \varepsilon^\mu(p, r)^* c^\dagger(p, r) e^{ip \cdot x} \right]. \quad (587)$$

For a charged vector field one would use independent particle and antiparticle operators, exactly as for the complex scalar field.

The equations obeyed by V^μ are now determined, not guessed. Since every plane wave has $p^2 = m^2$,

$$(\partial^2 + m^2)V^\mu(x) = 0. \quad (588)$$

Since every polarization is transverse,

$$\partial_\mu V^\mu(x) = 0. \quad (589)$$

Combining these two equations gives

$$\partial_\nu(\partial^\nu V^\mu - \partial^\mu V^\nu) + m^2 V^\mu = 0. \quad (590)$$

This is the Proca equation, but it has appeared here as a consequence of the massive spin-one field construction.

7.2 Causality and the choice of commutators

Spin one is an integer spin. The causal local algebra should therefore be bosonic: spacelike-separated vector fields must commute. The oscillator algebra is

$$[c(p, r), c^\dagger(p', s)] = (2\pi)^3 2p^0 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{rs}, \quad (591)$$

with all other commutators vanishing. Using Eq. (586), one obtains

$$[V^\mu(x), V^\nu(y)] = \left(-\eta^{\mu\nu} - \frac{1}{m^2} \partial^\mu \partial^\nu \right) \Delta(x - y), \quad (592)$$

where Δ is the Pauli-Jordan function of Eq. (105). The derivative term appears because $p^\mu p^\nu$ in the polarization sum becomes $-\partial^\mu \partial^\nu$ acting on the plane waves. Since $\Delta(z)$ vanishes for spacelike z , so do its derivatives. Hence

$$[V^\mu(x), V^\nu(y)] = 0, \quad (x - y)^2 < 0. \quad (593)$$

If instead we used anticommutators for the spin-one oscillators, the same calculation would contain the even Hadamard function $\Delta^{(1)}(x - y)$ rather than $\Delta(x - y)$. As in the scalar case, $\Delta^{(1)}$ is not zero at spacelike separation. Thus causality selects commutators, not anticommutators, for the massive vector field.

7.3 The Proca action and the scalar-vector split

We now introduce the action. The representation-theoretic construction has already told us which field equation the massive spin-one field must satisfy. The Proca action is the local Lorentz-invariant action whose Euler-Lagrange equation is precisely Eq. (590):

$$S_P = \int d^4x \mathcal{L}_P, \quad \mathcal{L}_P = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 V_\mu V^\mu, \quad (594)$$

where

$$F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu. \quad (595)$$

Varying V_μ gives

$$\partial_\nu F^{\nu\mu} + m^2 V^\mu = 0. \quad (596)$$

Taking the divergence gives

$$m^2 \partial_\mu V^\mu = 0. \quad (597)$$

For $m \neq 0$ this implies

$$\partial_\mu V^\mu = 0. \quad (598)$$

Substituting this back into Eq. (596) gives $(\partial^2 + m^2)V^\mu = 0$.

It is useful to separate the scalar and vector pieces under spatial rotations. A Lorentz vector decomposes as

$$V^\mu = (V^0, \mathbf{V}), \quad (599)$$

where V^0 is a scalar under rotations and $\mathbf{V}$ is a three-vector. The Weinberg rest-frame construction selected the spatial vector part: $\varepsilon^0(k, r) = 0$. The Proca action implements the same fact dynamically. There is no time derivative of V^0 in the action, so V^0 is not an independent propagating oscillator. It is fixed by the constraint (598).

The spatial vector itself may be decomposed as

$$\mathbf{V} = \mathbf{V}_T + \nabla\chi, \quad \nabla \cdot \mathbf{V}_T = 0. \quad (600)$$

The two components of $\mathbf{V}_T$ are the transverse polarizations. The scalar function χ describes the longitudinal polarization. The nondynamical field V^0 is tied to this longitudinal mode by the constraint. Therefore the massive vector field has three propagating degrees of freedom, not four: two transverse polarizations and one longitudinal polarization.

7.4 Hamiltonian analysis and Dirac brackets

The Proca action is a constrained Hamiltonian system. This is the simplest place to learn why Dirac brackets are useful.

Write the canonical variables as V_0, V_i and their conjugate momenta. From Eq. (594),

$$\Pi^0 = \frac{\partial \mathcal{L}_P}{\partial \dot{V}_0} = 0, \quad \Pi^i = \frac{\partial \mathcal{L}_P}{\partial \dot{V}_i} = F_{0i} = \dot{V}_i - \partial_i V_0. \quad (601)$$

The equation $\Pi^0 = 0$ is not an equation of motion. It is a primary constraint:

$$\chi_1(x) := \Pi^0(x) \approx 0. \quad (602)$$

The canonical Hamiltonian density is

$$\mathcal{H}_c = \frac{1}{2} \Pi_i \Pi_i + \frac{1}{4} F_{ij} F_{ij} + \frac{1}{2} m^2 V_i V_i - \frac{1}{2} m^2 V_0^2 - V_0 \partial_i \Pi_i, \quad (603)$$

where a spatial integration by parts has been used and surface terms are dropped. Requiring the primary constraint to be preserved in time gives the secondary constraint

$$\chi_2(x) := \partial_i \Pi_i(x) + m^2 V_0(x) \approx 0. \quad (604)$$

This is the Hamiltonian version of the nondynamical equation for V_0 .

The canonical Poisson brackets are

$$\{V_\mu(t, \mathbf{x}), \Pi^\nu(t, \mathbf{y})\} = \delta_\mu^\nu \delta^3(\mathbf{x} - \mathbf{y}). \quad (605)$$

The constraints have a nonzero Poisson bracket,

$$\{\chi_1(\mathbf{x}), \chi_2(\mathbf{y})\} = -m^2 \delta^3(\mathbf{x} - \mathbf{y}). \quad (606)$$

They are therefore second-class constraints. For second-class constraints, ordinary Poisson brackets are not the correct brackets on the constrained phase space. Dirac's prescription is to define

$$\{F, G\}_D = \{F, G\} - \int d^3\mathbf{u} d^3\mathbf{v} \{F, \chi_a(\mathbf{u})\} C_{ab}^{-1}(\mathbf{u}, \mathbf{v}) \{\chi_b(\mathbf{v}), G\}, \quad (607)$$

where

$$C_{ab}(\mathbf{x}, \mathbf{y}) := \{\chi_a(\mathbf{x}), \chi_b(\mathbf{y})\}. \quad (608)$$

This definition is possible only because C_{ab} is invertible. This is the essential distinction between first-class and second-class constraints. A first-class constraint has vanishing Poisson brackets with all constraints, at least on the constraint surface; it therefore generates a gauge direction and gives a degenerate matrix C_{ab} . There is then no inverse C_{ab}^{-1} and hence no Dirac bracket. A second-class set has no such gauge null direction: the constraints really remove phase-space directions. Precisely because C_{ab} is invertible, the Dirac bracket projects the ordinary Poisson bracket onto the surface where the second-class constraints hold.

The point of the Dirac bracket is that the constraints can now be imposed strongly: after replacing $\{, \}$ by $\{, \}_D$, one may set $\chi_1 = \chi_2 = 0$ inside all formulae. Indeed,

$$\{F, \chi_a\}_D = \{F, \chi_a\} - \{F, \chi_b\} C_{bc}^{-1} \{\chi_c, \chi_a\} \quad (609)$$

$$= \{F, \chi_a\} - \{F, \chi_b\} C_{bc}^{-1} C_{ca} = 0, \quad (610)$$

where the spatial integrations are suppressed in the second line. Thus the constraints have zero Dirac bracket with every observable: in the reduced bracket they are identities, not additional equations to be remembered.

There is also a useful geometric way to say the same thing. The original phase space has symplectic form Ω . The surface Σ defined by $\chi_a = 0$ inherits the pulled-back two-form Ω_Σ . If the constraints are second class, then Ω_Σ is nondegenerate. Therefore it has an inverse bivector, and that inverse is exactly the Dirac bracket. In coordinates z^A on phase space, with ordinary Poisson bivector $\Pi^{AB} = \{z^A, z^B\}$, the Dirac bivector is

$$\Pi_D^{AB} = \Pi^{AB} - \Pi^{AC} \partial_C \chi_a C_{ab}^{-1} \partial_D \chi_b \Pi^{DB}. \quad (611)$$

It obeys

$$\Pi_D^{AB} \partial_B \chi_a = 0, \quad (612)$$

so it has no component normal to the constraint surface. Restricted to tangent vectors on Σ , it is the inverse of Ω_Σ . This proves at once that the Dirac bracket is antisymmetric, satisfies the Leibniz rule, and satisfies the Jacobi identity: it is an ordinary Poisson bracket, but on the reduced phase space. Equivalently,

$$\boxed{\text{Dirac bracket} = (\Omega_\Sigma)^{-1}}. \quad (613)$$

This is the clean conceptual reason why the formal expression above is consistent.

Toy example: removing one canonical pair. Consider a four-dimensional phase space with canonical variables

$$(q_1, p_1, q_2, p_2), \quad \{q_i, p_j\} = \delta_{ij}.$$

Impose the two constraints

$$\chi_1 = q_2 \approx 0, \quad \chi_2 = p_2 \approx 0.$$

Their constraint matrix is

$$C_{ab} = \{\chi_a, \chi_b\} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C_{ab}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Hence the constraints are second class. The Dirac bracket gives

$$\{q_1, p_1\}_D = 1, \quad \{q_2, F\}_D = \{p_2, F\}_D = 0$$

for every observable F . The constrained pair (q_2, p_2) has disappeared, while the physical pair (q_1, p_1) keeps its usual bracket. Geometrically, we have restricted the symplectic form $dp_1 \wedge dq_1 + dp_2 \wedge dq_2$ to the surface $q_2 = p_2 = 0$, where it becomes simply $dp_1 \wedge dq_1$. If we imposed only $p_2 \approx 0$, the constraint matrix would be zero. This is a first-class constraint: it generates the gauge shift $q_2 \rightarrow q_2 + \epsilon$. There is no inverse constraint matrix and therefore no Dirac bracket until we add a gauge-fixing condition such as $q_2 \approx 0$, which turns the first-class constraint plus gauge condition into a second-class pair.

For the physical spatial variables, the Dirac bracket is simple:

$$\{V_i(t, \mathbf{x}), \Pi_j(t, \mathbf{y})\}_D = \delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}), \quad \{V_i, V_j\}_D = \{\Pi_i, \Pi_j\}_D = 0. \quad (614)$$

The constraint (604) determines

$$V_0 = -\frac{1}{m^2} \partial_i \Pi_i. \quad (615)$$

Substituting this into the Hamiltonian gives the reduced Hamiltonian

$$H_{\text{red}} = \int d^3\mathbf{x} \left[\frac{1}{2} \Pi_i \Pi_i + \frac{1}{4} F_{ij} F_{ij} + \frac{1}{2} m^2 V_i V_i + \frac{1}{2m^2} (\partial_i \Pi_i)^2 \right]. \quad (616)$$

Now compute the Hamilton equations with the Dirac bracket:

$$\dot{V}_i = \{V_i, H_{\text{red}}\}_D = \Pi_i - \frac{1}{m^2} \partial_i \partial_j \Pi_j = \Pi_i + \partial_i V_0, \quad (617)$$

$$\dot{\Pi}_i = \{\Pi_i, H_{\text{red}}\}_D = \partial_j F_{ji} - m^2 V_i. \quad (618)$$

The first equation is just the definition $\Pi_i = \dot{V}_i - \partial_i V_0$. The second equation, together with the constraint for V_0 , is equivalent to

$$\partial_\nu F^{\nu i} + m^2 V^i = 0. \quad (619)$$

The missing $\mu = 0$ component is precisely the secondary constraint

$$\partial_i \Pi_i + m^2 V_0 = 0, \quad (620)$$

which is the same as $\partial_\nu F^{\nu 0} + m^2 V^0 = 0$. Thus the Dirac-bracket Hamiltonian formalism reproduces the full Proca equations.

Upon quantization, the Dirac brackets of the physical variables become equal-time commutators,

$$[\hat{V}_i(t, \mathbf{x}), \hat{\Pi}_j(t, \mathbf{y})] = i \delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}), \quad (621)$$

with the constrained variables V_0 and Π^0 eliminated. The Hamilton equations are now equivalently

$$i\dot{V}_i = [V_i, H_{\text{red}}], \quad i\dot{\Pi}_i = [\Pi_i, H_{\text{red}}], \quad (622)$$

which are the operator version of Eqs. (617)–(618).

This is also consistent with the covariant oscillator commutators used earlier. Lowering indices in Eq. (592), write

$$[V_\mu(x), V_\nu(y)] = \left(-\eta_{\mu\nu} - \frac{1}{m^2} \partial_\mu \partial_\nu \right) \Delta(x - y), \quad (623)$$

where derivatives act on $z = x - y$. Since $\Pi_j = F_{0j} = \partial_0 V_j - \partial_j V_0$,

$$[V_i(x), \Pi_j(y)] = \partial_{y^0} [V_i(x), V_j(y)] - \partial_{y^j} [V_i(x), V_0(y)] \quad (624)$$

$$= -\partial_{z^0} \left(\delta_{ij} \Delta - \frac{1}{m^2} \partial_i \partial_j \Delta \right) + \partial_{z^j} \left(-\frac{1}{m^2} \partial_i \partial_0 \Delta \right) \quad (625)$$

$$= -\delta_{ij} \partial_{z^0} \Delta. \quad (626)$$

The longitudinal derivative terms cancel. Using Eq. (134) gives

$$[V_i(t, \mathbf{x}), \Pi_j(t, \mathbf{y})] = i\delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}). \quad (627)$$

Thus the equal-time commutator obtained from the Dirac-bracket Hamiltonian formalism is exactly the equal-time derivative of the Lorentz-covariant Proca commutator.

The reduced Hamiltonian in Eq. (616) is therefore diagonal in the massive spin-one oscillator basis. Substituting the mode expansion (587), using the transversality and completeness relations (585)–(586), and normal ordering gives

$$: H_{\text{red}} := \sum_{r=1}^3 \int d\Pi_p p^0 c^\dagger(p, r) c(p, r). \quad (628)$$

This is the expected result: the massive vector field contains three positive-energy harmonic-oscillator modes at each momentum, corresponding to the three spin-one polarizations. For a charged vector field one would add the identical antiparticle contribution.

7.5 Discrete symmetries and CPT for the massive vector field

For a neutral real vector field, charge conjugation can only multiply the field by an intrinsic sign:

$$C V^\mu(x) C^{-1} = \eta_C^V V^\mu(x), \quad \eta_C^V = \pm 1. \quad (629)$$

For a charged complex vector field W^μ , charge conjugation exchanges the charged field with its Hermitian conjugate:

$$C W^\mu(x) C^{-1} = \eta_C^W W^{\mu\dagger}(x). \quad (630)$$

Parity acts with the vector matrix

$$P^\mu{}_\nu = \text{diag}(1, -1, -1, -1), \quad (631)$$

so that

$$P V^\mu(t, \mathbf{x}) P^{-1} = \eta_P^V P^\mu{}_\nu V^\nu(t, -\mathbf{x}). \quad (632)$$

Time reversal is antiunitary. With the same convention used for the Dirac field,

$$T V^\mu(t, \mathbf{x}) T^{-1} = \eta_T^V P^\mu{}_\nu V^\nu(-t, \mathbf{x}), \quad (633)$$

where antiunitarity complex-conjugates all c-number coefficients in matrix elements. Combining the three transformations gives

$$\text{CPT } V^\mu(x) (\text{CPT})^{-1} = \eta_{CPT}^V V^\mu(-x) \quad (634)$$

for a neutral field, and the corresponding Hermitian-conjugate field for a charged vector field.

The CPT theorem for local interactions involving massive vector fields follows the same pattern as in Section 5.5. The elementary vector transformation above sends each vector index at x to the corresponding vector index at $-x$. In an ordinary Lorentz scalar, all vector indices are contracted with $\eta_{\mu\nu}$, derivatives, or with other tensor structures in such a way that the signs occur pairwise. Therefore any Hermitian local ordinary scalar built from scalar, Dirac, and massive vector fields, with the usual spin-statistics assignments, is invariant under CPT up to the reversal $x \mapsto -x$ and the antiunitary complex conjugation of coefficients. Pseudoscalars require the same extra care discussed earlier: they acquire the orientation sign associated with improper Lorentz transformations.

8 Massless spin-one field – the photon: Why a local theory must have gauge symmetry and how Maxwell’s equations arise from quantum field theory

We now repeat the same logic for a massless spin-one particle. The massive spin-one field had a rest frame and therefore a little group $SO(3)$. A massless particle has no rest frame. This changes the representation theory in an essential way and is the reason why gauge invariance enters.

8.1 Massless one-particle states and the little group

Choose the standard null momentum

$$k^\mu = (\kappa, 0, 0, \kappa), \quad \kappa > 0. \quad (635)$$

The little group is the subgroup of the Lorentz group that leaves k^μ fixed:

$$W^\mu{}_\nu k^\nu = k^\mu. \quad (636)$$

To find its Lie algebra, write an infinitesimal Lorentz transformation as

$$W^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu}. \quad (637)$$

The condition $\omega^\mu{}_\nu k^\nu = 0$ gives

$$\omega^\mu{}_0 + \omega^\mu{}_3 = 0. \quad (638)$$

Solving these equations leaves three independent generators. They may be chosen as

$$J_3, \quad N_1 := K_1 + J_2, \quad N_2 := K_2 - J_1, \quad (639)$$

where J_i are rotations and K_i are boosts. Using the Lorentz algebra,

$$[J_i, J_j] = i\epsilon_{ijk} J_k, \quad [J_i, K_j] = i\epsilon_{ijk} K_k, \quad [K_i, K_j] = -i\epsilon_{ijk} J_k, \quad (640)$$

we obtain

$$[J_3, N_1] = iN_2, \quad [J_3, N_2] = -iN_1, \quad [N_1, N_2] = 0. \quad (641)$$

This is the Lie algebra of the two-dimensional Euclidean group $ISO(2)$: J_3 is the rotation in the plane transverse to the momentum, while N_1, N_2 are translation-like generators.

A general unitary representation of $ISO(2)$ may allow N_1, N_2 to act nontrivially. These are the continuous-spin representations. The photon is not such a particle. For ordinary massless particles of fixed helicity, the translation-like part acts trivially:

$$N_1 |k, h\rangle = N_2 |k, h\rangle = 0, \quad J_3 |k, h\rangle = h |k, h\rangle. \quad (642)$$

The number h is called the helicity. For a massless spin-one particle,

$$h = +1 \quad \text{or} \quad h = -1. \quad (643)$$

Thus the physical photon has two one-particle states for each momentum, not three as a massive spin-one particle has.

For an arbitrary null momentum p^μ , choose a standard Lorentz transformation $L(p)$ such that $p = L(p)k$. Define

$$|p, h\rangle = U(L(p)) |k, h\rangle. \quad (644)$$

For a general Lorentz transformation,

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p) \quad (645)$$

belongs to the massless little group. Since the translation-like part acts trivially on helicity states, only the rotation angle $\theta(\Lambda, p)$ matters:

$$U(\Lambda) |p, h\rangle = e^{ih\theta(\Lambda, p)} |\Lambda p, h\rangle. \quad (646)$$

This is the massless analogue of the Wigner transformation law for massive spin.

8.2 The field-particle intertwiner, gauge symmetry, and Maxwell equations

The next question is the same one we asked in Section 3: given the one-particle transformation law, can we construct a local Lorentz-covariant field whose creation part creates precisely those one-particle states? For a massless spin-one particle the answer is subtle. We can construct a local four-vector potential, but only up to a gauge transformation. The point is not a peculiarity of electromagnetism; it follows directly from the representation theory of the massless little group.

Let a candidate vector potential have the form

$$A^\mu(x) = \sum_{h=\pm 1} \int \frac{d^3 \mathbf{p}}{(2\pi)^3 2|\mathbf{p}|} \left[\epsilon^\mu(p, h) a(p, h) e^{-ip \cdot x} + \epsilon^\mu(p, h)^* a^\dagger(p, h) e^{ip \cdot x} \right], \quad p^0 = |\mathbf{p}|. \quad (647)$$

The functions $\epsilon^\mu(p, h)$ are the massless analogues of the coefficients $u_\ell(p, \sigma)$ in Section 3. Since the field carries a Lorentz vector index, the finite-dimensional field representation is

$$S^\mu{}_\nu(\Lambda) = \Lambda^\mu{}_\nu. \quad (648)$$

The particle states carry the helicity representation of the little group,

$$D_{h'h}(W(\Lambda, p)) = \delta_{h'h} e^{ih\theta(\Lambda, p)}. \quad (649)$$

If the vector potential transformed as an honest vector field without any gauge term, the coefficients would have to obey the strict intertwiner equation

$$\Lambda^\mu{}_\nu \epsilon^\nu(p, h) = e^{ih\theta(\Lambda, p)} \epsilon^\mu(\Lambda p, h). \quad (650)$$

This is the massless version of the general relation in Eq. (85). The problem is that Eq. (650) has no nonzero solution in the four-vector representation for helicity $h = \pm 1$. To see this, it is enough to examine the standard momentum $k = (\kappa, 0, 0, \kappa)$ and the little group itself.

Choose the transverse standard polarization vectors

$$\epsilon^\mu(k, h) = \frac{1}{\sqrt{2}}(0, 1, -ih, 0), \quad h = \pm 1. \quad (651)$$

They satisfy

$$k_\mu \epsilon^\mu(k, h) = 0, \quad \epsilon(k, h)^* \cdot \epsilon(k, h') = -\delta_{hh'}. \quad (652)$$

Under a rotation about the z -axis,

$$R_z(\theta)^\mu{}_\nu \epsilon^\nu(k, h) = e^{ih\theta} \epsilon^\mu(k, h), \quad (653)$$

so these vectors have the desired helicities. The obstruction comes from the translation-like part of the little group. A convenient finite representative of a little-group translation is

$$W(\alpha, \beta) = \begin{pmatrix} 1 + \frac{\alpha^2 + \beta^2}{2} & \alpha & \beta & -\frac{\alpha^2 + \beta^2}{2} \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \frac{\alpha^2 + \beta^2}{2} & \alpha & \beta & 1 - \frac{\alpha^2 + \beta^2}{2} \end{pmatrix}, \quad W(\alpha, \beta)k = k. \quad (654)$$

Acting on Eq. (651), we find

$$W(\alpha, \beta)^\mu{}_\nu \epsilon^\nu(k, h) = \epsilon^\mu(k, h) + \frac{\alpha - ih\beta}{\sqrt{2}\kappa} k^\mu. \quad (655)$$

Thus the little-group translations do not act trivially on the four-vector ϵ^μ , although they act trivially on the physical helicity state. They move ϵ^μ by a vector proportional to k^μ . This is the crucial point. The physical massless spin-one representation is not realized on ordinary four-vectors themselves; it is realized on equivalence classes

$$\epsilon^\mu(k, h) \sim \epsilon^\mu(k, h) + c_h k^\mu. \quad (656)$$

On these equivalence classes, Eq. (655) reduces to the helicity representation.

For general momentum $p = L(p)k$, define

$$\epsilon^\mu(p, h) = L(p)^\mu{}_\nu \epsilon^\nu(k, h), \quad p_\mu \epsilon^\mu(p, h) = 0. \quad (657)$$

This definition is not unique, because we may always add a multiple of p^μ . Using

$$\Lambda L(p) = L(\Lambda p)W(\Lambda, p), \quad (658)$$

and Eq. (655), the true intertwiner equation is

$$\Lambda^\mu{}_\nu \epsilon^\nu(p, h) = e^{ih\theta(\Lambda, p)} \epsilon^\mu(\Lambda p, h) + \zeta_h(\Lambda, p)(\Lambda p)^\mu. \quad (659)$$

Here $\zeta_h(\Lambda, p)$ is determined by the translation part of the Wigner little-group element $W(\Lambda, p)$. Eq. (659) is the precise field-particle connection for a massless spin-one vector potential. It says that Lorentz covariance closes only modulo a term proportional to momentum.

In the field expansion, a term proportional to p^μ is a derivative. Therefore Eq. (659) implies that under a Lorentz transformation the vector potential transforms as

$$U(\Lambda)A^\mu(x)U(\Lambda)^{-1} = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x) + \partial^\mu \Omega_\Lambda(x), \quad (660)$$

where $\Omega_\Lambda(x)$ is an operator-valued gauge function built from the same creation and annihilation operators. The detailed form of Ω_Λ is not important. The important point is that the extra term is forced by the little-group translations. This is the Weinberg origin of the gauge principle:

$$A_\mu(x) \sim A_\mu(x) + \partial_\mu \alpha(x). \quad (661)$$

The vector potential itself is therefore not the invariant local observable. The gauge-invariant local field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (662)$$

The gauge term in Eq. (660) cancels in $F_{\mu\nu}$, so

$$U(\Lambda)F_{\mu\nu}(x)U(\Lambda)^{-1} = \Lambda_\mu{}^\rho \Lambda_\nu{}^\sigma F_{\rho\sigma}(\Lambda^{-1}x). \quad (663)$$

Thus the field strength is a genuine Lorentz tensor even though the vector potential is a Lorentz vector only up to gauge.

Now Maxwell's equations follow already from this Weinberg construction, before any action has been introduced. The mode coefficient of $F_{\mu\nu}$ is

$$f_{\mu\nu}(p, h) = -i(p_\mu \epsilon_\nu(p, h) - p_\nu \epsilon_\mu(p, h)), \quad (664)$$

so that

$$F_{\mu\nu}(x) = \sum_{h=\pm 1} \int \frac{d^3\mathbf{p}}{(2\pi)^3 2|\mathbf{p}|} \left[f_{\mu\nu}(p, h) a(p, h) e^{-ip \cdot x} + f_{\mu\nu}(p, h)^* a^\dagger(p, h) e^{ip \cdot x} \right]. \quad (665)$$

Because the photon is massless and the polarization is transverse,

$$p^2 = 0, \quad p_\mu \epsilon^\mu(p, h) = 0. \quad (666)$$

Therefore

$$p_\mu f^{\mu\nu}(p, h) = -i(p^2 \epsilon^\nu(p, h) - p^\nu p_\mu \epsilon^\mu(p, h)) = 0. \quad (667)$$

It follows mode by mode that

$$\partial_\mu F^{\mu\nu}(x) = 0. \quad (668)$$

Also, since $F_{\mu\nu}$ was constructed as the curl of A_μ , it automatically satisfies

$$\partial_{[\rho} F_{\mu\nu]} = 0. \quad (669)$$

Thus the source-free Maxwell equations have emerged from the Lorentz transformation law of a massless helicity- ± 1 particle and the demand for a local covariant field. They have not yet been postulated as Euler-Lagrange equations.

We can now rederive the same equations from an action. We ask for a free local action with the following properties: Lorentz invariance, locality, at most two derivatives, and invariance under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$. The only ordinary quadratic term with these properties is

$$S_{\text{Maxwell}} = -\frac{1}{4} \int d^4x F_{\mu\nu} F^{\mu\nu}. \quad (670)$$

The possible term $F_{\mu\nu} \tilde{F}^{\mu\nu}$ is a total derivative in the free theory and does not change the local equations of motion. A mass term $m^2 A_\mu A^\mu / 2$ is forbidden because it is not gauge invariant. Varying the Maxwell action gives

$$\delta S_{\text{Maxwell}} = -\frac{1}{2} \int d^4x F^{\mu\nu} \delta F_{\mu\nu} \quad (671)$$

$$= - \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu = \int d^4x (\partial_\mu F^{\mu\nu}) \delta A_\nu, \quad (672)$$

where we integrated by parts and discarded the boundary term. Hence

$$\partial_\mu F^{\mu\nu} = 0. \quad (673)$$

This is the same equation obtained above from the Weinberg field-particle construction. The action formulation is therefore not the starting point of the massless spin-one field in this approach; it is the local Lagrangian realization of the same representation-theoretic result.

8.3 Quantization and microcausality

In a physical, or radiation, gauge one keeps only the transverse modes. The field may be written as

$$A_i^T(x) = \sum_{h=\pm 1} \int \frac{d^3\mathbf{p}}{(2\pi)^3 2|\mathbf{p}|} \left[\epsilon_i(\mathbf{p}, h) a(\mathbf{p}, h) e^{-ip \cdot x} + \epsilon_i(\mathbf{p}, h)^* a^\dagger(\mathbf{p}, h) e^{ip \cdot x} \right], \quad p^0 = |\mathbf{p}|, \quad (674)$$

with

$$\mathbf{p} \cdot \boldsymbol{\epsilon}(\mathbf{p}, h) = 0. \quad (675)$$

The physical oscillator algebra is bosonic:

$$[a(\mathbf{p}, h), a^\dagger(\mathbf{p}', h')] = (2\pi)^3 2|\mathbf{p}| \delta^3(\mathbf{p} - \mathbf{p}') \delta_{hh'}. \quad (676)$$

We use commutators, not anticommutators, for the same spin-statistics reason as for the scalar and massive vector fields: the causal commutator is built from the Pauli-Jordan function, which vanishes outside the light cone. Anticommutators would instead produce the positive-frequency plus negative-frequency sum, which does not vanish at spacelike separation.

The vector potential is gauge dependent, so the clean local statement is made for the gauge-invariant field strength. In a covariant gauge the commutator has the schematic form

$$[A_\mu(x), A_\nu(y)] = -i\eta_{\mu\nu}\Delta(x-y) + \text{gauge-dependent derivative terms}, \quad (677)$$

where Δ is the massless Pauli-Jordan function. Therefore

$$[F_{\mu\nu}(x), F_{\rho\sigma}(y)] = -i \left(\eta_{\nu\sigma} \partial_\mu \partial_\rho - \eta_{\nu\rho} \partial_\mu \partial_\sigma - \eta_{\mu\sigma} \partial_\nu \partial_\rho + \eta_{\mu\rho} \partial_\nu \partial_\sigma \right) \Delta(x-y). \quad (678)$$

The right-hand side vanishes for spacelike separation because $\Delta(x-y)$ vanishes there. This is how microcausality is protected: the gauge-dependent potential may require a gauge choice, but the gauge-invariant local field strength has a causal commutator.

8.4 Dirac's Hamiltonian formalism for Maxwell theory

The Maxwell Lagrangian is singular. This means that the canonical momenta do not allow us to solve for all velocities. Dirac's formalism is designed precisely for such systems.

The canonical momentum conjugate to A_μ is

$$\Pi^\mu := \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)}. \quad (679)$$

For Maxwell theory,

$$\Pi^0 = 0, \quad \Pi^i = F^{i0}. \quad (680)$$

The equation $\Pi^0 = 0$ is not an equation of motion. It is a primary constraint:

$$\phi_1(x) := \Pi^0(x) \approx 0. \quad (681)$$

The symbol $\approx$ means weak equality: the equation holds on the constraint surface, but one should not impose it before computing Poisson brackets.

The canonical Hamiltonian is

$$H_c = \int d^3\mathbf{x} \left[\frac{1}{2} \boldsymbol{\Pi}^2 + \frac{1}{2} \mathbf{B}^2 - A_0 \partial_i \Pi^i \right], \quad (682)$$

up to a surface term. Preservation of the primary constraint gives

$$\dot{\Pi}^0(x) = \{\Pi^0(x), H_c\} = \partial_i \Pi^i(x) \approx 0. \quad (683)$$

Thus there is a secondary constraint,

$$\phi_2(x) := \partial_i \Pi^i(x) \approx 0. \quad (684)$$

This is Gauss's law in the free theory.

We now have to distinguish carefully between two kinds of constraints. A constraint is *first class* if its Poisson bracket with every constraint vanishes on the constraint surface. First-class constraints indicate gauge redundancy: they do not remove a physical canonical pair by themselves; rather, they say that some directions in phase space describe the same physical state. A constraint set is *second class* if the matrix of constraint brackets is invertible on the constraint surface. Second-class constraints remove phase-space directions directly and lead to a nondegenerate reduced symplectic form. Only such a second-class set can define a Dirac bracket, because the definition requires the inverse matrix C_{ab}^{-1} .

In Maxwell theory, before gauge fixing,

$$\{\phi_1(x), \phi_2(y)\} = 0, \quad \{\phi_1(x), \phi_1(y)\} = 0, \quad \{\phi_2(x), \phi_2(y)\} = 0. \quad (685)$$

Thus Π^0 and $\partial_i \Pi^i$ are first-class constraints. Their constraint matrix is not merely singular; it vanishes. Therefore there is no Dirac bracket at this stage. What we have instead is gauge freedom. First-class constraints generate gauge transformations. Indeed, with

$$G[\alpha] = \int d^3 \mathbf{x} [-\dot{\alpha} \Pi^0 + \alpha \partial_i \Pi^i], \quad (686)$$

one obtains

$$\delta A_\mu = \{A_\mu, G[\alpha]\} = \partial_\mu \alpha, \quad \delta \Pi^\mu = 0. \quad (687)$$

This is exactly the gauge redundancy found from the massless little group.

To get a nondegenerate Hamiltonian system, we fix a gauge. A convenient choice is temporal plus Coulomb gauge,

$$A_0 \approx 0, \quad \partial_i A^i \approx 0. \quad (688)$$

Together with $\Pi^0 \approx 0$ and $\partial_i \Pi^i \approx 0$, these gauge conditions form a second-class set. This is the same phenomenon as in the boxed example of Section 7.4: a first-class constraint plus a gauge-fixing condition becomes a second-class pair. The first-class constraint removes a gauge direction; the gauge condition chooses one representative on each gauge orbit. The combined set now has an invertible constraint matrix.

For second-class constraints $\chi_a \approx 0$, define

$$C_{ab}(x, y) := \{\chi_a(x), \chi_b(y)\}. \quad (689)$$

If C_{ab} is invertible, the Dirac bracket is

$$\{F, G\}_D = \{F, G\} - \int d^3 x d^3 y \{F, \chi_a(x)\} C_{ab}^{-1}(x, y) \{\chi_b(y), G\}. \quad (690)$$

This is not an arbitrary correction term. It is precisely the inverse of the symplectic form restricted to the gauge-fixed constraint surface. Hence it is a genuine Poisson bracket on the reduced phase space. In particular it is antisymmetric, obeys the Leibniz rule, and satisfies the Jacobi identity. It also kills all constraints:

$$\{F, \chi_a\}_D = 0. \quad (691)$$

Thus, after constructing the bracket, one may impose

$$A_0 = 0, \quad \Pi^0 = 0, \quad \partial_i A^i = 0, \quad \partial_i \Pi^i = 0 \quad (692)$$

strongly. This is the rigorous Hamiltonian meaning of quantizing only the two transverse photon polarizations.

For the transverse Maxwell variables, the result is

$$\{A_i(\mathbf{x}), \Pi^j(\mathbf{y})\}_D = \left(\delta_i^j - \frac{\partial_i \partial^j}{\nabla^2} \right) \delta^3(\mathbf{x} - \mathbf{y}) =: P_i^{jT} \delta^3(\mathbf{x} - \mathbf{y}), \quad (693)$$

with

$$\{A_i, A_j\}_D = 0, \quad \{\Pi^i, \Pi^j\}_D = 0. \quad (694)$$

The operator P^T projects any vector onto its transverse part.

On the reduced phase space the Hamiltonian is simply

$$H_{\text{phys}} = \frac{1}{2} \int d^3\mathbf{x} [\Pi_T^2 + \mathbf{B}^2]. \quad (695)$$

In the transverse oscillator basis of Eq. (674), this is again just a sum of harmonic-oscillator Hamiltonians. Using $\Pi_i^T = \dot{A}_i^T$, $\mathbf{B} = \nabla \times \mathbf{A}^T$, and the transverse polarization completeness relation, normal ordering gives

$$: H_{\text{phys}} := \sum_{h=\pm 1} \int \frac{d^3\mathbf{p}}{(2\pi)^3 2|\mathbf{p}|} |\mathbf{p}| a^\dagger(\mathbf{p}, h) a(\mathbf{p}, h). \quad (696)$$

Thus the free photon has two oscillator modes at each momentum, the two helicities $h = \pm 1$. Hamilton's equations with the Dirac bracket give

$$\dot{A}_i^T = \{A_i^T, H_{\text{phys}}\}_D = \Pi_i^T, \quad \dot{\Pi}_i^T = \{\Pi_i^T, H_{\text{phys}}\}_D = \nabla^2 A_i^T. \quad (697)$$

Therefore

$$(\partial_t^2 - \nabla^2) A_i^T = 0, \quad \partial_i A_i^T = 0. \quad (698)$$

These are precisely the physical Maxwell equations in radiation gauge. Quantization is now performed by replacing $i\{\cdot, \cdot\}_D$ with the commutator of operators:

$$[A_i^T(t, \mathbf{x}), \Pi_j^T(t, \mathbf{y})] = i P_i^{jT} \delta^3(\mathbf{x} - \mathbf{y}). \quad (699)$$

Equivalently, the Hamilton equations may be written as

$$i\dot{A}_i^T = [A_i^T, H_{\text{phys}}], \quad i\dot{\Pi}_i^T = [\Pi_i^T, H_{\text{phys}}]. \quad (700)$$

Again this equal-time commutator is the equal-time derivative of the covariant causal commutator. In radiation gauge the transverse potential has the schematic commutator

$$[A_i^T(x), A_j^T(y)] = P_{ij}^T \Delta_0(x - y), \quad (701)$$

where Δ_0 is the massless Pauli–Jordan function and $P_{ij}^T = \delta_{ij} - \partial_i \partial_j / \nabla^2$ is the transverse projector. Since $\Pi_j^T = \dot{A}_j^T$,

$$[A_i^T(t, \mathbf{x}), \Pi_j^T(t, \mathbf{y})] = \partial_{y^0} [A_i^T(x), A_j^T(y)]|_{x^0=y^0=t} \quad (702)$$

$$= -P_{ij}^T \partial_{z^0} \Delta_0(z)|_{z^0=0} = i P_{ij}^T \delta^3(\mathbf{x} - \mathbf{y}). \quad (703)$$

This is the precise Hamiltonian counterpart of the microcausal field-strength commutator in Eq. (678): after gauge fixing, the Dirac bracket quantizes only the transverse physical degrees of freedom, and their commutator is still inherited from the Pauli–Jordan distribution.

8.5 Discrete symmetries, CPT, and QED

For the photon,

$$CA_\mu(x)C^{-1} = -A_\mu(x). \quad (704)$$

This sign is required because the photon couples to the electric current, and the current is odd under charge conjugation. Parity and time reversal act as vector transformations:

$$PA^\mu(t, \mathbf{x})P^{-1} = P^\mu{}_\nu A^\nu(t, -\mathbf{x}), \quad (705)$$

$$TA^\mu(t, \mathbf{x})T^{-1} = P^\mu{}_\nu A^\nu(-t, \mathbf{x}), \quad (706)$$

with T antiunitary. These rules imply the familiar transformations of the electric and magnetic fields:

$$P: \quad \mathbf{E} \rightarrow -\mathbf{E}, \quad \mathbf{B} \rightarrow \mathbf{B}, \quad T: \quad \mathbf{E} \rightarrow \mathbf{E}, \quad \mathbf{B} \rightarrow -\mathbf{B}, \quad C: \quad \mathbf{E} \rightarrow -\mathbf{E}, \quad \mathbf{B} \rightarrow -\mathbf{B}. \quad (707)$$

The free Maxwell term is therefore invariant under C , P , and T separately:

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2). \quad (708)$$

Now couple the photon to the Dirac field:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu\psi A_\mu. \quad (709)$$

The current

$$j^\mu = \bar{\psi}\gamma^\mu\psi \quad (710)$$

transforms as follows. Under charge conjugation, $j^\mu \rightarrow -j^\mu$, while $A_\mu \rightarrow -A_\mu$, so $j^\mu A_\mu$ is invariant. Under parity, both j^μ and A^μ transform as vectors, so their contraction is invariant. Under time reversal, both transform as vectors with $t \rightarrow -t$, and the antiunitarity of T is already accounted for in the Dirac kinetic term. Thus QED is invariant under C , P , and T separately, with the usual convention that no theta-like pseudoscalar term is included.

Finally, CPT follows as before. The photon is a local bosonic field, its gauge-invariant field strength has causal commutators, and the Dirac field has causal anticommutators. Therefore every Hermitian local ordinary scalar built from $F_{\mu\nu}$, ψ , $\bar{\psi}$, and covariant derivatives is invariant under CPT, with $x \mapsto -x$ and complex conjugation of numerical coefficients from antiunitarity.

9 How Einstein's gravity comes from QFT

The lesson of the previous section was that massless spin one is not obtained by first writing Maxwell's action. Rather, one begins with the massless little group and asks for a local Lorentz-covariant field that creates the helicity states. The little-group translations then force the vector potential to be identified up to a gauge transformation. The same logic becomes even more constraining for massless spin two.

A massless spin-two particle has one-particle states

$$|p, h\rangle, \quad p^2 = 0, \quad h = \pm 2. \quad (711)$$

The standard covariant field that can create such states is a symmetric tensor potential $h_{\mu\nu}(x)$. Its polarization tensors may be written as products of spin-one helicity polarizations,

$$\epsilon_{\mu\nu}(p, \pm 2) = \epsilon_\mu(p, \pm 1)\epsilon_\nu(p, \pm 1), \quad p^\mu\epsilon_{\mu\nu}(p, h) = 0, \quad \eta^{\mu\nu}\epsilon_{\mu\nu}(p, h) = 0. \quad (712)$$

Under a little-group translation, the spin-one polarization shifts by a multiple of p_μ . Therefore the spin-two polarization shifts by

$$\epsilon_{\mu\nu}(p, h) \longrightarrow \epsilon_{\mu\nu}(p, h) + p_\mu \xi_\nu(p, h) + p_\nu \xi_\mu(p, h), \quad (713)$$

up to terms proportional to $p_\mu p_\nu$, which are contained in the same linearized gauge orbit. In position space this means that a local covariant spin-two potential cannot be a strictly invariant tensor field. It must be identified modulo

$$h_{\mu\nu}(x) \sim h_{\mu\nu}(x) + \partial_\mu \xi_\nu(x) + \partial_\nu \xi_\mu(x). \quad (714)$$

This is the spin-two analogue of $A_\mu \sim A_\mu + \partial_\mu \alpha$ for the photon.

Equation (714) is precisely the linearized form of a diffeomorphism. To see this, write the full metric as

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}. \quad (715)$$

An infinitesimal coordinate transformation generated by a vector field $\xi^\mu(x)$ changes the metric by its Lie derivative,

$$\delta_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu. \quad (716)$$

At lowest order around flat space this becomes

$$\delta_\xi h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu. \quad (717)$$

Thus the gauge redundancy forced by the massless spin-two little group is the linearized version of diffeomorphism invariance.

The free gauge-invariant curvature is the linearized Riemann tensor,

$$R_{\mu\nu\rho\sigma}^{(1)}[h] = \frac{1}{2}(\partial_\rho \partial_\nu h_{\mu\sigma} + \partial_\sigma \partial_\mu h_{\nu\rho} - \partial_\rho \partial_\mu h_{\nu\sigma} - \partial_\sigma \partial_\nu h_{\mu\rho}), \quad (718)$$

and the simplest free equations are the linearized Einstein equations. Once interactions are turned on, the gauge symmetry must also be deformed. The natural nonlinear completion of the linearized symmetry is full diffeomorphism invariance, and the simplest local two-derivative action with this symmetry is Einstein-Hilbert:

$$S_{\text{EH}} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} (R - 2\Lambda) + S_{\text{matter}}[g, \Phi]. \quad (719)$$

Therefore, in the same sense in which quantum mechanics plus Lorentz covariance pushes massless spin one toward gauge theory, it pushes massless spin two toward linearized diffeomorphism invariance, whose simplest nonlinear completion is Einstein's gravity.

This is not a derivation of all of general relativity from nothing. The point is more precise: if a local Lorentz-covariant QFT contains a massless helicity ± 2 particle that couples consistently at long distances, then the local field used to describe it must be gauge-redundant, and the required gauge redundancy has the form of diffeomorphism invariance.

10 The Weinberg-Witten theorem

The previous section may suggest that massless particles of spin one and spin two are just ordinary particles described by special covariant fields. The Weinberg-Witten theorem explains why this is too naive. It says that, under standard assumptions, sufficiently high-helicity massless particles cannot carry ordinary Lorentz-covariant conserved currents or ordinary Lorentz-covariant energy-momentum.

The theorem is useful pedagogically because it isolates the obstruction very sharply. The obstruction is not simply “high spin is hard”. The obstruction is that a physical helicity state transforms with little-group phases, whereas a local covariant current or stress tensor has only finitely many tensor components with definite spin under ordinary rotations.

10.1 Assumptions and notation

We assume the following.

- (i) The theory is a four-dimensional Poincare-invariant quantum theory with a unitary representation $U(\Lambda, a)$ on the Hilbert space.
- (ii) The theory contains massless one-particle states $|p, h, a\rangle$ with $p^2 = 0$ and helicity h . The label a denotes any internal degeneracy.
- (iii) There is either a conserved Lorentz-covariant current $J^\mu(x)$ or a conserved Lorentz-covariant stress tensor $T^{\mu\nu}(x)$:

$$U(\Lambda)J^\mu(x)U(\Lambda)^{-1} = \Lambda^\mu{}_\nu J^\nu(\Lambda x), \quad (720)$$

and similarly for $T^{\mu\nu}$ with two Lorentz indices.

- (iv) The current or stress tensor is a genuine local observable. If gauge symmetries are present, this means gauge-invariant as well as Lorentz-covariant.
- (v) The relevant charge or momentum is obtained by integrating the local density:

$$Q = \int d^3\mathbf{x} J^0(t, \mathbf{x}), \quad P^\mu = \int d^3\mathbf{x} T^{0\mu}(t, \mathbf{x}). \quad (721)$$

The corresponding forward matrix elements are assumed to be obtained as the limit of matrix elements with spacelike momentum transfer.

The last assumption is the physically important one. Charges and momenta are measured by long-wavelength probes. Therefore the matrix element at zero momentum transfer is defined as a limit of matrix elements with small spacelike momentum transfer.

10.2 A helicity selection rule

The key step in both parts of the theorem is the same. Consider two massless momenta p and p' with spacelike momentum transfer $q = p' - p$. Since $q^2 < 0$, there is a Breit frame in which

$$p^\mu = (E, 0, 0, E), \quad p'^\mu = (E, 0, 0, -E). \quad (722)$$

Now rotate around the z -axis by an angle θ . This rotation leaves both momenta unchanged, but the helicity phases of the two states are opposite, because the second momentum points along $-z$:

$$U(R_z(\theta))|p, h\rangle = e^{-ih\theta}|p, h\rangle, \quad U(R_z(\theta))|p', h\rangle = e^{+ih\theta}|p', h\rangle. \quad (723)$$

Therefore a matrix element between equal-helicity states carries the net phase

$$\langle p', h | \mathcal{O}(0) | p, h \rangle \longrightarrow e^{-2ih\theta} \langle p', h | \mathcal{O}(0) | p, h \rangle. \quad (724)$$

where the remaining transformation comes from the Lorentz indices of the local operator $\mathcal{O}$ itself.

Let $M^\mu = \langle p', h | J^\mu(0) | p, h \rangle$. Under $R_z(\theta)$, the components of a four-vector decompose into rotation weights

$$M^0, M^3 : m = 0, \quad M^1 \pm iM^2 : m = \pm 1. \quad (725)$$

Covariance and (724) imply that a component with rotation weight m can be nonzero only if

$$m = 2h. \quad (726)$$

Since a four-vector has only $m = 0, \pm 1$, all components of M^μ vanish for $|h| > 1/2$.

Similarly, let

$$M^{\mu\nu} = \langle p', h | T^{\mu\nu}(0) | p, h \rangle. \quad (727)$$

A rank-two Lorentz tensor has rotation weights

$$m = 0, \pm 1, \pm 2. \quad (728)$$

Thus covariance implies that a component can be nonzero only if

$$m = 2h. \quad (729)$$

For $|h| > 1$, no component can be nonzero.

This selection rule is the heart of the theorem.

10.3 The current condition: no charged massless particles with $j > 1/2$

Theorem 1 (Weinberg-Witten current condition). A Poincare-invariant QFT that contains a conserved Lorentz-covariant current J^μ cannot contain massless particles of spin $j > 1/2$ that carry a nonzero charge associated with that current.

Proof. Let $Q = \int d^3\mathbf{x} J^0(x)$ be the charge. Translational covariance gives

$$\langle p', h, b | J^0(t, \mathbf{x}) | p, h, a \rangle = e^{i(p' - p) \cdot x} \langle p', h, b | J^0(0) | p, h, a \rangle. \quad (730)$$

Integrating over $\mathbf{x}$ at fixed time gives

$$\langle p', h, b | Q | p, h, a \rangle = (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \langle p', h, b | J^0(0) | p, h, a \rangle. \quad (731)$$

If the particle carries a nonzero charge, we may diagonalize the finite charge matrix in the degeneracy labels and choose a charged one-particle state for which

$$Q | p, h, a \rangle = q_a | p, h, a \rangle, \quad q_a \neq 0. \quad (732)$$

Equivalently, in the forward limit,

$$\lim_{p' \rightarrow p} \langle p', h, a | J^0(0) | p, h, a \rangle \neq 0. \quad (733)$$

But for spacelike momentum transfer the selection rule (726) says that every component of $\langle p', h | J^\mu(0) | p, h \rangle$ vanishes if $|h| > 1/2$. Taking the spacelike momentum transfer to zero then gives a vanishing forward matrix element, contradicting (733). Therefore a massless particle with $|h| = j > 1/2$ cannot carry the charge of a Lorentz-covariant conserved current. $\square$

10.4 The energy-momentum condition: no ordinary stress tensor for massless particles with $j > 1$

Theorem 2 (Weinberg-Witten energy-momentum condition). A Poincare-invariant QFT that contains a conserved Lorentz-covariant energy-momentum tensor $T^{\mu\nu}$ cannot contain massless particles of spin $j > 1$ that carry nonzero energy and momentum.

Proof. The four-momentum is

$$P^\nu = \int d^3\mathbf{x} T^{0\nu}(t, \mathbf{x}). \quad (734)$$

By translational covariance,

$$\langle p', h, b | P^\nu | p, h, a \rangle = (2\pi)^3 \delta^3(\mathbf{p}' - \mathbf{p}) \langle p', h, b | T^{0\nu}(0) | p, h, a \rangle. \quad (735)$$

A one-particle momentum eigenstate has

$$P^\nu |p, h, a\rangle = p^\nu |p, h, a\rangle, \quad (736)$$

so its forward matrix element of $T^{0\nu}$ is nonzero; in particular its energy is positive.

However, for spacelike momentum transfer the selection rule (729) says that all components of $\langle p', h | T^{\mu\nu}(0) | p, h \rangle$ vanish when $|h| > 1$. Taking the spacelike momentum transfer to zero would then make the forward matrix element of $T^{0\nu}$ vanish. This contradicts the fact that the particle carries nonzero four-momentum. Hence a massless particle with $j = |h| > 1$ cannot coexist with a conserved Lorentz-covariant energy-momentum tensor whose integral gives its four-momentum. $\square$

10.5 Why this does not contradict photons, gluons, or gravitons

The assumptions are as important as the conclusion.

First, the photon is a massless spin-one particle, but it is not electrically charged. Its helicities $h = \pm 1$ are compatible with the existence of the Lorentz-covariant electromagnetic current because the photon carries zero charge under that current. A physical massless spin-one particle with nonzero charge under an ordinary gauge-invariant Lorentz-covariant current would be excluded by the current condition. This is why one should be careful with the phrase “charged massless vector particle”. The physical charged weak bosons $W^\pm$ are massive in the broken electroweak phase, so the Weinberg-Witten obstruction to massless charged spin-one particles does not apply to them. In the unbroken gauge theory, the weak gauge bosons are gauge fields rather than ordinary gauge-invariant charged particle states, so the assumptions of the theorem are also not applied in the naive way.

Second, non-abelian gauge bosons, such as gluons, carry gauge labels. The corresponding color charge is not represented by a gauge-invariant local Lorentz-covariant current acting on isolated physical one-particle gluon states. Therefore the theorem does not forbid non-abelian gauge theory. The theorem forbids massless spin-one particles carrying an ordinary global charge measured by a genuine local covariant current.

Third, the graviton is a massless spin-two particle and certainly carries energy. This seems to contradict the second theorem, but the escape is exactly what we learned in the previous section. In a diffeomorphism-invariant theory there is no gauge-invariant local Lorentz-covariant stress tensor for the gravitational field itself. Gravitational energy is encoded in boundary charges or in coordinate-dependent pseudotensors, not in a local gauge-invariant tensor $T_{\text{grav}}^{\mu\nu}(x)$. Therefore general relativity violates one of the assumptions of the theorem, not the theorem itself.

The same logic explains why a massless spin- $\frac{3}{2}$ gravitino can exist in supergravity. A gravitino is not an ordinary matter particle of a flat-space QFT with a gauge-invariant local stress tensor. It is the gauge field of local supersymmetry and has its own gauge redundancy,

$$\psi_\mu(x) \longrightarrow \psi_\mu(x) + \partial_\mu \epsilon(x) \quad \text{at the linearized level.} \quad (737)$$

The physical helicities are only $h = \pm \frac{3}{2}$; the lower spin components of ψ_μ are gauge. If one tried to regard such a particle as an ordinary massless spin- $\frac{3}{2}$ excitation carrying energy measured by a genuine local Lorentz-covariant stress tensor, the energy-momentum part of the Weinberg-Witten theorem would apply because $\frac{3}{2} > 1$. Supergravity evades this in the same way as the graviton: once gravity is dynamical, the total energy-momentum is not represented by a gauge-invariant local Lorentz-covariant tensor for the gravitational and gravitino fields. It is represented by constraints and boundary charges. Hence the theorem does not say that a gravitino cannot exist; it says that a massless spin- $\frac{3}{2}$ particle cannot be an ordinary charged excitation of a Lorentz-invariant local QFT with a standard covariant stress tensor carrying its energy.

10.6 Consequences for emergent gravity

The Weinberg-Witten theorem strongly constrains emergent-graviton proposals. An ordinary Lorentz-invariant QFT in flat spacetime has a conserved, local, Lorentz-covariant stress tensor. If such a theory also had a composite or emergent massless spin-two particle that carried the energy and momentum measured by this stress tensor, the energy-momentum condition would forbid it. Thus an emergent graviton cannot arise in a completely ordinary way from a flat-space QFT with a standard gauge-invariant local stress tensor.

The theorem can be evaded only by leaving at least one assumption. Examples include:

- (i) the absence of exact Lorentz invariance in the microscopic theory;
- (ii) the absence of a gauge-invariant local stress tensor for the would-be graviton;
- (iii) holographic emergence, where the graviton is not a particle of the same flat-space QFT in which the boundary stress tensor lives;
- (iv) nonlocality or the absence of ordinary asymptotic one-particle states;
- (v) theories with diffeomorphism gauge redundancy, where gravitational energy is not represented by a local covariant stress tensor.

Therefore the correct conclusion is not that gravity cannot be emergent in any sense. The correct conclusion is sharper: a massless spin-two graviton cannot emerge as an ordinary charged excitation of a Lorentz-invariant local QFT that possesses a genuine local Lorentz-covariant stress tensor carrying the graviton's energy and momentum.